

Siemens PTI Report R078-21

MISO DPP 2019 West Area Phase 2 Study

Prepared for

MISO

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Executive Summary

This report presents results of a System Impact Study (SIS) performed to evaluate the interconnection of the DPP 2019 Phase 2 West Area Group (DPP West Area) generating facilities. The results for 2024 scenario are summarized below.

1.1 Project List

The DPP West Area study group has thirty-six (36) generation projects with a combined nameplate rating of 5698 MW. The DPP West Area generating facilities are listed in Table ES-1. The modeling details and projects' slider diagrams are shown in Appendix B.

Table ES-1: Generating Facilities in DPP 2019 West Area Group

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW
J1190	NRIS	ITCM	Clinton	IA	Calamus East 161 kV	Solar	200	200	0	200
J1218	NRIS	MEC	Muscatine	IA	Sub 92 345 kV	Solar	200	200	0	200
J1221	NRIS	GRE	Wadena	MN	Wing River-Riverton 230 kV	Solar	150	150	0	150
J1229	NRIS	MEC	Muscatine	IA	Sub 92 345 kV	Battery	50	50	±50	50
J1270	NRIS	MEC	Palo Alto	IA	Kossuth - O'Brien 345 kV	Wind	200	200	200	31.2
J1298	NRIS	SMPMPA	Freeborn	MN	Hayward - Murphy Creek 161 kV	Solar	150	150	0	150
J1312	NRIS	XCEL	McCook	SD	Grant 115 kV	Solar	50	50	0	50
J1313	NRIS	ITCM	Greene	IA	Karma 161 kV	Solar	100	100	0	100
J1314	NRIS	DPC	Polk	WI	Apple River 161 kV	Solar	100	100	0	100
J1315	NRIS	XCEL	Cottonwood, Murray, Redwood	MN	Lyon County - Cedar Mountain 345 kV	Wind	600	600	600	93.6
J1321	NRIS	MEC	Palo Alto	IA	O'Brien - Kossuth 345 kV	Battery	65	65	±65	65
J1325	NRIS	ITCM	Rock	MN	Magnolia 161 kV	Solar	80	80	0	80
J1328	NRIS	OTP	Grant	SD	Big Stone South 345 kV	Solar	100	100	0	100
J1349	NRIS	XCEL	Cottonwood	MN	Crandall 345 kV	Hybrid	335	335	275	102.74
J1359	NRIS	ITCM	Linn	IA	Oakridge 69 kV	Gas	95	95	0	95
J1365	NRIS	ITCM	Tama	IA	Traer 161 kV	Wind	169.4	169.4	169.4	26.426
J1371	NRIS	OTP	Yellow Medicine	MN	Astoria 345 kV	Wind	150	150	150	23.4
J1395	NRIS	XCEL	Lyon	MN	Nest Lake - Lyon County 345 kV	Solar	200	200	0	200

MISO Project #	Service Type	TO	County	State	Point of Interconnection	Fuel Type	ERIS Output	NRIS Output	SH MW	SPK MW
J1413	NRIS	ITCM	Howard, Mitchell	IA	Hazleton - Mitchell County 345 kV	Solar	235	215	0	235
J1416	NRIS	ITCM	Worth	IA	Worth County - Glenworth 161 kV	Solar	149	149	0	149
J1417	NRIS	ITCM	Worth	IA	Ledyard - Colby 345 kV	Battery	75	75	±75	75
J1426	NRIS	GRE	Benton	MN	Benton County 115 kV	Solar	100	100	0	100
J1432	NRIS	ITCM	Clinton	IA	Salem - Rock Creek 345 kV	Wind	250	250	250	39
J1438	NRIS	ITCM	Fayette	IA	Windsor - Hazleton 161 kV	Solar	100	100	0	100
J1444	NRIS	ITCM	Linn	IA	Duane Arnold 345 kV	Solar	200	200	0	200
J1445	NRIS	XCEL	Benton	MN	Mayhew Lake 115 kV	Solar	100	100	0	100
J1446	NRIS	XCEL	Wright	MN	Lake Pulaski 115 kV	Battery	150	150	±150	150
J1456	NRIS	OTP	Sheridan	ND	Harvey - Underwood 230 kV	Wind	300	300	300	46.8
J1468	NRIS	XCEL	Dakota	MN	West Hastings 115 kV	Battery	100	100	±100	100
J1471	NRIS	ITCM	Lee	IA	Wever - Burlington Generator 161 kV	Solar	200	200	0	200
J1474	NRIS	XCEL	Chippewa	WI	Pine Lake - Eagle Point 115 kV	Solar	200	200	0	200
J1477	NRIS	MEC	Scott	IA	Sub 56 161 kV (J1131)	Battery	50	50	±50	50
J1478	NRIS	ITCM	Des Moines	IA	Huntwoods 69 kV (J1135)	Battery	25	25	±25	25
J1479	NRIS	ITCM	Des Moines	IA	Burlington 161 kV	Battery	100	100	±100	100
J1485	ERIS	OTP	Deuel	SD	Astoria 345 kV (J493/J510)	Wind	250	0	250	39
J1487	NRIS	ITCM	Freeborn	MN	Freeborn 161 kV	Wind	120	120	120	18.72

1.2 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

Per FERC Order 827, non-synchronous generation projects in the DPP 2019 West Area study group are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

The reactive power requirement analysis results are summarized as following:

- J1190, J1218, J1221, J1270, J1298, J1365, J1395, and J1485 do not meet the reactive power requirements per FERC Order 827.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection Service as of the System Impact Study report date. The total cost of network upgrades in the interconnection plan required for each generation project is listed in Table ES-2. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies.

Table ES-2: Total Cost of Network Upgrades for DPP 2019 West Area Generation Projects

Project Num	ERIS Network Upgrades (\$)													NRIS Network Upgrades (\$)	Inter- connection Substation TO NUs (\$)	TO's Inter- connection Facilities (TOIF)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)	M2 Received (\$)	M3 Received (\$)	M4 (\$)
	Base Case NUs	MWEX Voltage Stability	MISO Thermal & Voltage	Transient Stability	Short Circuit	GRE LPC	OTP LPC	DPC LPC	CIPCO AFS	MPC AFS	PJM AFS	AECI AFS	SPP AFS								
J1190	\$92,819	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$144,940	\$770,994	\$0	\$237,759	\$800,000	\$0	\$0
J1218	\$92,819	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$195,760	\$0	\$712,410	\$16,500,000	\$1,900,000	\$0	\$17,305,229	\$800,000	\$478,388	\$2,182,658
J1221	\$0	\$0	\$0	\$0	\$0	\$3,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$1,084,499	\$6,906,864	\$6,748,843	\$1,135,462	\$0	\$16,655,707	\$600,000	\$2,168,684	\$562,457
J1229	\$92,819	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$48,940	\$0	\$178,103	\$0	\$50,000	\$0	\$270,922	\$200,000	\$119,597	\$0
J1270	\$7,425,527	\$0	\$125,000	\$0	\$0	\$0	\$532,454	\$0	\$0	\$0	\$0	\$0	\$5,776,838	\$10,236,763	\$14,500,000	\$1,250,000	\$0	\$32,819,744	\$800,000	\$1,041,364	\$4,722,585
J1298	\$92,819	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$20,190,626	\$14,936,059	\$4,532,750	\$652,600	\$0	\$19,561,628	\$600,000	\$886,712	\$2,425,614
J1312	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$109,371,245	\$1,797,327	\$0	\$0	\$0	\$1,797,327	\$200,000	\$330,043	\$0
J1313	\$92,819	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$922,138	\$71,027	\$533,838	\$410,696	\$0	\$697,684	\$400,000	\$0	\$0
J1314	\$0	\$0	\$2,636,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$3,774,900	\$3,065,758	\$3,568,598	\$702,331	\$0	\$9,270,356	\$400,000	\$16,267	\$1,437,804
J1315	\$81,161,005	\$0	\$49,289,716	\$0	\$0	\$0	\$248,479	\$0	\$0	\$0	\$0	\$543,195	\$56,854,448	\$22,809,983	\$12,702,000	\$1,904,000	\$0	\$166,211,183	\$2,400,000	\$134,587	\$30,707,650
J1321	\$0	\$0	\$0	\$0	\$0	\$0	\$141,988	\$0	\$0	\$0	\$0	\$0	\$1,855,281	\$2,873,328	\$0	\$50,000	\$0	\$3,015,316	\$260,000	\$0	\$343,063
J1325	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$10,236,825	\$15,764,873	\$486,788	\$417,240	\$0	\$16,251,661	\$800,000	\$9,785,471	\$0
J1328	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$11,320,665	\$4,150,891	\$3,164,471	\$9,797,117	\$0	\$7,315,362	\$400,000	\$0	\$1,063,072
J1349	\$4,455,316	\$0	\$61,847,619	\$0	\$0	\$0	\$461,460	\$0	\$0	\$0	\$0	\$0	\$19,776,906	\$17,886,745	\$3,470,000	\$1,956,000	\$0	\$88,121,141	\$1,340,000	\$105,002	\$16,179,226
J1359	\$92,819	\$0	\$450,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$380,782	\$338,253	\$704,588	\$616,079	\$0	\$1,585,660	\$380,000	\$0	\$0
J1365	\$1,021,010	\$0	\$22,059	\$0	\$0	\$0	\$70,994	\$0	\$0	\$0	\$0	\$0	\$1,429,545	\$25,475,219	\$1,511,232	\$444,318	\$0	\$28,100,514	\$680,000	\$979,708	\$3,960,395
J1371	\$22,740,675	\$0	\$1,117,647	\$0	\$0	\$0	\$10,294,118	\$0	\$0	\$0	\$0	\$0	\$18,538,228	\$6,225,759	\$1,705,426	\$938,201	\$0	\$42,083,624	\$600,000	\$0	\$7,816,725
J1395	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$21,573,969	\$8,980,434	\$12,761,000	\$1,930,000	\$0	\$21,741,434	\$800,000	\$612,342	\$2,935,945
J1413	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$2,055,099	\$10,426,350	\$11,622,272	\$800,418	\$0	\$22,048,622	\$1,200,000	\$1,232,794	\$1,976,930
J1416	\$92,819	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$27,189,536	\$2,788,541	\$7,329,141	\$548,670	\$0	\$10,210,502	\$596,000	\$921,718	\$524,382
J1417	\$0	\$0	\$0	\$0	\$0	\$0	\$70,994	\$0	\$0	\$0	\$0	\$0	\$10,347,897	\$2,111,900	\$0	\$15,000	\$0	\$2,182,894	\$300,000	\$802,170	\$0
J1426	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$4,887,081	\$5,198,145	\$1,362,037	\$1,024,943	\$0	\$6,560,182	\$400,000	\$72,804	\$839,232
J1432	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$56,780,000	\$0	\$0	\$700,083	\$13,522,712	\$761,244	\$0	\$14,222,795	\$1,000,000	\$158,085	\$1,686,474
J1438	\$92,819	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$919,467	\$384,556	\$7,106,241	\$548,670	\$0	\$7,583,616	\$400,000	\$646,056	\$470,667
J1444	\$92,819	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$826,139	\$964,321	\$8,003,452	\$730,222	\$0	\$9,060,592	\$800,000	\$0	\$1,012,118
J1445	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$4,924,724	\$5,183,191	\$2,413,000	\$1,369,000	\$0	\$7,596,191	\$400,000	\$161,548	\$957,690
J1446	\$19,120,731	\$0	\$5,341,338	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$7,117,973	\$8,075,275	\$10,544,000	\$1,311,000	\$0	\$43,081,345	\$600,000	\$25,326	\$7,990,943

Project Num	ERIS Network Upgrades (\$)													NRIS Network Upgrades (\$)	Inter-connection Substation TO NUs (\$)	TO's Inter-connection Facilities (TOIF)	SNU (\$)	Total Network Upgrade Cost (Exclude TOIF & Affected System) (\$)	M2 Received (\$)	M3 Received (\$)	M4 (\$)
	Base Case NUs	MWEX Voltage Stability	MISO Thermal & Voltage	Transient Stability	Short Circuit	GRE LPC	OTP LPC	DPC LPC	CIPCO AFS	MPC AFS	PJM AFS	AECI AFS	SPP AFS								
J1456	\$29,145,192	\$0	\$1,911,765	\$0	\$0	\$0	\$6,992,901	\$0	\$0	\$60,000,000	\$0	\$271,588	\$2,476,406	\$11,463,212	\$7,267,428	\$413,590	\$0	\$56,780,497	\$1,200,000	\$0	\$10,156,099
J1468	\$9,281,908	\$0	\$19,029,899	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$4,426,235	\$0	\$4,664,000	\$698,000	\$0	\$32,975,807	\$600,000	\$0	\$5,995,161
J1471	\$92,819	\$0	\$14,250,603	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$532,261	\$0	\$11,881,166	\$551,224	\$503,193	\$0	\$26,775,812	\$800,000	\$807,805	\$3,747,357
J1474	\$92,819	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$6,805,294	\$8,828,114	\$8,257,000	\$1,248,000	\$0	\$17,177,933	\$800,000	\$268,331	\$2,367,256
J1477	\$185,638	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$160,255	\$0	\$50,000	\$0	\$345,894	\$200,000	\$0	\$0
J1478	\$0	\$0	\$884,281	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$61,450	\$0	\$1,150,033	\$0	\$558,384	\$0	\$2,034,314	\$100,000	\$7,972	\$298,891
J1479	\$0	\$0	\$6,070,116	\$0	\$0	\$0	\$35,497	\$0	\$0	\$0	\$0	\$258,529	\$0	\$6,608,614	\$0	\$0	\$0	\$12,714,227	\$400,000	\$904,630	\$1,238,215
J1485	\$35,271,251	\$0	\$1,867,647	\$0	\$0	\$0	\$16,151,116	\$0	\$0	\$0	\$0	\$0	\$25,842,417	\$0	\$1,705,426	\$814,313	\$0	\$54,995,440	\$1,000,000	\$0	\$9,999,088
J1487	\$0	\$0	\$25,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$17,399,893	\$6,161,603	\$579,344	\$11,268,840	\$0	\$31,740,947	\$480,000	\$330,031	\$5,538,158
Total (\$)	\$210,829,263	\$0	\$189,843,690	\$0	\$0	\$3,000,000	\$35,000,000	\$0	\$0	\$60,000,000	\$56,780,000	\$1,911,723	\$398,305,056	\$224,495,157	\$167,961,751	\$47,588,525	\$0	\$831,129,861	\$23,736,000	\$22,997,435	\$129,135,858

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models, and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

1.4 Per Project Summary

This section provides estimated cost of Network Upgrades on a per project basis for the 2024 scenario. The shared cost of Network Upgrades for all the generation projects are listed below.

The Interconnection Customers are required to mitigate the constraints observed from the 2024 scenario.

1.4.1 J1190 Summary

Network Upgrade	Cost	J1190	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$92,819		

1.4.2 J1218 Summary

Network Upgrade	Cost	J1218	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$112,486	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$599,924	NRIS	Yes
AECI AFS Network Upgrade	\$1,911,723	\$195,760	AECI AFS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,000,989		

1.4.3 J1221 Summary

Network Upgrade	Cost	J1221	NUs Type	Self Funding?
Wing River 230-115-13.8 kV xfmr	\$3,000,000	\$3,000,000	J1221 GRE LPC	No
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$906,944	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$40,040	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$184,583	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$227,543	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$27,466	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$22,647	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$27,382	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$37,076	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$4,845,363	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$587,821	NRIS	Yes
SPP AFS NRIS Network Upgrade	\$122,704,700	\$1,084,499	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$10,991,363		

1.4.4 J1229 Summary

Network Upgrade	Cost	J1229	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$28,121	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$149,981	NRIS	Yes
AECI AFS Network Upgrade	\$1,911,723	\$48,940	AECI AFS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$319,862		

1.4.5 J1270 Summary

Network Upgrade	Cost	J1270	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$7,425,527	Base Case NU	Yes
Add 2x40 Mvar switched capacitor at Audubon 230 kV (620336)	\$4,000,000	\$125,000	MISO Voltage	Yes
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$532,454	OTP LPC	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$2,631,648	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$373,266	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$417,405	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$191,752	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$34,966	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$29,799	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$35,836	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$90,456	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$1,838,174	NRIS	Yes
Install 2nd 345/161 kV Webster Transformer	\$10,000,000	\$4,098,102	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$495,358	NRIS	Yes
SPP AFS ERS Network Upgrade	\$275,600,356	\$2,205,552	SPP AFS ERS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$3,571,286	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$24,096,582		

1.4.6 J1298 Summary

Network Upgrade	Cost	J1298	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$46,191	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$246,353	NRIS	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$971,563	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$57,637	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$166,384	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$387,832	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$46,163	NRIS	Yes

Network Upgrade	Cost	J1298	NUs Type	Self Funding?
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$36,993	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$45,290	NRIS	Yes
J1181 POI-Hazleton 345 Structure Remediation	\$15,500,000	\$1,001,899	NRIS	Yes
Austin-Adams 161 kV Line Rebuild	\$15,750,000	\$11,929,753	NRIS	ITCM: Yes SMMPA: No
SPP AFS ERS Network Upgrade	\$275,600,356	\$7,876,897	SPP AFS ERS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$12,313,729	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$35,219,504		

1.4.7 J1312 Summary

Network Upgrade	Cost	J1312	NUs Type	Self Funding?
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$360,888	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$89,258	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$48,972	NRIS	Yes
Hazleton-Arnold 345 kV Structure Replacements	\$800,000	\$7,222	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$6,128	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$7,375	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$20,531	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$804,425	NRIS	Yes
J1181 POI-Hazleton 345 Structure Remediation	\$15,500,000	\$126,510	NRIS	Yes
Split Rock 230/115 kV Transformer Upgrade	\$5,000,000	\$326,018	NRIS	Yes
SPP AFS ERS Network Upgrade	\$275,600,356	\$81,710,893	SPP AFS ERS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$27,660,352	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$111,168,572		

1.4.8 J1313 Summary

Network Upgrade	Cost	J1313	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$71,027	NRIS	Yes
SPP AFS NRIS Network Upgrade	\$122,704,700	\$922,138	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,085,984		

1.4.9 J1314 Summary

Network Upgrade	Cost	J1314	NUs Type	Self Funding?
Apple River 161-69 kV xfmr #2	\$2,636,000	\$2,636,000	PK Discharge	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$503,354	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$28,641	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$98,646	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$97,943	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$17,951	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$14,867	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$17,878	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$2,033,458	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$253,020	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$3,193,893	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$581,007	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$9,476,658		

1.4.10 J1315 Summary

Network Upgrade	Cost	J1315	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$81,161,005	Base Case NU	Yes
Blue Lake-Scott Co 345 kV	\$20,250,000	\$16,451,057	SH Charge, SH Discharge	Yes
Blue Lake-Scott Co 345 kV (P2-3)	\$820,000	\$721,894	SH Charge, SH Discharge	No
Lyon County-Hazel Creek 345 kV	\$5,000,000	\$5,000,000	SH Charge, SH Discharge	Yes
Scott County 345-115-34.5 kV xfmr #9	\$15,000,000	\$15,000,000	SH Charge	Yes
Crystal Foods Tap-Arlington 69 kV	\$4,830,000	\$4,830,000	SH Charge, SH Discharge	Yes
Gibbon-Cairo 69 kV	\$7,000,000	\$7,000,000	SH Charge, SH Discharge	Yes
Add 2x40 Mvar switched capacitor at Audubon 230 kV (620336)	\$4,000,000	\$286,765	MISO Voltage	Yes
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$248,479	OTP LPC	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$4,244,496	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$149,271	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$876,985	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$884,590	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$112,346	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$92,867	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$111,309	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$179,550	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$12,200,748	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$2,285,190	NRIS	Yes
Split Rock 230/115 kV Transformer Upgrade	\$5,000,000	\$1,672,631	NRIS	Yes
AECI AFS Network Upgrade	\$1,911,723	\$543,195	AECI AFS	No
SPP AFS ERIS Network Upgrade	\$275,600,356	\$46,029,408	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$10,825,040	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$210,906,826		

1.4.11 J1321 Summary

Network Upgrade	Cost	J1321	NUs Type	Self Funding?
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$141,988	OTP LPC	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$873,462	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$121,312	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$130,399	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$61,983	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$11,621	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$9,910	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$11,917	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$29,580	NRIS	Yes
Install 2nd 345/161 kV Webster Transformer	\$10,000,000	\$1,463,019	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$160,124	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$702,286	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$1,152,995	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$4,870,597		

1.4.12 J1325 Summary

Network Upgrade	Cost	J1325	NUs Type	Self Funding?
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$689,882	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$28,508	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$151,941	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$84,143	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$13,094	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$11,100	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$13,352	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$32,433	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$1,181,018	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$217,370	NRIS	Yes
Split Rock 230/115 kV Transformer Upgrade	\$5,000,000	\$642,030	NRIS	Yes
Magnolia-Elk 161 kV Rebuild	\$12,700,000	\$12,700,000	NRIS	Yes

Network Upgrade	Cost	J1325	NUs Type	Self Funding?
SPP AFS ERIS Network Upgrade	\$275,600,356	\$8,284,359	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$1,952,466	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$26,001,698		

1.4.13 J1328 Summary

Network Upgrade	Cost	J1328	NUs Type	Self Funding?
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$692,300	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$153,819	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$127,274	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$16,854	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$14,057	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$16,987	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$33,570	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$2,440,150	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$328,792	NRIS	Yes
Split Rock 230/115 kV Transformer Upgrade	\$5,000,000	\$327,087	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$9,446,750	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$1,873,915	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$15,471,556		

1.4.14 J1349 Summary

Network Upgrade	Cost	J1349	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$4,455,316	Base Case NU	Yes
Wilmarth-Sheas Lake 345 kV	\$61,750,000	\$47,375,417	SH Charge	Yes
Helena-Sheas Lake 345 kV	\$18,750,000	\$14,361,908	SH Charge	Yes
Add 2x40 Mvar switched capacitor at Audubon 230 kV (620336)	\$4,000,000	\$110,294	MISO Voltage	Yes
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$461,460	OTP LPC	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$3,215,493	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$267,380	NRIS	Yes

Network Upgrade	Cost	J1349	NUs Type	Self Funding?
Raun-S3451 345 kV Rebuild	\$5,000,000	\$611,578	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$409,488	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$62,727	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$52,570	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$63,474	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$126,015	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$5,497,682	NRIS	Yes
Install 2nd 345/161 kV Webster Transformer	\$10,000,000	\$3,673,117	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$1,057,844	NRIS	Yes
Split Rock 230/115 kV Transformer Upgrade	\$5,000,000	\$849,378	NRIS	Yes
Lakefield-Lakefield Junction 345 kV Structure Remediation	\$2,000,000	\$2,000,000	NRIS	Yes
SPP AFS ERS Network Upgrade	\$275,600,356	\$13,472,065	SPP AFS ERS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$6,304,841	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$104,428,047		

1.4.15 J1359 Summary

Network Upgrade	Cost	J1359	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
PCI J2-Oakridge 69 kV	\$450,000	\$450,000	PK Discharge	ITCM: Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$46,642	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$248,755	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$42,857	NRIS	Yes
SPP AFS NRIS Network Upgrade	\$122,704,700	\$380,782	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,261,854		

1.4.16 J1365 Summary

Network Upgrade	Cost	J1365	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$1,021,010	Base Case NU	Yes
Add 2×40 Mvar switched capacitor at Audubon 230 kV (620336)	\$4,000,000	\$22,059	MISO Voltage	Yes
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$70,994	OTP LPC	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$67,126	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$358,005	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$50,089	NRIS	Yes
Laurel-Jasper 161 kV Rebuild	\$12,500,000	\$12,500,000	NRIS	Yes
M-TOWN-Laurel 161 kV Rebuild	\$12,500,000	\$12,500,000	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$420,151	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$1,009,394	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$28,018,827		

1.4.17 J1371 Summary

Network Upgrade	Cost	J1371	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$22,740,675	Base Case NU	Yes
Canby-Granites Falls 115 kV	\$2,000,000	\$750,000	SH Charge, SH Discharge	OTP: Yes
Add 2×40 Mvar switched capacitor at Audubon 230 kV (620336)	\$4,000,000	\$367,647	MISO Voltage	Yes
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$10,294,118	OTP LPC	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$1,053,188	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$235,278	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$189,943	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$25,227	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$21,061	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$25,422	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$51,017	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$3,598,414	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$490,685	NRIS	Yes
Split Rock 230/115 kV Transformer Upgrade	\$5,000,000	\$535,524	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$15,505,450	SPP AFS ERIS	No

Network Upgrade	Cost	J1371	NUs Type	Self Funding?
SPP AFS NRIS Network Upgrade	\$122,704,700	\$3,032,778	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$58,916,426		

1.4.18 J1395 Summary

Network Upgrade	Cost	J1395	NUs Type	Self Funding?
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$1,417,855	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$54,627	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$303,883	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$276,774	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$35,794	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$29,700	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$35,915	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$63,696	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$5,399,861	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$714,998	NRIS	Yes
Split Rock 230/115 kV Transformer Upgrade	\$5,000,000	\$647,332	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$17,757,399	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$3,816,570	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$30,554,403		

1.4.19 J1413 Summary

Network Upgrade	Cost	J1413	NUs Type	Self Funding?
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$73,328	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$391,082	NRIS	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$957,090	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$61,579	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$181,347	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$1,156,511	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$90,917	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$68,792	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$86,114	NRIS	Yes

Network Upgrade	Cost	J1413	NUs Type	Self Funding?
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$4,371,935	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$2,987,654	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$1,028,155	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$1,026,944	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$12,481,449		

1.4.20 J1416 Summary

Network Upgrade	Cost	J1416	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$46,056	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$245,634	NRIS	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$1,136,257	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$59,693	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$172,807	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$267,997	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$46,043	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$37,682	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$45,843	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$38,201	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$692,327	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$8,306,024	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$18,883,512	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$30,070,897		

1.4.21 J1417 Summary

Network Upgrade	Cost	J1417	NUs Type	Self Funding?
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$70,994	OTP LPC	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$793,576	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$47,344	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$108,540	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$86,928	NRIS	Yes

Network Upgrade	Cost	J1417	NUs Type	Self Funding?
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$20,465	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$16,874	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$20,507	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$27,341	NRIS	Yes
Install 2nd 345/161 kV Webster Transformer	\$10,000,000	\$765,763	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$224,564	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$3,002,249	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$7,345,648	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$12,530,791		

1.4.22 J1426 Summary

Network Upgrade	Cost	J1426	NUs Type	Self Funding?
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$611,431	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$32,138	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$120,022	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$167,976	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$19,714	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$16,172	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$19,581	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$21,993	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$3,755,179	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$433,939	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$4,202,720	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$684,361	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$10,085,226		

1.4.23 J1432 Summary

Network Upgrade	Cost	J1432	NUs Type	Self Funding?
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$110,539	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$589,543	NRIS	Yes
GARDEN PR; R - SILVER LK; R 345 kV	\$32,400,000	\$1,580,000	PJM AFS	No
CHERRY VA; B - GARDEN PR; R 345 kV	\$165,000,000	\$41,540,000	PJM AFS	No
NELSON ; B - ELECT JCT; B 345 kV	\$10,500,000	\$1,740,000	PJM AFS	No
BYRON ; B - CHERRY VA; B 345 kV	\$12,200,000	\$970,000	PJM AFS	No
BYRON ; B - CHERRY VA; B 345 kV	\$23,400,000	\$10,950,000	PJM AFS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$57,480,083		

1.4.24 J1438 Summary

Network Upgrade	Cost	J1438	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$39,568	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$211,027	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$46,370	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$39,975	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$47,616	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$441,237	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$478,230	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,396,842		

1.4.25 J1444 Summary

Network Upgrade	Cost	J1444	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$96,450	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$514,398	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$149,625	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$203,849	NRIS	Yes
SPP AFS NRIS Network Upgrade	\$122,704,700	\$826,139	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,883,279		

1.4.26 J1445 Summary

Network Upgrade	Cost	J1445	NUs Type	Self Funding?
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$612,187	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$32,006	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$120,455	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$167,976	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$19,696	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$16,320	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$19,363	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$22,193	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$3,739,054	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$433,939	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$4,239,986	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$684,738	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$10,107,915		

1.4.27 J1446 Summary

Network Upgrade	Cost	J1446	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$19,120,731	Base Case NU	Yes
Blue Lake-Scott Co 345 kV	\$20,250,000	\$3,798,943	SH Charge, SH Discharge	Yes
Pulaski-Dickinson 115 kV	\$1,000,000	\$1,000,000	SH Charge	XEL: Yes GRE: No
Dickinson-Dickinson S 115 kV	\$152,690	\$152,690	SH Charge	No
Add 2×40 Mvar switched capacitor at Audubon 230 kV (620336)	\$4,000,000	\$389,706	MISO Voltage	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$931,884	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$36,986	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$181,116	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$260,299	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$30,353	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$24,852	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$30,144	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$32,569	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$5,874,633	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$672,439	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$6,092,136	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$1,025,837	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$39,655,318		

1.4.28 J1456 Summary

Network Upgrade	Cost	J1456	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$29,145,192	Base Case NU	Yes
Add 2×40 Mvar switched capacitor at Audubon 230 kV (620336)	\$4,000,000	\$1,911,765	MISO Voltage	Yes
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$6,992,901	OTP LPC	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$1,718,658	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$364,399	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$380,660	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$48,403	NRIS	Yes

Network Upgrade	Cost	J1456	NUs Type	Self Funding?
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$40,239	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$48,646	NRIS	Yes
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$92,900	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$7,685,934	NRIS	Yes
J1181 POI-Hazleton 345 Structure Remediation	\$15,500,000	\$983,373	NRIS	Yes
Merricourt-Tatanka North Structure Remediation	\$100,000	\$100,000	NRIS	Yes
AECI AFS Network Upgrade	\$1,911,723	\$271,588	AECI AFS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$2,476,406	SPP AFS NRIS	No
Center-Roughrider 230 kV (MPC)	\$500,000	\$500,000	MPC AFS	No
Drayton 230-115 kV xfmr (MPC)	\$2,000,000	\$2,000,000	MPC AFS	No
Prairie-Lake Ardoch 230 kV (MPC)	\$500,000	\$500,000	MPC AFS	No
+50/-0 MVar SVC at Winger 230 kV (MPC)	\$15,000,000	\$15,000,000	MPC AFS	No
2x30 MVar MSC at Winger 230 kV (MPC)	\$2,000,000	\$2,000,000	MPC AFS	No
+50/-0 MVar SVC at Prairie 230 kV (MPC)	\$15,000,000	\$15,000,000	MPC AFS	No
5x30 MVar MSC at Prairie 230 kV (MPC)	\$5,000,000	\$5,000,000	MPC AFS	No
+100/-0 MVar SVC at Frontier 230 kV (MPC)	\$20,000,000	\$20,000,000	MPC AFS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$112,261,063		

1.4.29 J1468 Summary

Network Upgrade	Cost	J1468	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$9,281,908	Base Case NU	Yes
Wilmarth-Sheas Lake 345 kV	\$61,750,000	\$14,374,583	SH Charge	Yes
Blue Lake-Scott Co 345 kV (P2-3)	\$820,000	\$98,106	SH Charge, SH Discharge	No
Helena-Sheas Lake 345 kV	\$18,750,000	\$4,388,092	SH Charge	Yes
Add 2x40 Mvar switched capacitor at Audubon 230 kV (620336)	\$4,000,000	\$169,118	MISO Voltage	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$3,767,009	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$659,226	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$32,738,042		

1.4.30 J1471 Summary

Network Upgrade	Cost	J1471	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
Amoco-Niota 69 kV	\$55,000	\$55,000	PK Discharge	Yes
Niota 161-69 kV xfmr	\$4,000,000	\$4,000,000	PK Discharge	Yes
Denmark-Burlington 161 kV #1	\$16,500,000	\$9,805,934	PK Discharge	Yes
Denmark-Burlington 161 kV #2	\$650,000	\$389,669	PK Discharge	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$150,368	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$801,965	NRIS	Yes
Denmark-Burlington 161 kV Line Rebuild	\$13,850,000	\$8,224,566	NRIS	Yes
Hamilton 138/69 kV Transformer Upgrade	\$4,000,000	\$2,539,987	NRIS	Yes
Macomb NE-Macomb W 138 kV Terminal Equipment Upgrade	\$250,000	\$164,281	NRIS	Yes
AECI AFS Network Upgrade	\$1,911,723	\$532,261	AECI AFS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$26,756,849		

1.4.31 J1474 Summary

Network Upgrade	Cost	J1474	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$92,819	Base Case NU	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$798,110	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$52,944	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$179,672	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$270,830	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$32,808	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$27,388	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$32,747	NRIS	Yes
Prairie Island-North Rochester 345 kV Rebuild	\$68,000,000	\$2,733,971	NRIS	Yes
J1181 POI-Hazelton 345 Structure Remediation	\$15,500,000	\$699,643	NRIS	Yes
Pine Lake 161/115 kV Transformer Upgrade	\$4,000,000	\$4,000,000	NRIS	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$5,722,187	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$1,083,107	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$15,726,227		

1.4.32 J1477 Summary

Network Upgrade	Cost	J1477	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$185,638	Base Case NU	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$25,303	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$134,952	NRIS	Yes
Total Cost Per Project for Actual NRIS Elections for each Project		\$345,894		

1.4.33 J1478 Summary

Network Upgrade	Cost	J1478	NUs Type	Self Funding?
Denmark-Burlington 161 kV #1	\$16,500,000	\$849,840	PK Discharge	Yes
Denmark-Burlington 161 kV #2	\$650,000	\$34,441	PK Discharge	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$18,215	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$97,147	NRIS	Yes
Denmark-Burlington 161 kV Line Rebuild	\$13,850,000	\$743,212	NRIS	Yes
Hamilton 138/69 kV Transformer Upgrade	\$4,000,000	\$277,185	NRIS	Yes
Macomb NE-Macomb W 138 kV Terminal Equipment Upgrade	\$250,000	\$14,275	NRIS	Yes
AECI AFS Network Upgrade	\$1,911,723	\$61,450	AECI AFS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,095,764		

1.4.34 J1479 Summary

Network Upgrade	Cost	J1479	NUs Type	Self Funding?
Denmark-Burlington 161 kV #1	\$16,500,000	\$5,844,226	PK Discharge	Yes
Denmark-Burlington 161 kV #2	\$650,000	\$225,890	PK Discharge	Yes
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$35,497	OTP LPC	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$74,545	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$397,574	NRIS	Yes
Denmark-Burlington 161 kV Line Rebuild	\$13,850,000	\$4,882,222	NRIS	Yes
Hamilton 138/69 kV Transformer Upgrade	\$4,000,000	\$1,182,829	NRIS	Yes

Network Upgrade	Cost	J1479	NUs Type	Self Funding?
Macomb NE-Macomb W 138 kV Terminal Equipment Upgrade	\$250,000	\$71,444	NRIS	Yes
AECI AFS Network Upgrade	\$1,911,723	\$258,529	AECI AFS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$12,972,756		

1.4.35 J1485 Summary

Network Upgrade	Cost	J1485	NUs Type	Self Funding?
Hazel Creek-Scott Co. 345 kV	\$210,829,263	\$35,271,251	Base Case NU	Yes
Canby-Granites Falls 115 kV	\$2,000,000	\$1,250,000	SH Charge, SH Discharge	OTP: Yes
Add 2×40 Mvar switched capacitor at Audubon 230 kV (620336)	\$4,000,000	\$617,647	MISO Voltage	Yes
Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000	\$16,151,116	OTP LPC	Yes
SPP AFS ERIS Network Upgrade	\$275,600,356	\$25,842,417	SPP AFS ERIS	No
Total Cost Per Project for Actual ERIS Elections for each Project		\$79,132,431		

1.4.36 J1487 Summary

Network Upgrade	Cost	J1487	NUs Type	Self Funding?
Rebuild Murphy Creek-Hayward 161 kV line	\$12,000,000	\$12,000,000	SH Charge, SH Discharge	No
Walters-Alden Junction 69 kV	\$6,000,000	\$6,000,000	SH Charge, SH Discharge	Yes
Alden Junction-Albert Lea Westside 69 kV	\$7,000,000	\$7,000,000	SH Charge, SH Discharge	Yes
J1363 POI-J1352 POI 345 kV Rebuild	\$3,000,000	\$35,349	NRIS	Yes
J1338 POI-J1363 POI 345 kV Rebuild	\$16,000,000	\$188,530	NRIS	Yes
BVR CRK-Grimes 345 Line Rebuild	\$26,000,000	\$879,735	NRIS	Yes
Raun-Remsen 345 kV Structure Replacements	\$1,500,000	\$56,627	NRIS	Yes
Raun-S3451 345 kV Rebuild	\$5,000,000	\$151,479	NRIS	MEC: Yes
J1413 POI-J1181 POI 345 kV Structure Replacements	\$6,000,000	\$252,585	NRIS	Yes
Hazelton-Arnold 345 kV Structure Replacements	\$800,000	\$33,801	NRIS	Yes
Morgan Valley-Tiffin 345 Structure Replacements	\$900,000	\$27,434	NRIS	Yes
Arnold-J1284 POI 345 kV Structure Replacements	\$1,000,000	\$33,452	NRIS	Yes

Network Upgrade	Cost	J1487	NUs Type	Self Funding?
Bondurant-Montezuma 345 kV Structure Replacements	\$1,000,000	\$29,853	NRIS	Yes
J1181 POI-Hazleton 345 Structure Remediation	\$15,500,000	\$652,511	NRIS	Yes
Austin-Adams 161 kV Line Rebuild	\$15,750,000	\$3,820,247	NRIS	ITCM: Yes SMPMA: No
SPP AFS ERIS Network Upgrade	\$275,600,356	\$6,351,133	SPP AFS ERIS	No
SPP AFS NRIS Network Upgrade	\$122,704,700	\$11,048,760	SPP AFS NRIS	No
Total Cost Per Project for Actual NRIS Elections for each Project		\$48,561,496		

1.5 Study Compliance with NERC FAC-002-3 Standard

This DPP 2019 West Area study was completed in compliance with NERC FAC-002-3:

R1.1: The reliability impact of the new interconnection, or materially modified existing interconnection, on affected system(s).

Section 4 covers summer peak steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 3 covers summer shoulder steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 5.1 covers reliability impact of the generating facilities per GRE Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.2 covers reliability impact of the generating facilities per OTP Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.3 covers reliability impact of the generating facilities per DPC Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 6.1 covers reliability impact of the new generating facilities in the CIPCO affected systems.

Section 6.2 covers reliability impact of the new generating facilities in the MPC affected systems.

Section 6.3 covers reliability impact of the new generating facilities in the PJM affected systems.

Section 6.4 covers reliability impact of the new generating facilities in the AECI affected systems.

Section 6.5 covers reliability impact of the new generating facilities in the SPP affected systems.

Section 7 covers transient stability analysis results.

Section 8 covers voltage stability (PV) analysis on the MWEX System Operating Limit (SOL). Network Upgrades required for MWEX voltage stability are identified.

Section 9 covers short circuit reliability impact of the new generating facilities.

Section 10 covers Deliverability reliability impact of the new NRIS generating facilities.

R1.2: Adherence to applicable NERC Reliability Standards; regional and Transmission Owner planning criteria; and Facility interconnection requirements.

Sections 2.2-2.4, Section 5, Section 6, Section 7, and Section 9 all cover NERC Reliability Standard TPL-001-4.

Section 5.1 covers GRE Local Planning Criteria (LPC).

Section 5.2 covers OTP LPC.

Section 5.3 covers DPC LPC.

Section 6.1 covers CIPCO system planning criteria.

Section 6.2 covers MPC system planning criteria.

Section 6.3 covers PJM system planning criteria.

Section 6.4 covers AECI system planning criteria.

Section 6.5 covers SPP system planning criteria.

Section 8 (voltage stability analysis) covers individual system planning criteria (ATC).

Section 10 covers MISO system planning criteria.

R1.3: Steady-state, short-circuit, and dynamics studies, as necessary, to evaluate system performance under both normal and contingency conditions.

Section 4 and Section 3 cover MISO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.1 covers GRE's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.2 covers OTP's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.3 covers DPC's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.1 covers CIPCO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.2 covers MPC steady-state and transient stability assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.3 covers PJM steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.4 covers AECI steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6.5 covers SPP steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 7 covers transient stability studies under NERC category P0 to P7 contingencies (TPL-001-4).

Section 8 covers steady-state voltage stability assessment.

Section 9 covers short circuit assessment.

Section 10 covers MISO deliverability study (steady-state assessment) including NERC category P0 to P1 contingencies (TPL-001-4).

R1.4: Study assumptions, system performance, alternatives considered, and coordinated recommendations. While these studies may be performed independently, the results shall be evaluated and coordinated by the entities involved.

Section 2.1, Section 2.2, Section 2.3, and Section 2.4, Section 7.2, Section 7.3, and Section 7.4 cover study assumptions and system performance criteria.

Jointly coordinated recommendations can be found in Section 5.1 (MISO and GRE), Section 5.2 (MISO and OTP), Section 5.3 (MISO and DPC), Section 6.1 (MISO and CIPCO), Section 6.2 (MISO and MPC), Section 6.3 (MISO and PJM), Section 6.4 (MISO and AECI), Section 6.5 (MISO and SPP), and Section 8 (MISO and ATC). Results in Sections 3, 4, 5, 6, 7, 8 and 10 have also been reviewed by PJM, SPP, AECI, CIPCO, MPC, and ATC.

Introduction

Thirty-six (36) generation projects, listed in Table A-1 (Appendix A.1), have requested to interconnect to the MISO transmission network in the West Area and have advanced to the Definitive Planning Phase (DPP) 2019 Phase 2 study (DPP West Area). J1485 has requested Energy Resource Interconnection Service (ERIS); All other generating facilities have requested both ERIS and Network Resource Interconnection Service (NRIS).

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generating facilities in the DPP West Area Phase 2 study.

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models, and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

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Model Development and Study Criteria

2.1 Model Development

2.1.1 Benchmark Cases

DPP 2019 West area power flow benchmark cases representing 2024 summer shoulder and summer peak conditions were developed from the MTEP19 models with LBA dispatch.

The benchmark cases for DPP 2019 study were created as follows:

- MISO prior queued generation projects were modeled, and their associated Network Upgrades (NU) up to DPP 2017 August cycle were also modeled. Appendix A.2 lists all DPP 2017 August West Area Phase 3 MISO ERS Network Upgrades included in the models.
- Network Upgrades required in DPP 2018 April West Area Phase 3 were not modeled since the study has not been completed yet.
- DPP 2019 generation projects in the West Area (DPP West Area, Table A-1), Central Area (Table A-3), Michigan Area (Table A-4), and ATC Area (Table A-5) were modeled with offline status.
- For MISO North generation projects, their output was sunk to the MISO North (Appendix A.4, Table A-8), where generation was scaled uniformly;
- PJM generation projects were modeled and dispatched. The generation output was sunk to the PJM market (Appendix A.5, Table A-9), where generation was scaled uniformly.
- SPP generation projects were modeled and dispatched. The generation output was sunk to the SPP market (Appendix A.6, Table A-10), where generation was scaled uniformly.
- Models were further reviewed by the Ad Hoc study members (transmission owners and customers). Model corrections and changes were made based on the comments and feedback. These modeling changes are listed in Appendix A.3.
- Adjusted Square Butte DC to match the total output of the Bison (Bison 1 to 5) and Oliver County (Oliver County 1 and 2) wind farms.
- Adjusted CU DC to match the total output of Coal Creek generation units #1 and #2.
- MHEX interface transfer level is at 1530 MW in summer shoulder and 1800 MW in summer peak cases.

2.1.2 Study Cases

Summer peak study discharging case was created by dispatching the DPP 2019 generation projects at the specified summer peak level from the benchmark case. Battery projects were dispatched in discharging mode.

Summer shoulder study discharging case was created by dispatching the DPP 2019 generation projects at the specified summer shoulder level from the benchmark case. Battery projects were dispatched in discharging mode, and PV projects were turned off.

Summer shoulder study charging case was created by dispatching the DPP 2019 generation projects at the specified summer shoulder level from the benchmark case. Battery projects were dispatched in charging mode, and PV projects were turned off.

For the DPP 2019 generation projects in MISO North, the MISO North was used for power balance, where generation was scaled uniformly.

Both study and benchmark power flow cases were solved with transformer tap adjustment enabled, area interchange disabled, phase shifter adjustment enabled and switched shunt adjustment enabled.

The interface transfer levels in the 2024 study cases are summarized in Table 2-1.

Table 2-1: Interface Transfer Levels in 2024 Study Cases

Interface	2024 SH Discharging Case (MW)	2024 SH Charging Case (MW)	2024 SPK Discharging Case (MW)
MHEX	1530	1530	1800
MWEX	964.7	966.8	87.4
Arrowhead PST	384.2	395.4	-79.8
J732 POI – Stone Lake 345 kV	632.7	667.3	358.6

2.2 Contingency Criteria

A variety of contingencies were considered for steady-state analysis:

- NERC Category P0 with system intact (no contingencies)
- NERC Category P1 contingencies
 - NERC Category P1 contingencies, at buses with a nominal voltage of 69 kV and above, in the following areas: CWLD (area 333), AMMO (area 356), AMIL (area 357), CWLP (area 360), SIPC (area 361), WEC (area 295), WEC MI (area 296), XCEL (area 600), MP (area 608), SMMPA (area 613), GRE (area 615), OTP (area 620), ITCM (area 627), MPW (area 633), MEC (area 635), MDU (area 661), BEPC-MISO (area 663), MHEB (area 667), DPC (area 680), ALTE (area 694), WPS (area 696), MGE (area 697), UPPC (area 698), CE(area 222), NPPD (area 640), OPPD (area 645), LES (area 650), WAPA (area 652), BEPC-SPP (area 659), AECI (area 330), MIPU(area 540), KCPL (area 541), KACY (area 542), INDN (area 545).

- Multiple-element NERC Category P1 contingencies, in Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P1 contingencies are listed in Appendix A.7.
- NERC Category P2-P7 contingencies
 - Selected NERC Category P2-P7 contingencies provided by the Ad Hoc Study Group, in the study region of Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P2-P7 contingencies are listed in Appendix A.7.

For all contingency and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

2.3 Monitored Elements

The study area is defined in Table 2-2. Facilities in the study area were monitored for system intact and contingency conditions. Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal (PSS[®]E rate A) rating. Under NERC category P1-P7 conditions, branches were monitored for loading as shown in the column labeled "Post-Disturbance Thermal Limits".

Table 2-2: Monitored Elements

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
AECI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
AMIL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
AMMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
ATCLLC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
BEPC-MISO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
BEPC-SPP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CBPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CMMPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.07/0.90
CWLD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
CE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
DPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GRE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92/0.90
INDN	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
ITCM	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.95	1.10/0.93
KACY	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
KCPL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
LES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MDU	57 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MEC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.96/0.95	1.05/0.96/0.95/0.94 /0.93 ³
MHEB	69 kV and above	100% of Rate A	100% of Rate B	1.12/1.1/1.07/1.05/1.04/0.99/0.97/0.96/0.95	1.15/1.10/0.94/0.90
MP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
MPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.97	1.10/0.92
MPW	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.06/0.92
MRES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.97	1.05/0.92
NPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OTP	40 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.97	1.10/0.92
PPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.075/0.90
RPU	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92
SIPC	69 kV and above	100% of Rate A	100% of Rate B	1.07/0.95	1.09/0.91
SMMPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WAPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WPPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
XEL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.92

Notes

- 1: PSS®E Rate A, Rate B or Rate C
- 2: Limits dependent on nominal bus voltage
- 3: For facilities in Cedar Falls Utilities or Ames Municipal Utilities, post-contingency voltage limits are 1.05/0.94 for >200 kV, and 1.05/0.93 for others.

2.4 Performance Criteria

A branch is considered as a thermal injection constraint if the branch is loaded above its applicable normal or emergency rating for the post-change case, and any of the following conditions are met:

- 1) the generator (NR/ER) has a larger than 20% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, or
- 2) the megawatt impact due to the generator is greater than or equal to 20% of the applicable rating (normal or emergency) of the overloaded facility, or

- 3) the overloaded facility or the overload-causing contingency is at generator's outlet, or
- 4) for any other constrained facility, where none of the study generators meet one of the above criteria in 1), 2), or 3), however, the cumulative megawatt impact of the group of study generators (NR/ER) is greater than 20% of the applicable rating, then only those study generators whose individual MW impact is greater than 5% of the applicable rating and has DF greater than 5% (OTDF or PTDF) will be responsible for mitigating the cumulative MW impact constraint.

A bus is considered a voltage constraint if both of the following conditions are met. All voltage constraints must be resolved before a project can receive interconnection service.

- 1) the bus voltage is outside of applicable normal or emergency limits for the post-change case, and
- 2) the change in bus voltage is greater than 0.01 per unit.

All DPP 2019 West Area study generators must mitigate thermal injection constraints and voltage constraints to obtain unconditional Interconnection Service.

Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the deliverability algorithm, to meet the system performance criteria for NERC category P0-P1 events, if the constraint demonstrates an incremental flow caused by the generator equal to or greater than 5% of the generator's maximum dispatch level in each case.

2.5 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

Per FERC Order 827, non-synchronous generation projects in the DPP 2019 West Area study group are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Collector system and shunt compensation of DPP West projects are modeled, which are listed in Appendix A.1, Table A-2. An analysis was performed to study the reactive power requirements (FERC Order 827) for the non-synchronous generation projects in the DPP 2019 West study group. The analysis was performed as follows:

Step 1: Verify whether total dynamic reactive power (reactive power from generators and dynamic compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the generator terminal bus. The verification in Step 1 was performed when generator data was submitted and modeled.

Step 2: Verify whether total reactive power (reactive power from generators, dynamic compensation devices, and static compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the high-side of the generator substation. The testing procedure in Step 2 is described in the following:

- Lock the high-side of the generator substation at 1.0 pu voltage by adding a fictitious SVC. This is to ensure that the test result is not affected by the system condition.
- Lock the reactive power output of the generator to the maximum limit (Q_{max}). Make sure all shunt compensation devices within the substation are at the maximum

- capacitive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 lagging requirement.
- Lock the reactive power output of the generator to the minimum limit (Q_{min}). Make sure all shunt compensation devices within the substation are at the maximum inductive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 leading requirement.

Appendix C lists reactive power requirement analysis results for the DPP West generation projects. The results are summarized as following:

- J1190, J1218, J1221, J1270, J1298, J1365, J1395, and J1485 do not meet the reactive power requirements per FERC Order 827.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

Section

3

Summer Shoulder Steady-State Analysis

Summer shoulder steady-state analysis was performed in summer shoulder scenario to identify thermal and voltage upgrades required for interconnecting the generating facilities in the DPP 2019 West Area group to the transmission system. Two conditions for discharging and charging of battery projects were studied for summer shoulder scenario.

3.1 Study Procedure

Summer shoulder power flow case (discharging and charging) was created in the procedure as described in Section 2.1. Due to post-contingent voltage collapse and thermal overloads in the initial power flow case, steady-state analysis was performed in the following three-step procedure:

- Step 1: Non-linear (AC) contingency analysis (Stage-1 ACCC) was performed on summer shoulder charging scenario to identify voltage collapse and thermal overloads. The summer shoulder charging scenario has the worst voltage violations.
- Step 2: Based on the identified voltage collapse and thermal overloads in the Stage-1 ACCC, project justification analysis was performed to determine Network Upgrades (NUs) required for interconnection of DPP West projects. These selected NUs are called Base Case NUs.
- Step 3: The Base Case NUs were added to the models. Stage-2 ACCC was performed to identify any remaining thermal and voltage constraints.

3.2 Step 1 – Stage-1 ACCC Analysis

Since the summer shoulder charging scenario has the worst voltage violations, AC contingency analysis was performed on summer shoulder charging scenario to identify voltage collapse and thermal overloads. Detailed Stage-1 ACCC results for summer shoulder charging scenario are listed in Appendix D.1.

3.2.1 Stage-1 Potential Voltage Collapse

Multiple contingencies caused potential voltage collapse (<0.9 p.u.) in 200 kV and above power systems. The contingencies which caused potential voltage collapse are listed in Table 3-1. The potential voltage collapses were identified in the following areas:

- Jamestown-Buffalo-Bison-Maple River 345 kV
- Hazel Creek-Minn Valley-Minn Valley Tap-Panther-McLeod-Blue Lake 230 kV

- Appledorn-Granite Falls-Willmar-Paynesville 230 kV
- Granite Falls-Morris 230 kV
- Wahpeton-Fergus Falls-Silver Lake-Henning-Inman-Wing River-J1221 POI 230 kV
- Alexandria-Riverview 345 kV
- Lake Park-Audubon-Erie Jct. 230 kV

Table 3-1: List of Contingencies which Caused Potential Voltage Collapse

CEII Redacted

3.2.2 Stage-1 Thermal Overloads in Systems 200 kV and Above

Table 3-2 lists worst thermal overloads in 200 kV and above systems. These overloads are on the following facilities:

- Wilmarth-Sheas Lake-Helena 345 kV
- Hazel Creek 345-230-13.8 kV xfmr
- Cedar Mountain 345-115-34.5 kV xfmr
- Minn Valley-Minn Valley Tap-Panther-McLeod 230 kV
- Granite Falls-Willmar-Paynesville 230 kV

Table 3-2: Worst Thermal Overloads in >200 kV Systems

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type
			(MVA)	(%)		
Wilmarth-Sheas Lake 345 kV	1515.8	XEL	1967.1	129.8	CEII Redacted	P2-P7
Blue Lake-Scott Co 345 kV	1515.8	XEL	1820.1	120.1	CEII Redacted	P2-P7
Helena-Sheas Lake 345 kV	1513.4	XEL	1970.6	130.2	CEII Redacted	P2-P7
Hazel Creek 345-230-13.8 kV xfmr #9	515.0	XEL	705.5	137.0	CEII Redacted	P2-P7

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type
			(MVA)	(%)		
Minn Valley-Minn Valley Tap 230 kV	506.3	XEL	598.1	118.1	CEII Redacted	P2-P7
Minn Valley Tap-Panther 230 kV	505.3	XEL GRE	689.3	136.4	CEII Redacted	P2-P7
Paynesville-Willmar 230 kV	398.0	XEL GRE	431.7	108.5	CEII Redacted	P2-P7
Panther-McLeod 230 kV	450.0	GRE	640.6	142.3	CEII Redacted	P2-P7
Cedar Mountain 345-115-34.5 kV xfmr	336.0	XEL	425.6	126.7	CEII Redacted	P2-P7
Willmar-Granite Falls 230 kV	264.9	GRE WAPA	443.2	167.3	CEII Redacted	P2-P7

3.3 Step 2 – Base Case NUs Justification Analysis

Based on the identified potential voltage collapse and voltage violations in the Stage-1 ACCC and the identified overloaded 200 kV and above facilities under post-contingent conditions, various transmission Network Upgrades (NUs) were tested. The Hazel Creek-Scott County 345 kV line was justified as Base Case NU due to the following performance advantages:

- 1) The new line carries 693 MVA significant flow under system intact condition.
- 2) The new line can effectively mitigate all potential voltage collapse identified in the Stage-1 study.

- 3) The new line can greatly improve voltages system widely.
- 4) The new line can mitigate or significantly reduce the following identified thermal overloads:
 - Wilmarth-Sheas Lake-Helena 345 kV
 - Hazel Creek 345-230-13.8 kV xfmr
 - Cedar Mountain 345-115-34.5 kV xfmr
 - Minn. Valley- Minn. Valley Tap-Panther-McLeod 230 kV
 - Granite Falls-Willmar-Paynesville 230 kV
 - Split Rock-White 345 kV

2×75 MVar capacitor bank at Cedar Mountain 345 kV (615647) is required Network Upgrade in DPP 2018 April Phase 3 study. With the Hazel Creek-Scott County 345 kV Base Case NU and 2×75 MVar capacitor bank at Cedar Mountain 345 kV bus, all potential voltage collapse and majority of the identified voltage constraints can be mitigated.

In summary, one justified Base Case NU and one DPP 2018 April Phase 3 NU (Table 3-3) are required for mitigating potential voltage collapse, majority of voltage constraints, and many >200 kV system thermal overloads identified in the Stage-1 ACCC analysis.

Table 3-3: Network Upgrades Required for Mitigating Voltage Collapse, Voltage Constraints, and Thermal Overloads

NUs	Needs	Miles	Cost (\$)
Hazel Creek-Scott County 345 kV	Base Case NU	115	\$210,829,263
2×75 MVar switched capacitor at Cedar Mountain 345 kV (615647)	DPP 2018 Apr. Ph3 NU		\$3,000,000 ¹

Note 1: The cost is currently assigned to projects in DPP 2018 Apr. Phase 3 cycle. The NU cost may be assigned to DPP 2019 projects if projects in DPP 2018 Apr. are withdrawn.

3.4 Step 3 – Stage-2 ACCC Analysis

The Hazel Creek-Scott County 345 kV Base Case NU and 2×75 MVar capacitor bank at Cedar Mountain 345 kV bus were added to create Stage-2 models. AC contingency analysis was performed in the Stage-2 models to identify any remaining thermal and voltage constraints.

In the summer shoulder scenario, two conditions for discharging and charging of battery projects were studied.

3.4.1 Stage-2 ACCC Analysis Results for Summer Shoulder Discharging Condition

The discharging condition has all DPP 2019 battery projects in maximum discharging mode. The ACCC analysis results are in Appendix D.2.

3.4.1.1 System Intact Conditions

For NERC category P0 (system intact) conditions, no thermal or voltage constraints were identified (Table D-11, Table D-12).

3.4.1.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All NERC Category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table D-13. No voltage constraints were identified (Table D-14).

Two category P2-P7 contingencies (Table D-17) were not converged in both the benchmark and study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For the non-converged contingencies in Table D-17, DC contingency analysis was performed to get the dc thermal results. The dc thermal results for non-converged contingencies are listed in Table D-18.

For P2-P7 converged contingencies, thermal constraints are listed in Table D-15. No voltage constraints were identified (Table D-16).

3.4.2 Contingency Analysis Results for Summer Shoulder Charging Condition

The charging condition has all DPP 2019 battery projects in maximum charging mode. The ACCC analysis results are in Appendix D.3.

3.4.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, no thermal or voltage constraints were identified (Table D-19, Table D-20).

3.4.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All NERC Category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table D-21, and voltage constraints are listed in Table D-22.

Two category P2-P7 contingencies (Table D-25) were not converged in both the benchmark and study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For the non-converged contingencies in Table D-25, DC contingency analysis was performed to get the dc thermal results. The dc thermal results for non-converged contingencies are listed in Table D-26.

For P2-P7 converged contingencies, thermal constraints are listed in Table D-23, and voltage constraints are listed in Table D-24.

3.4.3 Summer Shoulder Worst Thermal Constraints in the Stage-2 ACCC

Table 3-4 lists worst thermal constraints and Network Upgrades identified in the Stage-2 ACCC for 2024 summer shoulder scenario.

Table 3-4: 2024 Summer Shoulder Thermal Constraints, Maximum Screened Loading, Stage-2 ACCC

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1270,J1321,J1349, J1365,J1371,J1417, J1485,J1487	Montgomery-J1145 POI 345 kV	1193.0	Ameren	1740.1	145.9	CEII Redacted	P1	Rebuild 28 mi of 345 kV line for 3000A SE
J1349,J1468,	Wilmarth-Sheas Lake 345 kV	1515.8	XEL	1574.7	103.9	CEII Redacted	P2-P7	Rebuild 24.7 mi of 345 kV to bundled 954 ACSS
J1315	Blue Lake-Scott Co 345 kV	1515.8	XEL	1749.7	115.4	CEII Redacted	P1	Rebuild 8.1 mi of 345 kV to bundled 954 ACSS
J1315,J1446,	Blue Lake-Scott Co 345 kV	1515.8	XEL	1769.9	116.8	CEII Redacted	P2-P7	Rebuild 8.1 mi of 345 kV to bundled 954 ACSS
J1315, J1468	Blue Lake-Scott Co 345 kV	1515.8	XEL	2039.2	134.5	CEII Redacted	P2-P7	Add an extra breaker at Chub Lake 345 kV to eliminate the P2-3 contingency
J1315	Lyon County-Hazel Creek 345 kV	1314.6	XEL	1407.2	107.0	CEII Redacted	P2-P7	Upgrade Substation Equipment

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1349,J1468	Helena-Sheas Lake 345 kV	1513.4	XEL	1566.8	103.5	CEII Redacted	P2-P7	Rebuild 7.5 mi of 345 kV to bundled 954 ACSS
J1446	Pulaski-Dickinson 115 kV	174.2	XEL GRE	211.7	121.5	CEII Redacted	P1	XEL: Upgrade Substation Equipment. \$1M GRE: GRE owns a switch at Dickinson rated to 239 MVA Summer/Winter. \$0
J1446	Pulaski-Dickinson 115 kV	174.2	XEL GRE	186.6	107.1	CEII Redacted	P2-P7	XEL: Upgrade Substation Equipment. \$1M GRE: GRE owns a switch at Dickinson rated to 239 MVA Summer/Winter. \$0
J1315	Scott County 345-115-34.5 kV xfmr #9	648.9	XEL	975.5	150.3	CEII Redacted	P2-P7	Add 3rd Scott County 345/115 672 MVA transformer
J1315	Crystal Foods Tap-Arlington 69 kV	35.9	XEL	46.8	130.3	CEII Redacted	P2-P7	Rebuild 6.9 miles to 477 ACSR
J1315	Gibbon-Cairo 69 kV	53.2	XEL	60.6	113.9	CEII Redacted	P2-P7	Rebuild 8.1 miles to 477 ACSR

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1487	Murphy Crk-J1298 POI	276.0	SMMPA	301.9	109.4	CEII Redacted	P1	Rebuild 16 miles of Murphy Creek-Hayward 161 kV line. New rating 406 MVA
J1487	Murphy Crk-J1298 POI	276.0	SMMPA	350.1	126.9	CEII Redacted	P2-P7	Rebuild 16 miles of Murphy Creek-Hayward 161 kV line. New rating 406 MVA
J1446	Dickinson-Dickinson S 115 kV	174.2	GRE	212.5	122.0	CEII Redacted	P1	Re-conductor to 795 DRAKE ACSS and replace wave trap
J1446	Dickinson-Dickinson S 115 kV	174.2	GRE	187.5	107.6	CEII Redacted	P2-P7	Re-conductor to 795 DRAKE ACSS and replace wave trap
J1371,J1485	Canby-Granites Falls 115 kV	112.5	OTP WAPA	125.6	111.7	CEII Redacted	P1	OTP: Structure replacements. \$2M WAPA: NU is not required unless it is identified as constraint in affected system study.

Summer Shoulder Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
J1371,J1485	Big Stone-Highway 12 115 kV	230.0	OTP	230.0	100.0	CEII Redacted	P2-P7	Loading does not exceed rate B. Not a constraint
J1371,J1485	Wahpeton-Fergus Falls 230 kV	379.1	OTP MRES	388.7	102.5	CEII Redacted	P1	OTP: Structure replacement. \$750,000. 2017 Aug. Ph3 OTP LPC upgrade MRES: Terminal rating at Fergus Falls is sufficient. \$0
J1487	Walters-Alden Junction 69 kV	48.0	ITCM	81.7	170.2	CEII Redacted	P1	Rebuild 6 miles of 69 kV. New rating of 157 MVA
J1487	Walters-Alden Junction 69 kV	48.0	ITCM	78.2	163.0	CEII Redacted	P2-P7	Rebuild 6 miles of 69 kV. New rating of 157 MVA
J1487	Alden Junction-Albert Lea Westside 69 kV	45.0	ITCM	81.6	181.4	CEII Redacted	P1	Rebuild 7 miles of 69 kV. New rating of 157 MVA
J1487	Alden Junction-Albert Lea Westside 69 kV	45.0	ITCM	78.2	173.7	CEII Redacted	P2-P7	Rebuild 7 miles of 69 kV. New rating of 157 MVA
J1487	Hayward-J1298 161 kV	276.0	ITCM SMMPA	302.0	109.4	CEII Redacted	P1	ITCM: Not a constraint. ITCM Facilities at 308 MVA. \$0

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation
				(MVA)	(%)			
								SMMPA: Rebuild 16 miles of Murphy Creek-Hayward 161 kV line. New rating 406 MVA. \$12M
J1487	Hayward-J1298 161 kV	276.0	ITCM SMMPA	350.2	126.9	CEII Redacted	P2-P7	ITCM: Not a constraint. ITCM Facilities at 308 MVA. \$0 SMMPA: Rebuild 16 miles of Murphy Creek-Hayward 161 kV line. New rating 406 MVA. \$12M
J1371,J1485	Granite Falls-Blair 230 kV	413.0	WAPA BEPC	465.2	112.6	CEII Redacted	P1	NU is not required unless it is identified as constraint in affected system study.
J1315	Granite Falls-Marshall 115 kV	103.0	WAPA BEPC	126.4	122.7	CEII Redacted	P2-P7	NU is not required unless it is identified as constraint in affected system study.
J1487	Wabaco-Rochester 161 kV	221.1	DPC	221.7	100.3	CEII Redacted	P1	A third party will fund the Rochester-Wabaco 161kV rebuild. \$11M. MTEP Appendix A project

3.5 Network Upgrades Identified in MISO ERS Analysis for 2024 Summer Shoulder Scenario

Based on the MISO 2024 summer shoulder steady state analyses, the MISO Base Case NU and one DPP 2018 April Phase 3 NU are listed in Table 3-5, which are required for mitigating potential voltage collapse, majority of voltage constraints, and many >200 kV system thermal overloads. Additional thermal NUs and cost are listed in Table 3-6, and additional reactive power NUs and cost are listed in Table 3-7.

Table 3-5: Network Upgrades Required for Mitigating Voltage Collapse, Voltage Constraints, and Thermal Overloads

NUs	Needs	Miles	Cost (\$)
Hazel Creek-Scott County 345 kV	Base Case NU	115	\$210,829,263
2x75 MVar switched capacitor at Cedar Mountain 345 kV (615647)	DPP 2018 Apr. Ph3 NU		\$3,000,000 ¹

Note 1: The cost is currently assigned to projects in DPP 2018 Apr. Phase 3 cycle. The NU cost may be assigned to DPP 2019 projects if projects in DPP 2018 Apr. are withdrawn.

Table 3-6: Additional Thermal NUs Required in Summer Shoulder Scenario

Constraint	Owner	Mitigation	Cost (\$)
Montgomery-J1145 POI 345 kV	Ameren	Rebuild 28 mi of 345 kV line for 3000A SE. Exclusively tied to H104/H105 through the TCA study process. \$42M	\$0 ¹
Wilmarth-Sheas Lake 345 kV	XEL	Rebuild 24.7 mi of 345 kV to bundled 954 ACSS	\$61,750,000
Blue Lake-Scott Co 345 kV	XEL	Rebuild 8.1 mi of 345 kV to bundled 954 ACSS	\$20,250,000
Blue Lake-Scott Co 345 kV (P2-3)	XEL	Add an extra breaker at Chub Lake 345 kV to eliminate the P2-3 contingency	\$820,000
Lyon County-Hazel Creek 345 kV	XEL	Upgrade Substation Equipment	\$5,000,000
Helena-Sheas Lake 345 kV	XEL	Rebuild 7.5 mi of 345 kV to bundled 954 ACSS	\$18,750,000
Pulaski-Dickinson 115 kV	XEL GRE	XEL: Upgrade Substation Equipment. \$1M GRE: GRE owns a switch at Dickinson rated to 239 MVA Summer/Winter. \$0	\$1,000,000
Scott County 345-115-34.5 kV xfmr #9	XEL	Add 3rd Scott County 345/115 672 MVA transformer	\$15,000,000
Crystal Foods Tap-Arlington 69 kV	XEL	Rebuild 6.9 miles to 477 ACSR	\$4,830,000
Gibbon-Cairo 69 kV	XEL	Rebuild 8.1 miles to 477 ACSR	\$7,000,000

Constraint	Owner	Mitigation	Cost (\$)
Murphy Crk-J1298 POI	SMMPA	Rebuild 16 miles of Murphy Creek-Hayward 161 kV line. New rating 406 MVA	\$12,000,000
Dickinson-Dickinson S 115 kV	GRE	Re-conductor to 795 DRAKE ACSS and replace wave trap	\$152,690
Canby-Granites Falls 115 kV	OTP WAPA	OTP: Structure replacements. \$2M WAPA: NU is not required unless it is identified as constraint in affected system study.	\$2,000,000
Big Stone-Highway 12 115 kV	OTP	Loading does not exceed rate B. Not a constraint	\$0
Wahpeton-Fergus Falls 230 kV	OTP MRES	OTP: Structure replacement. \$750,000. 2017 Aug. Ph3 OTP LPC upgrade MRES: Terminal rating at Fergus Falls is sufficient. \$0	\$0 ²
Walters-Alden Junction 69 kV	ITCM	Rebuild 6 miles of 69 kV. New rating of 157 MVA	\$6,000,000
Alden Junction-Albert Lea Westside 69 kV	ITCM	Rebuild 7 miles of 69 kV. New rating of 157 MVA	\$7,000,000
Hayward-J1298 161 kV	ITCM SMMPA	ITCM: Not a constraint. ITCM Facilities at 308 MVA. \$0 SMMPA: Rebuild 16 miles of Murphy Creek-Hayward 161 kV line. New rating 406 MVA. \$12M	Incl.
Granite Falls-Blair 230 kV	WAPA BEPC	NU is not required unless it is identified as constraint in affected system study.	\$0
Granite Falls-Marshall 115 kV	WAPA BEPC	NU is not required unless it is identified as constraint in affected system study.	\$0
Wabaco-Rochester 161 kV	DPC	A third party will fund the Rochester-Wabaco 161kV rebuild. \$11M. MTEP Appendix A project	\$0

Note 1: This Network Upgrade is exclusively tied to H104/H105 through the TCA study process.

Note 2: Costs of Network Upgrades required in DPP 2017 Aug West Ph3 OTP LPC are currently assigned to projects in DPP 2017 Aug. Phase 3 cycle. The NUs cost may be assigned to DPP 2019 projects if projects in DPP 2017 Aug. are withdrawn.

Table 3-7: Additional Reactive Power NUs

Network Upgrades	Owner	Cost (\$)
Add 2×40 Mvar switched capacitor at Audubon 230 kV (620336)	OTP	\$4,000,000

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Summer Peak Steady-State Analysis

Summer peak steady-state analysis was performed in summer peak scenario to identify thermal and voltage upgrades required for interconnecting the generating facilities in the DPP 2019 West Area group to the transmission system. Battery projects are in discharging mode in the summer peak scenario.

4.1 Study Procedure

Summer peak power flow cases were created in the procedure as described in Section 2.1. For consistency with summer shoulder power flow cases, the Hazel Creek-Scott County 345 kV Base Case NU and 2×75 MVar capacitor bank at Cedar Mountain 345 kV bus were also included in the summer peak cases. Nonlinear (AC) contingency analysis was performed on the benchmark and study cases, and incremental impact of the DPP West Area generating facilities was evaluated by comparing the steady-state performance of the transmission system in the benchmark and study cases. Network upgrades were identified to mitigate any summer peak constraints.

4.2 Contingency Analysis Results for Summer Peak Discharging Condition

The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows and voltages between benchmark case (without DPP 2019 projects) and study case (with DPP 2019 projects). All battery projects in DPP 2019 cycle were in discharging condition. The contingency analysis results are in Appendix E.1.

4.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, no thermal constraints (Table E-1) or voltage constraints (Table E-2) were identified.

4.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All category P1 contingencies were converged.

For P1 contingencies, thermal constraints are listed in Table E-3. No voltage constraints were identified (Table E-4).

Two category P2-P7 contingencies (Table E-7) were not converged, and their dc thermal results are listed in Table E-8. The contingency was not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these contingencies.

For P2-P7 converged contingencies, thermal constraints are listed in Table E-5. No voltage constraints were identified (Table E-6).

4.3 Summer Peak Worst Thermal Constraints

Table 4-1 lists worst thermal constraints and Network Upgrades in the 2024 summer peak discharging scenario.

Table 4-1: 2024 Summer Peak Thermal Constraints and Network Upgrades, Maximum Screened Loading

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation	Cost (\$)
				(MVA)	(%)				
J1471	Denmark-Spring Grove 69 kV	70.0	AECI	70.5	100.7	CEII Redacted	P2-P7	NU is not required unless it is identified as constraint in affected system study.	\$0
J1479	Henry Co. 161-69 kV xfmr	56.0	AECI	64.0	114.3	CEII Redacted	P1	NU is not required unless it is identified as constraint in affected system study.	\$0
J1471	Amoco-Niota 69 kV	72.0	Ameren	75.1	104.3	CEII Redacted	P1	Replace (1) 72 MVA switch with 1200 A switch.	\$55,000
J1471	Amoco-Niota 69 kV	72.0	Ameren	75.0	104.1	CEII Redacted	P2-P7	Replace (1) 72 MVA switch with 1200 A switch.	
J1471	Niota 161-69 kV xfmr	84.0	Ameren	102.4	121.9	CEII Redacted	P1	Replace with new 112 MVA 161/69 kV Transformer.	\$4,000,000
J1471	Niota 161-69 kV xfmr	84.0	Ameren	101.9	121.3	CEII Redacted	P2-P7	Replace with new 112 MVA 161/69 kV Transformer.	
J1359	PCI J2-Oakridge 69 kV	109.0	ITCM CIPCO	128.9	118.3	CEII Redacted	P1	CIPCO: Not a constraint. \$0 ITCM: Relay thermal, replace sub taps (.08 miles), new rating 143 MVA. \$450K	\$450,000

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation	Cost (\$)
				(MVA)	(%)				
J1359	PCI J2-Oakridge 69 kV	109.0	ITCM CIPCO	123.8	113.5	CEII Redacted	P2-P7	CIPCO: Not a constraint. \$0 ITCM: Relay thermal, replace sub taps (.08 miles), new rating 143 MVA. \$450K	
J1471,J1479	MPW Split-Newport 161 kV	175.0	ITCM MEC CIPCO	182.9	104.5	CEII Redacted	P2-P7	ITCM: Not a constraint. ITCM sag limit at 186 MVA. \$0 MEC: No MEC mitigation required. MEC only rating projected to be 203 MVA after previously planned maintenance activity. \$0 CIPCO: Not a constraint. Rating has been increased to 248 SUM / 321 WIN. \$0	\$0
J1190,J1432	Grand Mound-Maquoketa 161 kV	200.0	CIPCO ITCM	225.5	112.7	CEII Redacted	P1	CIPCO: Not a constraint, rating can be increased to 237 MVA at no cost. ITCM: Not an ITCM constraint. ITCM facilities rated at 366 MVA. \$0	\$0
J1190	Grand Mound-Maquoketa 161 kV	200.0	CIPCO ITCM	225.7	112.8	CEII Redacted	P2-P7	CIPCO: Not a constraint, rating can be increased to 237 MVA at no cost. ITCM: Not an ITCM constraint. ITCM facilities rated at 366 MVA. \$0	
J1471,J1478, J1479	Denmark-Burlington 161 kV #1	240.0	ITCM	250.0	104.2	CEII Redacted	P1	Line is sag limited. Replace 4 spans to increase rating to 262 MVA. \$1.25M	
J1471,J1478, J1479	Denmark-Burlington 161 kV #1	240.0	ITCM	266.5	111.1	CEII Redacted	P2-P7	Rebuild 13 miles of 161 kV and bus conductor at Denmark and Burlington. New line rating would be 371 MVA	\$16,500,000

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation	Cost (\$)
				(MVA)	(%)				
J1471,J1478, J1479	Denmark-Burlington 161 kV #2	254.0	ITCM	254.8	100.3	CEII Redacted	P1	Line is sag limited. Replace 2 spans to increase rating to 257 MVA	\$650,000
J1471	Wever-Lee 69 kV	84.0	ITCM AECI	96.5	114.9	CEII Redacted	P1	ITCM: AECI constraint AECI: NU is not required unless it is identified as constraint in affected system study.	\$0
J1471	Wever-Lee 69 kV	84.0	ITCM AECI	97.4	115.9	CEII Redacted	P2-P7	ITCM: AECI constraint AECI: NU is not required unless it is identified as constraint in affected system study.	\$0
J1314	Apple River 161-69 kV xfmr #2	82.0	DPC	84.8	103.4	CEII Redacted	P2-P7	Upgrade Apple River Transformer #2	\$2,636,000
J1298,J1413	Wabaco-Rochester 161 kV	221.1	DPC	235.4	106.5	CEII Redacted	P2-P7	A third party will fund the Rochester-Wabaco 161kV rebuild. \$11M. MTEP Appendix A project	\$0

4.4 Network Upgrades Identified in MISO ERIIS Analysis for 2024 Summer Peak Scenario

Based on the MISO 2024 summer peak discharging steady state analyses, the MISO thermal NUs and cost are listed in Table 4-2. No reactive power NUs are required.

Table 4-2: Summer Peak Discharging Thermal NUs

Constraint	Owner	Mitigation	Cost (\$)
Denmark-Spring Grove 69 kV	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
Henry Co. 161-69 kV xfmr	AECI	NU is not required unless it is identified as constraint in affected system study.	\$0
Amoco-Niota 69 kV	Ameren	Replace (1) 72 MVA switch with 1200 A switch.	\$55,000
Niota 161-69 kV xfmr	Ameren	Replace with new 112 MVA 161/69 kV Transformer.	\$4,000,000
PCI J2-Oakridge 69 kV	ITCM CIPCO	CIPCO: Not a constraint. \$0 ITCM: Relay thermal, replace sub taps (.08 miles), new rating 143 MVA. \$450K	\$450,000
MPW Split-Newport 161 kV	ITCM MEC CIPCO	ITCM: Not a constraint. ITCM sag limit at 186 MVA. \$0 MEC: No MEC mitigation required. MEC only rating projected to be 203 MVA after previously planned maintenance activity. \$0 CIPCO: Not a constraint. Rating has been increased to 248 SUM / 321 WIN. \$0	\$0
Grand Mound-Maquoketa 161 kV	CIPCO ITCM	CIPCO: Not a constraint, rating can be increased to 237 MVA at no cost. ITCM: Not an ITCM constraint. ITCM facilities rated at 366 MVA. \$0	\$0
Denmark-Burlington 161 kV #1	ITCM	Rebuild 13 miles of 161 kV and bus conductor at Denmark and Burlington. New line rating would be 371 MVA	\$16,500,000
Denmark-Burlington 161 kV #2	ITCM	Line is sag limited. Replace 2 spans to increase rating to 257 MVA	\$650,000
Wever-Lee 69 kV	ITCM AECI	ITCM: AECI constraint AECI: NU is not required unless it is identified as constraint in affected system study.	\$0
Apple River 161-69 kV xfmr #2	DPC	Upgrade Apple River Transformer #2	\$2,636,000
Wabaco-Rochester 161 kV	DPC	A third party will fund the Rochester-Wabaco 161kV rebuild. \$11M. MTEP Appendix A project	\$0

Section

5

Local Planning Criteria Analysis

Local Planning Criteria (LPC) analyses were performed to identify additional constraints per Transmission Owning Companies' LPC.

5.1 GRE Local Planning Criteria Analysis

Great River Energy (GRE) determined with provided rationale that the GRE LPC should be applied to the projects listed in Table 5-1. The GRE LPC analysis consisted of steady-state contingency analysis and stability analysis for summer shoulder condition.

Table 5-1: DPP Projects with GRE LPC Applicable

MISO Project	Pmax (MW)	Fuel type	POI	Rationale
J1221	150	Solar/Battery	Wing River-Riverton 230 kV	GRE is the Transmission Owner of the portion of the transmission line at the POI
J1426	100	Solar	Belton County 115 kV	GRE is the Transmission Owner of the substation

Based on geographic locations of the projects' Point of Interconnection (POI), the projects were separated into the following two groups for GRE LPC study:

- J1221 GRE LPC group: J1221
- J1426 GRE LPC group: J1426

Siemens PTI performed the GRE local planning criteria analysis based on GRE's LPC. The J1221 GRE local planning criteria analysis details can be found in Appendix F.1.1, and the J1426 GRE local planning criteria analysis details can be found in Appendix F.1.2.

5.1.1 Additional Network Upgrades Identified in J1221 GRE LPC Analysis

Additional Network Upgrades required in the J1221 GRE LPC study are listed in Table 5-2.

Table 5-2: Additional Network Upgrades Required in the J1221 GRE LPC Study

Constraint	Owner	Mitigation	Cost (\$)
Wing River 230-115-13.8 kV xfmr	GRE	Replacement of the Wing River xfmr	\$3,000,000

5.1.2 Additional Network Upgrades Identified in J1426 GRE LPC Analysis

No thermal constraints or voltage constraints were identified in the J1426 GRE LPC analysis.

No stability violations were identified in the J1426 GRE LPC stability analysis.

No additional Network Upgrades are required in the J1426 GRE LPC study.

5.2 OTP Local Planning Criteria Analysis

OTP's LPC is being performed due to interconnections J1328, J1371, J1456, and J1485 requesting interconnection into OTP's transmission system. The OTP LPC analysis consisted of steady-state contingency analysis for summer shoulder, summer peak, and Light Load No Wind conditions.

Three scenarios were analyzed in the OTP LPC study:

- a. OTP LPC Summer Peak (SPK)
- b. OTP LPC Summer Shoulder (SH)
- c. OTP LPC Light Load No Wind (LLNW)

Any constraints or violation of system performance criteria that occurs in the LPC Study Cases and does not occur in the LPC Benchmark Cases should be identified, allocated and mitigated by the 2019 DPP generators.

Siemens PTI performed the local planning criteria analysis based on OTP's LPC. The OTP local planning criteria analysis details can be found in Appendix F.2.

5.2.1 Additional Network Upgrades Identified in OTP LPC Analysis

No thermal or voltage constraints were identified in OTP LPC summer peak study.

Voltage collapse, thermal and voltage constraints were identified in OTP LPC summer shoulder study. The identified voltage collapse, thermal and voltage constraints can be mitigated by building a 2nd Astoria-Brookings County 345 kV line.

In OTP LPC Light Load No Wind (LLNW) condition, high voltage violation was identified at J1456 POI 230 kV bus. J1456 is responsible for mitigating the identified high voltage violation.

Additional Network Upgrades required in the OTP LPC study are listed in Table 5-3.

Table 5-3: Additional Network Upgrades for Constraints Identified in OTP LPC Analysis

Constraint	Owner	Mitigation	Cost (\$)
Voltage collapse, thermal and voltage constraints in OTP	OTP	Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000
High voltage at J1456 POI 230 kV	OTP	1×15 Mvar switched reactor at J1456 POI 230 kV	\$2,500,000 ¹

Note 1: The cost of \$2.5 M of the reactor is accounted for in the interconnection substation.

5.3 DPC Local Planning Criteria Analysis

The DPC LPC study is performed to adhere to DPC Local Planning Criteria. This DPC LPC analysis is required for J1314 project in addition to MISO's standard DPP analysis.

Two additional scenarios are analyzed in this DPC LPC study:

1. DPC LPC Summer Shoulder
2. DPC LPC Summer Peak

Siemens PTI performed the local planning criteria analysis based on DPC's LPC. The DPC local planning criteria analysis details can be found in Appendix F.3.

5.3.1 Additional Network Upgrades Identified in DPC LPC Analysis

No thermal or voltage constraints were identified in the DPC LPC summer shoulder or summer peak analysis.

No additional Network Upgrades are required in the DPC LPC study.

Affected System Steady-State Analysis

Steady state analyses were performed to identify constraints in affected systems.

6.1 Affected System Analysis for CIPCO Company

Per CIPCO Affected System Planning Criteria, a CIPCO transmission facility is a constraint if it satisfies all three of the following conditions:

1. the branch is loaded above its applicable normal or emergency rating for the post-change case, and
2. the generator has a larger than 3% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, and
3. the loading increase of the overloaded facility is greater than 1 MVA compared with that in the pre-change case under system intact or contingency conditions.

AC contingency analysis was performed for this CIPCO affected system analysis, using the following benchmark and study cases:

- Summer peak discharging benchmark and study cases
- Summer shoulder charging benchmark and study cases
- Summer shoulder discharging benchmark and study cases

All NERC category P0-P7 contingencies described in Section 2.2 were simulated. The CIPCO affected system was monitored.

CIPCO affected system results are listed in Appendix G.1. No CIPCO thermal constraints were identified.

6.2 MPC Affected System Analysis

The MPC affected system analysis details can be found in Appendix G.2.

6.2.1 Study Summary

Minnkota Power Cooperative (MPC) performed an Affected System Analysis (ASA) to determine impacts of generators in the MISO DPP 2019 Phase 2 study cycle on MPC facilities and any network upgrades required to mitigate those impacts. Steady-state power flow analysis, steady-state contingency analysis, and dynamic stability analysis were performed for three DPP generating facility.

- J1456, NRIS, 300 MW wind at Harvey – Underwood 230 kV line

6.2.2 Network Upgrades

The Network Upgrades required to mitigate constraints identified in the Minnkota ASA are listed in Table 6-1 through Table 6-3. Costs are planning level estimates and subject to revision in the facility studies.

Table 6-1. Minnkota Network Upgrades Allocated to DPP 2019 Projects

Constraint	Owner	Highest Loading (MVA)	Mitigation	Cost (\$)	Generators
Center-Roughrider 230	MPC	632	Resag to 100°	\$500,000	J1456
Drayton 230-115 kV xfmr	MPC	224	Replace transformer.	\$2,000,000	J1456
Prairie-Lake Ardoch 230	OTP MPC	409	MPC Resag to 100° OTP: Mitigation not required for ASA	\$500,000	J1456
Transient Instability	MPC	-	+50/-0 MVar SVC at Winger 230	\$15,000,000	J1456
Transient Instability	MPC	-	2x30 MVar MSC at Winger 230	\$2,000,000	J1456
Transient Instability	MPC	-	+50/-0 MVar SVC at Prairie 230	\$15,000,000	J1456
Transient Instability	MPC	-	5x30 MVar MSC at Prairie 230	\$5,000,000	J1456
Transient Instability	MPC	-	+100/-0 MVar SVC at Frontier 230	\$20,000,000	J1456
Total				\$60,000,000	

Table 6-2: Contingent Facilities: Minnkota Network Upgrades Allocated to DPP 2017 August Projects

Constraint	Owner	Highest Loading (MVA)	Mitigation	Generators
Voltage collapse	-	-	Install 2x75 MVar MSC at Bison 345 (601067)	J1456
Voltage collapse	-	-	Install 2x30 MVar MSC at Winger 230 (657758)	J1456

**Table 6-3: Contingent Facilities: Minnkota Network Upgrades
Allocated to DPP 2018 April Projects**

Constraint	Owner	Highest Loading (MVA)	Mitigation	Generators
MPC Steady State Voltage	-	-	Additional 1x75 MVar MSC at Bison 345 (601067)	J1456
MPC Steady State Voltage	-	-	2x40 MVar MSC at Maple River 230 (657754)	J1456
Transient Instability	-	-	+200/-0 MVar SVC at Winger 230 (657758)	J1456
Transient Instability	-	-	+50/-0 MVar SVC at Frontier 230 (657750)	J1456
Transient Instability	-	-	2x30 MVar MSC at Frontier 230 (657750)	J1456

6.3 PJM Affected System Analysis

The PJM affected system analysis details can be found in Appendix G.3.

6.3.1 Study Results

6.3.1.1 Overload on GARDEN PR; R 345 kV - SILVER LK; R 345 kV line

The upgrade will be to re-conductor the line, mitigate sag limits, upgrade relay package, replace station conductor at both line terminals, replace 2-345 kV breakers and CT's at TSS 138 Silver Lake. A preliminary estimate for the upgrade is \$32.4 M.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$ 32.4 M)
J850	25.30	6.6%	\$2.14
J870	17.20	4.5%	\$1.45
AD2-102	47.10	12.3%	\$3.98
AE1-114	27.00	7.0%	\$2.28
AF1-012	18.85	4.9%	\$1.59
AF1-280	32.70	8.5%	\$2.76
AF1-296	21.97	5.7%	\$1.86
AF2-182	29.50	7.7%	\$2.49
AF2-349	89.80	23.4%	\$7.59
J1214	38.66	10.1%	\$3.27
J1377	16.74	4.4%	\$1.41
J1432	18.64	4.9%	\$1.58

6.3.1.2 Overload on CHERRY VA; B 345 kV - GARDEN PR; R 345 kV line

The upgrade will be to install a new 345kV line from the AB1-089 interconnection substation to TSS 144 Wayne station. A preliminary estimate for the upgrade is \$165 M.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$ 165 M)
J1214	38.66	52.22%	\$86.15
J1377	16.74	22.61%	\$37.31
J1432	18.64	25.18%	\$41.54

6.3.1.3 Overload on NELSON ; B 345 kV - ELECT JCT; B 345 kV line

The upgrade will be to perform sag mitigation on a portion of the line section along with re-conductoring on a different section, upgrade station conductor at both line terminals, replace both line motor operated disconnect switches and line current transformers, upgrade line relay schemes at both terminals. A preliminary estimate for the upgrade is \$10.5 M.

The cost allocation is as follows:

Queue	MW Contribution	Percentage of Cost	Cost (\$10.5 M)
AF2-199	59.4	23.57%	\$2.48
AF2-200	118.8	47.15%	\$4.95
AF2-346	32.1	12.74%	\$1.34
J1432	41.68	16.54%	\$1.74

6.3.1.4 Overload on BYRON ; B 345 kV - CHERRY VA; B 345 kV line

The upgrade will be to mitigate sag on the line and replace a 345kV circuit breaker at Cherry Valley. A preliminary estimate for the upgrade is \$12.2 M.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$ 12.2 M)
AF1-280	55.12	19.37%	\$2.36
AF1-296	25.97	9.13%	\$1.11
AF2-041	30.1	10.58%	\$1.29
AF2-182	83.0	29.17%	\$3.56
AF2-183	22.1	7.77%	\$0.95
AF2-200	20.1	7.07%	\$0.86
AF2-392	25.6	9.00%	\$1.10
J1432	22.51	7.91%	\$0.97

The upgrade will be to reductor the line, station conductor, relay scheme upgrade and replace 3-345kV circuit breakers at Cherry Valley. A preliminary estimate for the upgrade is \$23.4 M.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$ 23.4 M)
AF2-392	25.6	53.21%	\$12.45
J1432	22.51	46.79%	\$10.95

6.3.2 Study Summary

Project J1432 in the MISO DPP 2019 West Area group contributes loading to the overloads in the PJM system. J1432 is responsible for the cost of Network Upgrades per PJM cost allocation rules.

6.4 AECI Affected System Analysis

The AECI affected system analysis details (Version 1) can be found in Appendix G.4.

6.4.1 Study Results

Associated Electric Cooperative Inc. (AECI), through coordination with the Midcontinent Independent System Operator (MISO), has identified several generator interconnection requests (GIRs) within the DPP-2019 Study Cycle (the “Study Cycle”) for an Affected System Study (AFS) evaluation on the AECI transmission system.

Steady state analysis was performed to confirm the reliability impacts on the AECI system under a variety of system conditions and outages. AECI's transmission system must be capable of operating within the applicable normal ratings, emergency ratings, and voltage limits of AECI planning criteria.

Steady state analysis results showed twenty (20) constraints reported on the AECI transmission system, which are attributed to the Study Cycle requests.

AECI developed non-binding, good faith estimates of the timing and cost estimates for upgrades needed as a result of the addition of the Study Cycle projects, as shown in Table 6-4. The associated cost allocation of the network upgrades to each of the Study Cycle projects is provided in Table 6-5.

Generation projects J1218, J1229, J1315, J1456, J1471, J1478, and J1479 in DPP 2019 West Phase 2 study cycle are responsible for the cost of some of the Network Upgrades identified in the AECI Affected System Study.

Table 6-4: Network Upgrade Costs

ID	Option / Description	Current Cost*	Year In Service
NU01	Rebuild 9.36 mile-long Boone to Millersburg 161 kV line to 1192 ACSR, replace jumpers at Boone and Millersburg 161 kV buses	\$4,055,000	2025
NU02	Upgrade jumpers at Millersburg 161 kV bus to 1192 ACSR	\$50,000	2025
NU03	Rebuild 8.07 mile-long Kingdom City to Millersburg 161 kV line to 1192 ACSR	\$3,410,000	2025
NU04	Replace Chillicothe 161/69 kV transformer 1 with 84/95 MVA unit	\$2,929,000	2025
NU05	Rebuild 2.05 mile-long Chillicothe to Chillicothe Rural 69 line to 336 ACSR	\$952,000	2025
NU06	Rebuild 10.23 mile-long Avalon to Hale 69 line to 336 ACSR	\$4,748,000	2025
NU07	Rebuild 10.9 mile-long 4/0 ACSR segment of Hale to Indian Grove 69 kV line to 336 ACSR at 100C	\$5,059,000	2025
NU08	Rebuild 0.1 mile-long Bevier to Bevier Tap 69 line to 795 ACSR, replace circuit breakers at Bevier 69 kV bus	\$186,000	2025
NU09	Rebuild 4.25 mile-long Macon Lake to Bevier Tap 69 line to 795 ACSR at 100C	\$3,655,000	2025
NU10	Rebuild 12.2 mile-long Love Lake to Macon Tap 69 line to 336 ACSR, replace circuit breakers at Love Lake 69 kV bus	\$10,520,000	2025
NU11	Rebuild 1.15 mile-long Axtell to Macon Lake 69 line to 795 ACSR	\$982,000	2025
NU12	Rebuild 1.05 mile-long Axtell to Macon Tap 69 kV line to 795 ACSR	\$897,000	2025
NU13	Replace Holden 161/69 kV transformer 1 with 112/127 MVA unit	\$3,661,000	2025

ID	Option / Description	Current Cost*	Year In Service
NU14	Uprate 3.1 mile-long 336 ACSR segment of Elm to Holden 69 kV line to from 75 C to 100C.	\$407,000	2025
NU15	Rebuild 4.36 mile-long Gobbler Knob to Poplar Bluff South 69 kV line to 336 ACSR at 100C	\$1,536,000	2025
NU16	Upgrade relays at St. Francis 161 kV bus	\$50,000	2025
NU17	Rebuild 9.93 mile-long St. Francis to Jim Hill 161 kV line to 1192 ACSR at 100C, replace jumpers at St. Francis with 1192 ACSR at 100C, replace disconnect switches at St. Francis 161 kV bus	\$7,276,000	2025
NU18	Replace Thomas Hill 161 kV CB 1292, replace 161 kV disconnect switches at Thomas Hill, upgrade jumpers at Thomas Hill to 1192 ACSR, adjust Thomas Hill relay settings	\$300,000	2025
NU19	Rebuild 7.0 mile-long 4/0 ACSR segment of Avalon - Chillicothe Industrial Park 69 kV line to 336 ACSR	\$3,249,000	2025
NU20	Uprate 336 ACSR 23 mile-long Thomas Hill to Indian Grove 69 kV line from 75C to 100C.	\$3,020,000	2025
*Includes engineering and contingencies		Total Cost:	\$56,942,000

Table 6-5: Network Upgrade Cost Allocation

Project	Study Group	POI	MW	Total Cost
J1191	Central	Bolstad 69 kV Substation	64	\$1,568,634
J1213	Central	Taum Sauk 138kV Switchyard, Bus 1 & 2	60	\$586,383
J1216	Central	Ashley 138 kV Substation	250	\$1,716,892
J1218	West	Sub 92 345 kV	200	\$195,760
J1222	Central	Spencer Creek 345 kV Substation	200	\$2,609,217
J1229	West	Sub 92 345 kV Substation	50	\$48,940
J1241	Central	West Mt Vernon - Xenia 345kV Line	200	\$1,204,440
J1268	Central	Pike - Troy 161kV Line	150	\$1,660,878
J1289	Central	Turner 345kV Switching Station	200	\$723,061
J1299	Central	Miner - Kelso 161kV Line	200	\$1,374,539
J1306	Central	Pickneyville - St John 230kV Line	200	\$1,496,606
J1315	West	Lyon County - Cedar Mountain 345 kV Line Tap	600	\$543,195
J1330	Central	Baldwin 345kV Substation	200	\$1,727,050
J1338	Central	Spencer Creek - Montgomery 345 kV Line Tap	150	\$1,998,304
J1352	Central	Montgomery 345kV Sub	100	\$1,488,969
J1354	Central	Decatur County 138kV Switching Station	146	\$402,255
J1363	Central	Spencer Creek - Montgomery 345 kV Line Tap	150	\$2,231,074
J1376	Central	Barnett - Georgetown Overton 161 kV Line Tap	100	\$1,696,644
J1383	Central	Ipava 138 kV Substation	150	\$435,397
J1408	Central	Viriden South 138kV Substation	100	\$345,983
J1437	South	AECC Dry Creek 161kV Switchyard Station	180	\$800,267
J1453	Central	Havana - Shockey 138 kV Line Tap	188	\$520,594

Project	Study Group	POI	MW	Total Cost
J1456	West	Harvey - Underwood 230 kV Line Tap	300	\$271,588
J1464	Central	Meredosia - Austin 345 kV Line tap	593	\$2,049,993
J1471	West	ITC 161kV Wever Substation	200	\$532,261
J1478	West	Huntwoods 69 kV substation (J1135)	25	\$61,450
J1479	West	Burlington 161 kV Substation	100	\$258,529
J1488	Central	McCredie - Montgomery 345 kV Line Tap	500	\$9,464,366
J1490	Central	McCredie - Montgomery 345 kV Line Tap	1000	\$18,928,732
			Total Cost	\$56,942,000

6.5 SPP Affected System AC Contingency Analysis

Southwest Power Pool (SPP) conducted an Affected System Impact Study (ASIS) to evaluate potential impacts to the SPP Transmission System related to the interconnection of generators on the Mid-Continent Independent System Operation (MISO) Transmission System.

A steady-state thermal and voltage analysis as well as Transfer Distribution Factor analysis was performed to determine the impact the MISO GIRs have on the SPP system.

Cost allocation of SPP Network Upgrades are listed in Table 6-6.

The SPP affected system analysis results (09/29/2021) for this study are in Appendix G.5.

Table 6-6: SPP Network Upgrades Cost Allocation

Interconnection Request	ERIS MW	NRIS MW	ERIS	NRIS	Total	ERIS Total	NRIS Total	Total
J1190	200	200	\$0	\$0	\$0	\$275,600,356	\$122,704,700	\$398,305,056
J1218	200	200	\$0	\$0	\$0			
J1221	150	150	\$0	\$1,084,499	\$1,084,499			
J1229	50	50	\$0	\$0	\$0			
J1270	200	200	\$2,205,552	\$3,571,286	\$5,776,838			
J1298	150	150	\$7,876,897	\$12,313,729	\$20,190,626			
J1312	50	50	\$81,710,893	\$27,660,352	\$109,371,245			
J1313	100	100	\$0	\$922,138	\$922,138			
J1314	100	100	\$3,193,893	\$581,007	\$3,774,900			
J1315	600	600	\$46,029,408	\$10,825,040	\$56,854,448			
J1321	65	65	\$702,286	\$1,152,995	\$1,855,281			
J1325	80	80	\$8,284,359	\$1,952,466	\$10,236,825			
J1328	100	100	\$9,446,750	\$1,873,915	\$11,320,665			
J1349	335	335	\$13,472,065	\$6,304,841	\$19,776,906			
J1359	95	95	\$0	\$380,782	\$380,782			
J1365	169.4	169.4	\$420,151	\$1,009,394	\$1,429,545			
J1371	150	150	\$15,505,450	\$3,032,778	\$18,538,228			
J1395	200	200	\$17,757,399	\$3,816,570	\$21,573,969			
J1413	235	215	\$1,028,155	\$1,026,944	\$2,055,099			
J1416	149	149	\$8,306,024	\$18,883,512	\$27,189,536			
J1417	75	75	\$3,002,249	\$7,345,648	\$10,347,897			
J1426	100	100	\$4,202,720	\$684,361	\$4,887,081			

Interconnection Request	ERIS MW	NRIS MW	ERIS	NRIS	Total	ERIS Total	NRIS Total	Total
J1432	250	250	\$0	\$0	\$0			
J1438	100	100	\$441,237	\$478,230	\$919,467			
J1444	200	200	\$0	\$826,139	\$826,139			
J1445	100	100	\$4,239,986	\$684,738	\$4,924,724			
J1446	150	150	\$6,092,136	\$1,025,837	\$7,117,973			
J1456	300	300	\$0	\$2,476,406	\$2,476,406			
J1468	100	100	\$3,767,009	\$659,226	\$4,426,235			
J1471	200	200	\$0	\$0	\$0			
J1474	200	200	\$5,722,187	\$1,083,107	\$6,805,294			
J1477	50	50	\$0	\$0	\$0			
J1478	25	25	\$0	\$0	\$0			
J1479	100	100	\$0	\$0	\$0			
J1485	250	0	\$25,842,417	\$0	\$25,842,417			
J1487	120	120	\$6,351,133	\$11,048,760	\$17,399,893			

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Stability Analysis

Stability analysis was performed to evaluate the transient stability and impact on the region of the generating facilities in the DPP 2019 West study cycle.

7.1 Procedure

7.1.1 Computer Programs

Stability analysis was performed using TSAT revision 19.0.

7.1.2 Study Methodology

A stability package representing 2024 summer peak (PK) and summer shoulder (SH) conditions with generating facilities in the DPP 2019 cycle was created from the MTEP19 stability package. Disturbances were simulated to evaluate the transient stability and impact on the region of the generating facilities. MISO transient stability criteria and local TOs' planning criteria specified in MTEP19 were adopted for checking stability violations.

7.2 Case Development

7.2.1 Summer Peak (PK) Stability Model

Summer peak stability model is the same as the summer peak steady state model. The stability model includes the Hazel Creek-Scott County 345 kV Base Case NU and 2×75 MVar capacitor bank at Cedar Mountain 345 kV bus (DPP 2018 April voltage NU).

7.2.2 Summer Shoulder (SH) Stability Model

Summer shoulder stability model is also the same as the summer shoulder steady state Stage-2 model. The stability model also includes the Hazel Creek-Scott County 345 kV Base Case NU and 2×75 MVar capacitor bank at Cedar Mountain 345 kV bus (DPP 2018 April voltage NU).

7.3 Disturbance Criteria

The stability simulations performed as part of this study considered all the regional and local contingencies listed in Table 7-1. Regional contingencies with pre-defined switching sequences were selected from the MISO MTEP19 study; switching sequences for local contingencies were developed based on the generic clearing times shown in Table 7-2. The admittance for local single line-to-ground (SLG) faults were estimated by assuming that the

Thevenin impedance of the positive, negative and zero sequence networks at the fault point are equal.

Table 7-1: Regional and Local Disturbance Descriptions

CEII Redacted

Table 7-2: Generic Clearing Time Assumption

Voltage Level (kV)	Primary Clearing Time (cycle)	Backup Clearing Time (cycle)
345 kV	4	11
230 kV	5	13
161/138 kV	6	18
115 kV	6	20
69 kV	8	24

7.4 Performance Criteria

MISO transient stability criteria and local TOs' planning criteria specified in MTEP19 were adopted. All generators must mitigate the stability constraints to obtain any type of Interconnection Service.

7.5 Summer Peak Stability Results

The contingencies listed in Table 7-1 were simulated using the summer peak stability case.

Appendix H.1.2 contains plots of generator rotor angles, generator power output, and bus voltages for each simulation. Simulations were performed with a 0.5 seconds steady-state run followed by the appropriate disturbance. Simulations were run for a 10-second duration.

Summer peak stability study results summary is in Appendix H.1.1, **Table H-1**.

The following stability related issues were identified in the summer peak stability study.

7.5.1 HVRT Violations

J1432 High Voltage Ride Through (HVRT) violations were identified under multiple faults listed in Table 7-3. J1432 was tripped by its high voltage relay (>1.15 per unit for 0.1 second). Mitigations are required for J1432 to meet HVRT requirement.

Table 7-3: J1432 HVRT Violations in Summer Peak

Contingency Name	Contingency Description	NERC Cat.	Violations
J1432_3PH_POI_SALEM_3_Fault	3PH fault on J1432 POI to Salem 345 kV line; Cleared at 4 cycles	P1-2	J1432 tripped by voltage relay (>1.15 pu for 0.1 s)

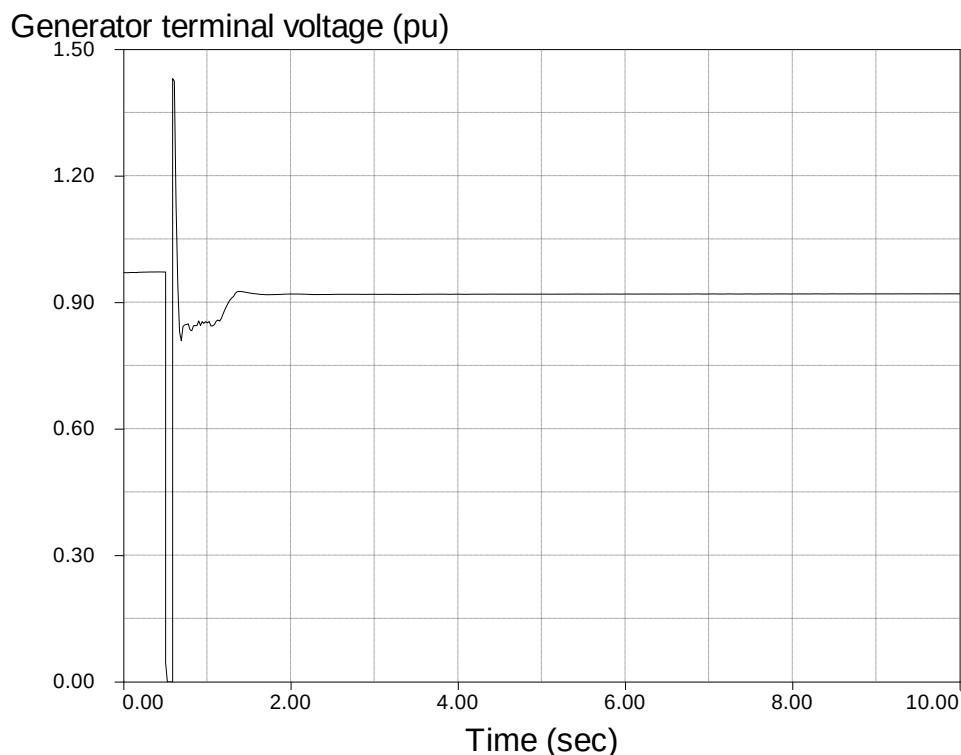
Contingency Name	Contingency Description	NERC Cat.	Violations
J1432_SLG_POI_SALEM_3_Fault	SLG fault on J1432 POI to Salem 345 kV line; Delayed cleared at 11 cycles	P5-2	J1432 tripped by voltage relay (>1.15 pu for 0.1 s)
J1432_3PH_POI_ROCK CK3_Fault	3PH fault on J1432 POI to Rock Creek 345 kV line; Cleared at 4 cycles	P1-2	J1432 tripped by voltage relay (>1.15 pu for 0.1 s)
J1432_SLG_POI_ROCK CK3_Fault	SLG fault on J1432 POI to Rock Creek 345 kV line; Delayed cleared at 11 cycles	P5-2	J1432 tripped by voltage relay (>1.15 pu for 0.1 s)
1619_w_itcm_p42	SLG fault at Rock Creek; Trip Quad City to Rock Creek after 5 cycles; Trip Rock Creek to J1432 POI (Salem) after 4 cycles; clear fault	P4-2	J1432 tripped by voltage relay (>1.15 pu for 0.1 s)

7.5.2 Transient High Voltage Violation at Harvey 230 kV Bus

Under the fault of “J1456_3PH_POI_UNDERWD4_Fault” (Table 7-4), transient voltage at Harvey 230 kV bus was above 1.2 p.u. for 0.017 second (>0.000 second threshold). This transient high voltage violation was caused by J1456 electrical control model. As shown in Figure 7-1, J1456 terminal voltage shot to 1.425 per unit immediately after the fault was cleared. J1456 project is responsible for mitigating this high voltage violation at Harvey 230 kV bus.

Table 7-4: Transient High Voltage Violation at Harvey 230 kV in Summer Peak

Contingency Name	NERC Cat.	Violations	Mitigations
J1456_3PH_POI_UNDERWD4_Fault	P1-2	Instantaneous Voltage (>1.2 p.u. for more than 0.0 s): 655612HARVEY -CP7115. 620290HARVEY 4 230. 620276NHARVEY7 115.	Harvey 230 kV bus voltage is above 1.2 p.u. for 0.017 sec. This is caused by J1456 Electrical Control model.



**Figure 7-1: J1456 Terminal Voltage under the Fault of
"J1456_3PH_POI_UNDERWD4_Fault"**

7.5.3 Transient Voltage Rise/Drop at Arnold 161 kV Bus

Under multiple faults listed in Table 7-5, transient voltage at Arnold 161 kV bus was above 1.041 p.u. for more than 1.0 second, and/or transient voltage at Arnold 161 kV bus dropped to below 0.992 p.u. for more than 1.0 second. The transient voltage rise/drop can be resolved by resetting generator's scheduled voltage, turning off capacitor at Arnold 161 kV bus, or adjusting transformer tap position.

**Table 7-5: Transient Voltage Rise/Drop at Arnold 161 kV Bus in
Summer Peak**

CEII Redacted

7.5.4 Transient High Voltage Violations

Under multiple faults listed in Table 7-6, transient high voltage violations were observed. These transient high voltage violations were either not higher than voltages in benchmark model by 0.01 per unit, or they could be resolved by resetting transformer tap position.

Table 7-6: Transient High Voltage Violations in Summer Peak

CEII Redacted

7.5.5 Stability Network Upgrades Identified in Summer Peak

In summary, J1432 is required to meet HVRT requirement, and J1456 is required to meet transient high voltage violation at Harvey 230 kV bus. There are no other stability Network Upgrades identified in summer peak stability study.

7.6 Summer Shoulder Stability Results

The contingencies listed in Table 7-1 were simulated using the summer shoulder stability case.

Appendix H.2.2 contains plots of generator rotor angles, generator power output, and bus voltages for each simulation. Simulations were performed with a 0.5 seconds steady-state run followed by the appropriate disturbance. Simulations were run for a 10-second duration.

Summer shoulder stability study results summary is in Appendix H.2.1, Table H-2.

The following stability related issues were identified in the summer shoulder stability study.

7.6.1 LVRT or HVRT Violations

J1432 High Voltage Ride Through (HVRT) violations and J1446 Low Voltage Ride Through (LVRT) violations were identified under multiple faults listed in Table 7-7. J1432 was tripped by its high voltage relay (>1.15 per unit for 0.1 second), and J1446 was tripped by its low voltage relay (<0.65 per unit for 0.3 second). Mitigations are required for J1432 to meet HVRT requirement, and mitigations are required for J1446 to meet LVRT requirement.

Table 7-7: J1432 HVRT Violations and J1446 LVRT Violations in Summer Shoulder

CEII Redacted

7.6.2 Transient Voltage Rise/Drop at Arnold 161 kV Bus

Under multiple faults listed in Table 7-8, transient voltage at Arnold 161 kV bus was above 1.041 p.u. for more than 1.0 second, and/or transient voltage at Arnold 161 kV bus dropped to below 0.992 p.u. for more than 1.0 second. The transient voltage rise/drop can be resolved by resetting generator's scheduled voltage, turning off capacitor at Arnold 161 kV bus, or adjusting transformer tap position.

Table 7-8: Transient Voltage Rise/Drop at Arnold 161 kV Bus in Summer Shoulder

CEII Redacted

7.6.3 Transient High Voltage Violations

Under multiple faults listed in Table 7-9, transient high voltage violations were observed. These transient high voltage violations can be resolved by resetting generator's scheduled voltage, switching off capacitor, and/or switching on reactor.

Table 7-9: Transient High Voltage Violations in Summer Shoulder

CEII Redacted

7.6.4 Transient Low Voltage Violations

Under the fault of “0888_w_xel_p12” (Table 7-10), transient voltage at Underwood 230 kV bus is less than 0.7 p.u. for 0.003 second (>0.000 second threshold). This transient low voltage violation can be resolved by switching on capacitors of J1456 and resetting J1456 scheduled voltage.

Table 7-10: Transient Low Voltage Violations in Summer Shoulder

CEII Redacted

7.6.5 Stability Network Upgrades Identified in Summer Shoulder

In summary, J1432 is required to meet HVRT requirement, and J1446 is required to meet LVRT requirement. There are no other stability Network Upgrades identified in summer shoulder stability study.

7.7 Additional Network Upgrades Identified in Stability Analysis

Additional mitigations required in the DPP 2019 Phase 2 stability analysis are listed below:

- Mitigation is required for J1432 to meet HVRT requirement
- Mitigation is required for J1446 to meet LVRT requirement
- Mitigate is required for J1456 to resolve transient high voltage violation at Harvey 230 kV.

MWEX Voltage Stability Study

ATC performed steady state voltage stability analysis. Voltage stability analysis is required to determine if the initial conditions of the DPP system models under study are in a stable state as defined by Power-Voltage (PV) curves of the Minnesota Wisconsin Export Interface (MWEX) for the worst contingency.

As shown in Table 8-1, the results of this analysis show that the Bench and Study scenarios are voltage stable and meet all other criteria, so Network Upgrades will not be assigned to the Interconnection Customers.

The MWEX voltage stability study details can be found in Appendix I.

Table 8-1: MWEX Margins to Collapse in the 2024SH Cases

	Real Power Flow (MW)							
	AHD-PAR ²	MWEX			Post-Contingent Margin to Nose ³		MWEX Control Point	
Case	N-0 Initial Condition	N-0 I.C. ⁴	N-1 I.C. ⁴	N-1 Nose	(MW)	(%)	(MW)	Notes
Bench	333	820	713	1314	601	46	1492	Voltage Stable, Sufficient Margin
Study (Charging)	395	966	835	1329	494	37	1507	Voltage Stable, Sufficient Margin

Notes:

- Based on 50 MW transfer intervals
- As described in the active MWEX Operating Guide, the AHD-SLK interface is a single element Power Transfer Distribution Factor (PTDF) interface measured at the Minnesota Power 230 kV side of the Arrowhead 230 kV phase shifter. Due to the addition of the Superior substation, the AHD-SLK interface has here been re-named to AHD-PAR to avoid confusion.
- Post-Contingent Margin to Nose is defined as:
 - "Post-Contingent Margin to Nose (MW)" = "MWEX N-1 Nose" – "N-1 Initial Condition"
 - "Post-Contingent Margin to Nose (%)" = "Post-Contingent Margin to Nose (MW)" / "MWEX N-1 Nose"
- Initial Condition flows were measured in the base cases with an intact system and the worst contingency, plus operation of various control systems as needed with all transformer taps, switched shunts, and PARs locked.
- ATC Planning Criteria requires:
 - The N-0 Initial Condition must be less than or equal to the Control Point (CP). The CP is 98% of the System Operating Limit (SOL). The SOL is the pre-contingent flow that results in a post-contingent flow that is 95% of the post-Contingency P-V curve nose.

- b. $V_{nose} < V_{min}$.

Short Circuit Analysis

Siemens PTI and several transmission owning companies performed short circuit analysis for the DPP 2019 West study cycle projects.

9.1 J1190 Short Circuit Study

The J1190 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1190. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.1.

9.2 J1218 Short Circuit Study

The J1218 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 19,352.5 (increased by 365.2 A) and SLG fault current is 15,492.6 A (increased by 1,407.2 A) at the Sub 92 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1218 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.2.

9.3 J1221 Short Circuit Study

The J1221 short circuit study was performed by GRE. The study results show that none of the circuit breaker interrupting capabilities at GRE Substations will be exceeded after the addition of J1221.

Study details can be found in Appendix J.3.

9.4 J1229 Short Circuit Study

The J1229 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 19,352.2 (increased by 94.70 A) and SLG fault current is 15,492.6 A (increased by 350.7 A) at the Sub 92 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1229 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.4.

9.5 J1270 Short Circuit Study

The J1270 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 11,981.8 (increased by 339.8 A) and SLG fault current is 10,890.6 A (increased by 856.4 A) at the J1270 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1270 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.5.

9.6 J1298 Short Circuit Study

The J1298 short circuit study was performed by SMMPA. Based on the results of the study, SMMPA and DPC equipment have adequate interrupting capability to accommodate the interconnection project J1298. Equipment not owned by SMMPA or DPC was not evaluated for interrupting capability.

Study details can be found in Appendix J.6.

9.7 J1312 Short Circuit Study

The J1312 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 4,627.9 (increased by 336.5 A) and SLG fault current is 4,726.7 A (increased by 1,753.3 A) at the Grant 115 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1312 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.7.

9.8 J1313 Short Circuit Study

The J1313 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1313. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.8.

9.9 J1314 Short Circuit Study

The J1314 short circuit study was performed by DPC. Based on the results of the study, DPC's equipment has adequate interrupting capability to accommodate the interconnection project J1314. Equipment not owned by DPC was not evaluated for interrupting capability.

Study details can be found in Appendix J.9.

9.10 J1315 Short Circuit Study

The J1315 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 12,105.6 (increased by 1,167.2 A) and SLG fault current is 12,633.2 A (increased by 3,689.3 A) at the J1315 POI 345 kV bus. Based on the Transmission

Owner's short circuit criteria, interconnection of the J1315 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.10.

9.11 J1321 Short Circuit Study

The J1321 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 11,769.8 (increased by 121.40 A) and SLG fault current is 10,533.0 A (increased by 359.8 A) at the FPA-345-BT 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1321 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.11.

9.12 J1325 Short Circuit Study

The J1325 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1325. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.12.

9.13 J1328 Short Circuit Study

The J1328 short circuit study was performed by OTP. Based on the analysis performed, the fault current ratings of the OTP's equipment in the area are not exceeded and there are no upgrades required. Equipment not owned by OTP was not evaluated for any impacts on the fault-current levels.

Study details can be found in Appendix J.13.

9.14 J1349 Short Circuit Study

The J1349 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 17,555.9 (increased by 558.4 A) and SLG fault current is 19,201.0 A (increased by 1,374.1 A) at the Lakefield Generation 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1349 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.14.

9.15 J1359 Short Circuit Study

The J1359 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1359. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.15.

9.16 J1365 Short Circuit Study

The J1365 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1365. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.16.

9.17 J1371 Short Circuit Study

The J1371 short circuit study was performed by OTP. Based on the analysis performed, the fault current ratings of the OTP's equipment in the area are not exceeded and there are no upgrades required. Equipment not owned by OTP was not evaluated for any impacts on the fault-current levels.

Study details can be found in Appendix J.13.

9.18 J1395 Short Circuit Study

The J1395 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 12,968.0 (increased by 453.8 A) and SLG fault current is 12,125.7 A (increased by 1,404.1 A) at the J1395 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1395 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.17.

9.19 J1413 Short Circuit Study

The J1413 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1413. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.18.

9.20 J1416 Short Circuit Study

The J1416 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1416. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.19.

9.21 J1417 Short Circuit Study

The J1417 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1417. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.20.

9.22 J1426 Short Circuit Study

The J1426 short circuit study was performed by GRE. The study results show that none of the circuit breaker interrupting capabilities at GRE Substations will be exceeded after the addition of J1426.

Study details can be found in Appendix J.21.

9.23 J1432 Short Circuit Study

The J1432 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1432. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.22.

9.24 J1438 Short Circuit Study

The J1438 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1438. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.23.

9.25 J1444 Short Circuit Study

The J1444 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1444. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.24.

9.26 J1445 Short Circuit Study

The J1445 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 11,328.8 (increased by 705.2 A) and SLG fault current is 9,439.7 A (increased by 2,684.3 A) at the Mayhew Lake 115 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1445 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.25.

9.27 J1446 Short Circuit Study

The J1446 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 15,133.4 (increased by 984.5 A) and SLG fault current is 14,032.9 A (increased by 3,255.6 A) at the Lake Pulaski 115 kV bus. Based on the Transmission

Owner's short circuit criteria, interconnection of the J1446 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.26.

9.28 J1456 Short Circuit Study

The J1456 short circuit study was performed by OTP. Based on the analysis performed, the fault current ratings of the OTP's equipment in the area are not exceeded and there are no upgrades required. Equipment not owned by OTP was not evaluated for any impacts on the fault-current levels.

Study details can be found in Appendix J.13.

9.29 J1468 Short Circuit Study

The J1468 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 16,849.7 (increased by 655.0 A) and SLG fault current is 14,218.0 A (increased by 2,257.3 A) at the West Hastings 115 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1468 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.27.

9.30 J1471 Short Circuit Study

The J1471 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1471. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.28.

9.31 J1474 Short Circuit Study

The J1474 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 8,830.0 (increased by 1,400.7 A) and SLG fault current is 9,479.3 A (increased by 3,766.5 A) at the J1474 POI 115 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1474 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.29.

9.32 J1477 Short Circuit Study

The J1477 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 17,720.9 (increased by 339.1 A) and SLG fault current is 12,560.6 A (increased by 237.1 A) at the J1131 SUB 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J1477 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix J.30.

9.33 J1478 Short Circuit Study

The J1478 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1478. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.31.

9.34 J1479 Short Circuit Study

The J1479 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1479. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.32.

9.35 J1485 Short Circuit Study

The J1485 short circuit study was performed by OTP. Based on the analysis performed, the fault current ratings of the OTP's equipment in the area are not exceeded and there are no upgrades required. Equipment not owned by OTP was not evaluated for any impacts on the fault-current levels.

Study details can be found in Appendix J.13.

9.36 J1487 Short Circuit Study

The J1487 short circuit study was performed by ITC Midwest. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J1487. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix J.33.

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Deliverability Study

10.1 Project Description

Interconnection requests requesting Network Resource Interconnection Services (NRIS) were considered for deliverability analysis.

10.2 Introduction

Generator interconnection projects must pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS).

If the generator is determined as not fully deliverable, the customer can choose either to change his project to an Energy Resource (ER) project or proceed with the system upgrades that will make the generator fully deliverable.

Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100 % of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

10.3 Study Methodology

MISO Generation Deliverability Study method can be found in Appendix C of the MISO Generation Interconnection Business Practices Manual BPM-015-r22.

10.4 2024 Deliverability Study Result

10.4.1 J1190

J1190 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)
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10.4.2 J1218

J1218 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	165.86	0.0968	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$112,486	\$3,000,000
J1338 POI - J1363 POI 345 kV Rebuild	200.00	0.0968	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$599,924	\$16,000,000

10.4.3 J1221

J1221 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	30.34 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
BVR CRK - Grimes 345 Line Rebuild	54.46	0.0800	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$906,944	\$26,000,000
Raun-Remsen 345 kV Structure Replacements	72.12	0.0603	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$40,040	\$1,500,000
J1181 POI - Hazelton 345 Structure Remediation	72.12	0.1174	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$587,821	\$15,500,000
Raun-S3451 345 kV Rebuild	75.25	0.0852	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$184,583	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	93.21	0.1174	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$227,543	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	102.96	0.1018	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$27,466	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	118.02	0.0914	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395,	\$22,647	\$900,000

J1221 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	30.34 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
			J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487		
Arnold - J1284 POI 345 kV Structure Replacements	145.52	0.0922	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$27,382	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	146.46	0.0617	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$37,076	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	150.00	0.1803	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$4,845,363	\$68,000,000

10.4.4 J1229

J1229 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	41.47	0.0968	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$28,121	\$3,000,000
J1338 POI - J1363 POI 345 kV Rebuild	50.00	0.0968	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$149,981	\$16,000,000

10.4.5 J1270

J1270 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Install 2nd 345/161 kV Webster Transformer	28.35	0.1168	J1270, J1321, J1349, J1417	\$4,098,102	\$10,000,000
Raun-Remsen 345 kV Structure Replacements	40.45	0.4216	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$373,266	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	96.16	0.1741	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$2,631,648	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	96.16	0.0742	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$495,358	\$15,500,000
Raun-S3451 345 kV Rebuild	100.33	0.1445	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$417,405	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	117.38	0.0742	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$191,752	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	152.52	0.0972	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$34,966	\$800,000

J1270 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Morgan Valley-Tiffin 345 Structure Replacements	157.36	0.0902	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$29,799	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	194.03	0.0905	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$35,836	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	195.48	0.1129	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$90,456	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	200.00	0.0513	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$1,838,174	\$68,000,000

10.4.6 J1298

J1298 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	21.26	0.0530	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$46,191	\$3,000,000
Raun-Remsen 345 kV Structure Replacements	30.34	0.0868	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$57,637	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	72.12	0.0857	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$971,563	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	72.12	0.2001	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$1,001,899	\$15,500,000
Raun-S3451 345 kV Rebuild	75.25	0.0768	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$166,384	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	88.03	0.2001	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$387,832	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	114.39	0.1711	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$46,163	\$800,000

J1298 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Morgan Valley-Tiffin 345 Structure Replacements	118.02	0.1493	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$36,993	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	124.40	0.1525	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$45,290	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	142.16	0.0530	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$246,353	\$16,000,000
Austin - Adams 161 kV Line Rebuild	150.00	0.2101	J1298, J1487	\$11,929,753	\$15,750,000

10.4.7 J1312

J1312 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	10.11 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
BVR CRK - Grimes 345 Line Rebuild	21.79	0.0955	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$360,888	\$26,000,000
Split Rock 230/115 kV Transformer Upgrade	24.04	0.1525	J1312, J1315, J1325, J1328, J1349, J1371, J1395	\$326,018	\$5,000,000
J1181 POI - Hazelton 345 Structure Remediation	24.04	0.0758	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$126,510	\$15,500,000
Raun-S3451 345 kV Rebuild	25.08	0.1236	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$89,258	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	31.07	0.0758	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$48,972	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	38.13	0.0803	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$7,222	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	39.34	0.0742	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$6,128	\$900,000

J1312 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	10.11 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Arnold - J1284 POI 345 kV Structure Replacements	48.51	0.0745	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$7,375	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	48.87	0.1025	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$20,531	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	50.00	0.0898	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$804,425	\$68,000,000

10.4.8 J1313

J1313 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	97.02 MW (97.0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Bondurant - Montezuma 345 kV Structure Replacements	100	0.1773	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$71,027	\$1,000,000

10.4.9 J1314

J1314 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	20.23 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
BVR CRK - Grimes 345 Line Rebuild	48.08	0.0666	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$503,354	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	48.08	0.0758	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$253,020	\$15,500,000
Raun-S3451 345 kV Rebuild	50.17	0.0683	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$98,646	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	54.46	0.0758	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$97,943	\$6,000,000
Raun-Remsen 345 kV Structure Replacements	62.14	0.0647	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$28,641	\$1,500,000
Hazelton - Arnold 345 kV Structure Replacements	68.64	0.0998	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$17,951	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	78.68	0.0900	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395,	\$14,867	\$900,000

J1314 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	20.23 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
			J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487		
Arnold - J1284 POI 345 kV Structure Replacements	97.74	0.0903	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$17,878	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	100.00	0.1135	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$2,033,458	\$68,000,000

10.4.10 J1315

J1315 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	85.02 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Raun-Remsen 345 kV Structure Replacements	261.54	0.0562	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$149,271	\$1,500,000
Split Rock 230/115 kV Transformer Upgrade	276.30	0.0652	J1312, J1315, J1325, J1328, J1349, J1371, J1395	\$1,672,631	\$5,000,000
BVR CRK - Grimes 345 Line Rebuild	288.48	0.0936	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$4,244,496	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	288.48	0.1141	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$2,285,190	\$15,500,000
Raun-S3451 345 kV Rebuild	301.02	0.1012	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$876,985	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	352.14	0.1141	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$884,590	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	457.56	0.1041	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$112,346	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	472.08	0.0937	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416,	\$92,867	\$900,000

J1315 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	85.02 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
			J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487		
Arnold - J1284 POI 345 kV Structure Replacements	582.12	0.0937	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$111,309	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	586.44	0.0747	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$179,550	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	600.00	0.1135	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$12,200,748	\$68,000,000

10.4.11 J1321

J1321 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Install 2nd 345/161 kV Webster Transformer	9.21	0.1283	J1270, J1321, J1349, J1417	\$1,463,019	\$10,000,000
Raun-Remsen 345 kV Structure Replacements	13.15	0.4216	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$121,312	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	31.25	0.1778	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$873,462	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	31.25	0.0738	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$160,124	\$15,500,000
Raun-S3451 345 kV Rebuild	32.61	0.1389	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$130,399	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	38.15	0.0738	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$61,983	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	49.57	0.0994	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$11,621	\$800,000

J1321 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Morgan Valley-Tiffin 345 Structure Replacements	51.14	0.0923	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$9,910	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	63.06	0.0926	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$11,917	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	65.00	0.1136	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$29,580	\$1,000,000

10.4.12 J1325

J1325 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	11.34 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Raun-Remsen 345 kV Structure Replacements	34.87	0.0805	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$28,508	\$1,500,000
Split Rock 230/115 kV Transformer Upgrade	38.46	0.1877	J1312, J1315, J1325, J1328, J1349, J1371, J1395	\$642,030	\$5,000,000
J1181 POI - Hazelton 345 Structure Remediation	38.46	0.0814	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$217,370	\$15,500,000
Raun-S3451 345 kV Rebuild	40.13	0.1315	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$151,941	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	44.63	0.0814	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$84,143	\$6,000,000
BVR CRK - Grimes 345 Line Rebuild	46.95	0.1141	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$689,882	\$26,000,000
Hazelton - Arnold 345 kV Structure Replacements	61.01	0.0910	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$13,094	\$800,000

J1325 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	11.34 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Morgan Valley-Tiffin 345 Structure Replacements	62.94	0.0840	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$11,100	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	77.39	0.0843	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$13,352	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	77.39	0.0824	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$1,181,018	\$68,000,000
Magnolia - Elk 161 kV Rebuild	77.61	0.8621	J1325	\$12,700,000	\$12,700,000
Bondurant - Montezuma 345 kV Structure Replacements	80.00	0.1012	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$32,433	\$1,000,000

10.4.13 J1328

J1328 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	20.23 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
BVR CRK - Grimes 345 Line Rebuild	43.59	0.0916	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$692,300	\$26,000,000
Split Rock 230/115 kV Transformer Upgrade	48.08	0.0765	J1312, J1315, J1325, J1328, J1349, J1371, J1395	\$327,087	\$5,000,000
J1181 POI - Hazelton 345 Structure Remediation	48.08	0.0985	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$328,792	\$15,500,000
Raun-S3451 345 kV Rebuild	50.17	0.1065	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$153,819	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	62.14	0.0985	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$127,274	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	76.26	0.0937	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$16,854	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	78.68	0.0851	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$14,057	\$900,000

J1328 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	20.23 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Arnold - J1284 POI 345 kV Structure Replacements	97.02	0.0858	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$16,987	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	97.74	0.0838	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$33,570	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	100.00	0.1362	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$2,440,150	\$68,000,000

10.4.14 J1349

J1349 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Install 2nd 345/161 kV Webster Transformer	42.58	0.0625	J1270, J1321, J1349, J1417	\$3,673,117	\$10,000,000
Raun-Remsen 345 kV Structure Replacements	63.25	0.1803	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$267,380	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	80.50	0.1270	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$3,215,493	\$26,000,000
Lakefield - Lakefield Junction 345 kV Structure Remediation	142.80	0.9976	J1349	\$2,000,000	\$2,000,000
Split Rock 230/115 kV Transformer Upgrade	158.13	0.0593	J1312, J1315, J1325, J1328, J1349, J1371, J1395	\$849,378	\$5,000,000
J1181 POI - Hazelton 345 Structure Remediation	158.13	0.0946	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$1,057,844	\$15,500,000
Raun-S3451 345 kV Rebuild	168.55	0.1264	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$611,578	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	194.24	0.0946	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$409,488	\$6,000,000

J1349 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Hazelton - Arnold 345 kV Structure Replacements	254.10	0.1041	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$62,727	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	262.30	0.0950	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$52,570	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	324.83	0.0957	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$63,474	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	327.28	0.0939	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$126,015	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	335.00	0.0916	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$5,497,682	\$68,000,000

10.4.15 J1359

J1359 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	44.24 MW (46.6%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	72.44	0.0845	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$46,642	\$3,000,000
Morgan Valley-Tiffin 345 Structure Replacements	86.89	0.2731	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$42,857	\$900,000
J1338 POI - J1363 POI 345 kV Rebuild	95.00	0.0845	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$248,755	\$16,000,000

10.4.16 J1365

J1365 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	0.00	0.0682	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$67,126	\$3,000,000
Laurel - Jasper 161 kV Rebuild	114.78	0.2044	J1365	\$12,500,000	\$12,500,000
M-TOWN - Laurel 161 kV Rebuild	129.19	0.2045	J1365	\$12,500,000	\$12,500,000
Morgan Valley-Tiffin 345 Structure Replacements	140.48	0.1790	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$50,089	\$900,000
J1338 POI - J1363 POI 345 kV Rebuild	169.40	0.0682	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$358,005	\$16,000,000

10.4.17 J1371

J1371 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	30.34 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
BVR CRK - Grimes 345 Line Rebuild	65.38	0.0929	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$1,053,188	\$26,000,000
Split Rock 230/115 kV Transformer Upgrade	72.12	0.0835	J1312, J1315, J1325, J1328, J1349, J1371, J1395	\$535,524	\$5,000,000
J1181 POI - Hazelton 345 Structure Remediation	72.12	0.0980	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$490,685	\$15,500,000
Raun-S3451 345 kV Rebuild	75.25	0.1086	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$235,278	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	93.21	0.0980	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$189,943	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	114.39	0.0935	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$25,227	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	118.02	0.0850	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$21,061	\$900,000

J1371 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	30.34 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Arnold - J1284 POI 345 kV Structure Replacements	145.52	0.0856	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$25,422	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	146.61	0.0849	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$51,017	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	150.00	0.1339	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$3,598,414	\$68,000,000

10.4.18 J1395

J1395 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	40.46 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
BVR CRK - Grimes 345 Line Rebuild	72.62	0.0938	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$1,417,855	\$26,000,000
Raun-Remsen 345 kV Structure Replacements	87.18	0.0617	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$54,627	\$1,500,000
Split Rock 230/115 kV Transformer Upgrade	96.16	0.0757	J1312, J1315, J1325, J1328, J1349, J1371, J1395	\$647,332	\$5,000,000
J1181 POI - Hazelton 345 Structure Remediation	96.16	0.1071	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$714,998	\$15,500,000
Raun-S3451 345 kV Rebuild	100.34	0.1052	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$303,883	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	124.28	0.1071	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$276,774	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	152.52	0.0995	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$35,794	\$800,000

J1395 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	40.46 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Morgan Valley-Tiffin 345 Structure Replacements	157.36	0.0899	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$29,700	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	168.64	0.0907	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$35,915	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	195.48	0.0795	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$63,696	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	200.00	0.1507	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$5,399,861	\$68,000,000

10.4.19 J1413

J1413 Deliverable (NRIS) Amount in 2023 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
(i.e. All upgrades must be made for 100% NRIS)					
J1363 POI - J1352 POI 345 kV Rebuild	43.48	0.0587	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$73,328	\$3,000,000
Raun-Remsen 345 kV Structure Replacements	77.42	0.0647	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$61,579	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	107.86	0.0589	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$957,090	\$26,000,000
Raun-S3451 345 kV Rebuild	108.86	0.0584	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$181,347	\$5,000,000
J1181 POI - Hazelton 345 Structure Remediation	117.61	0.4163	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$2,987,654	\$15,500,000
Hazelton - Arnold 345 kV Structure Replacements	133.60	0.2351	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$90,917	\$800,000
J1413 POI - J1181 POI 345 kV Structure Replacements	177.14	0.4163	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$1,156,511	\$6,000,000

J1413 Deliverable (NRIS) Amount in 2023 case: (Conditional on ERI and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1338 POI - J1363 POI 345 kV Rebuild	182.83	0.0587	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$391,082	\$16,000,000
Morgan Valley-Tiffin 345 Structure Replacements	193.68	0.1937	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$68,792	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	212.74	0.2023	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$86,114	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	215.00	0.1135	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$4,371,935	\$68,000,000

10.4.20 J1416

J1416 Deliverable (NRIS) Amount in 2023 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	30.14	0.0532	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$46,056	\$3,000,000
BVR CRK - Grimes 345 Line Rebuild	54.10	0.1009	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$1,136,257	\$26,000,000
Raun-Remsen 345 kV Structure Replacements	71.64	0.0905	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$59,693	\$1,500,000
J1181 POI - Hazelton 345 Structure Remediation	71.64	0.1392	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$692,327	\$15,500,000
Raun-S3451 345 kV Rebuild	74.75	0.0803	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$172,807	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	92.59	0.1392	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$267,997	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	102.28	0.1718	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$46,043	\$800,000

J1416 Deliverable (NRIS) Amount in 2023 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Morgan Valley-Tiffin 345 Structure Replacements	117.23	0.1531	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$37,682	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	123.57	0.1554	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$45,843	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	144.55	0.0532	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$245,634	\$16,000,000
Bondurant - Montezuma 345 kV Structure Replacements	149.00	0.0640	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$38,201	\$1,000,000

10.4.21 J1417

J1417 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Install 2nd 345/161 kV Webster Transformer	10.63	0.0582	J1270, J1321, J1349, J1417	\$765,763	\$10,000,000
Raun-Remsen 345 kV Structure Replacements	15.17	0.1426	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$47,344	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	36.06	0.1400	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$793,576	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	36.06	0.0897	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$224,564	\$15,500,000
Raun-S3451 345 kV Rebuild	37.62	0.1002	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$108,540	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	44.02	0.0897	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$86,928	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	57.20	0.1517	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$20,465	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	59.01	0.1362	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416,	\$16,874	\$900,000

J1417 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
			J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487		
Arnold - J1284 POI 345 kV Structure Replacements	72.76	0.1381	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$20,507	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	75.00	0.0910	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$27,341	\$1,000,000

10.4.22 J1426

J1426 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	14.17 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Raun-Remsen 345 kV Structure Replacements	20.23	0.0726	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$32,138	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	48.08	0.0809	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$611,431	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	48.08	0.1300	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$433,939	\$15,500,000
Raun-S3451 345 kV Rebuild	50.17	0.0831	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$120,022	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	58.69	0.1300	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$167,976	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	76.26	0.1096	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$19,714	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	78.68	0.0979	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416,	\$16,172	\$900,000

J1426 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	14.17 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
			J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487		
Arnold - J1284 POI 345 kV Structure Replacements	97.02	0.0989	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$19,581	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	97.74	0.0549	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$21,993	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	100.00	0.2096	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$3,755,179	\$68,000,000

10.4.23 J1432

J1432 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	15.38 MW (6.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	207.33	0.0761	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$110,539	\$3,000,000
J1338 POI - J1363 POI 345 kV Rebuild	250.00	0.0761	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$589,543	\$16,000,000

10.4.24 J1438

J1438 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	62.14	0.0681	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$39,568	\$3,000,000
Hazelton - Arnold 345 kV Structure Replacements	76.26	0.2578	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$46,370	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	78.68	0.2420	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$39,975	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	82.93	0.2405	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$47,616	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	100.00	0.0681	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$211,027	\$16,000,000

10.4.25 J1444

J1444 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	152.52	0.0830	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$96,450	\$3,000,000
Morgan Valley-Tiffin 345 Structure Replacements	157.36	0.4529	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$149,625	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	165.86	0.5148	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$203,849	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	200.00	0.0830	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$514,398	\$16,000,000

10.4.26 J1445

J1445 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	14.17 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Raun-Remsen 345 kV Structure Replacements	20.23	0.0723	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$32,006	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	48.08	0.0810	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$612,187	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	48.08	0.1300	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$433,939	\$15,500,000
Raun-S3451 345 kV Rebuild	50.17	0.0834	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$120,455	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	58.69	0.1300	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$167,976	\$6,000,000
Morgan Valley-Tiffin 345 Structure Replacements	76.26	0.0988	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$16,320	\$900,000

J1445 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	14.17 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Hazleton - Arnold 345 kV Structure Replacements	78.68	0.1095	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$19,696	\$800,000
Arnold - J1284 POI 345 kV Structure Replacements	97.02	0.0978	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$19,363	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	97.74	0.0554	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$22,193	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	100.00	0.2087	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$3,739,054	\$68,000,000

10.4.27 J1446

J1446 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	21.23 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Raun-Remsen 345 kV Structure Replacements	30.34	0.0557	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$36,986	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	72.12	0.0822	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$931,884	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	72.12	0.1343	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$672,439	\$15,500,000
Raun-S3451 345 kV Rebuild	75.25	0.0836	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$181,116	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	88.03	0.1343	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$260,299	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	114.39	0.1125	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$30,353	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	118.02	0.1003	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395,	\$24,852	\$900,000

J1446 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	21.23 MW (14.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
			J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487		
Arnold - J1284 POI 345 kV Structure Replacements	145.52	0.1015	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$30,144	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	146.61	0.0542	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$32,569	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	150.00	0.2186	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$5,874,633	\$68,000,000

10.4.28 J1456

J1456 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	60.69 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Raun-S3451 345 kV Rebuild	100.34	0.0841	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$364,399	\$5,000,000
BVR CRK - Grimes 345 Line Rebuild	144.24	0.0758	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$1,718,658	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	144.24	0.0982	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$983,373	\$15,500,000
J1413 POI - J1181 POI 345 kV Structure Replacements	186.42	0.0982	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$380,660	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	228.78	0.0897	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$48,403	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	234.30	0.0812	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$40,239	\$900,000
Merricourt - Tatanka North Structure Remediation	236.04	0.0705	J1456	\$100,000	\$100,000

J1456 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	60.69 MW (20.2%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Arnold - J1284 POI 345 kV Structure Replacements	291.06	0.0819	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$48,646	\$1,000,000
Bondurant - Montezuma 345 kV Structure Replacements	293.22	0.0773	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$92,900	\$1,000,000
Prairie Island - North Rochester 345 kV Rebuild	300.00	0.1430	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$7,685,934	\$100,000

10.4.29 J1468

J1468 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	100 MW (100%)
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10.4.30 J1471

J1471 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	146.57	0.1294	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$150,368	\$3,000,000
J1338 POI - J1363 POI 345 kV Rebuild	161.00	0.1294	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$801,965	\$16,000,000
Denmark - Burlington 161 kV Line Rebuild	161.95	0.4385	J1471, J1478, J1479	\$8,224,566	\$13,850,000
Hamilton 138/69 kV Transformer Upgrade	177.58	0.1457	J1471, J1478, J1479	\$2,539,987	\$4,000,000
Macomb NE - Macomb W 138 kV Terminal Equipment Upgrade	200.00	0.2949	J1471, J1478, J1479	\$164,281	\$250,000

10.4.31 J1474

J1474 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	40.45 MW (20.25%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
BVR CRK - Grimes 345 Line Rebuild	72.61	0.0528	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$798,110	\$26,000,000
Raun-Remsen 345 kV Structure Replacements	96.16	0.0598	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$52,944	\$1,500,000
J1181 POI - Hazelton 345 Structure Remediation	96.16	0.1048	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$699,643	\$15,500,000
Raun-S3451 345 kV Rebuild	100.33	0.0622	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$179,672	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	124.28	0.1048	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$270,830	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	137.28	0.0912	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$32,808	\$800,000
Morgan Valley-Tiffin 345 Structure Replacements	157.36	0.0829	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395,	\$27,388	\$900,000

J1474 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	40.45 MW (20.25%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
			J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487		
Arnold - J1284 POI 345 kV Structure Replacements	190.99	0.0827	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$32,747	\$1,000,000
Pine Lake 161/115 kV Transformer Upgrade	195.48	0.4984	J1474	\$4,000,000	\$4,000,000
Prairie Island - North Rochester 345 kV Rebuild	200.00	0.0763	J1221, J1270, J1312, J1314, J1315, J1325, J1328, J1349, J1371, J1395, J1413, J1426, J1445, J1446, J1456, J1474	\$2,733,971	\$68,000,000

10.4.32 J1477

J1477 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	41.47	0.0871	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$25,303	\$3,000,000
J1338 POI - J1363 POI 345 kV Rebuild	50.00	0.0871	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$134,952	\$16,000,000

10.4.33 J1478

J1478 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	20.73	0.1254	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$18,215	\$3,000,000
J1338 POI - J1363 POI 345 kV Rebuild	18.32	0.1254	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$97,147	\$16,000,000
Denmark - Burlington 161 kV Line Rebuild	20.24	0.3170	J1471, J1478, J1479	\$743,212	\$13,850,000
Hamilton 138/69 kV Transformer Upgrade	22.20	0.1272	J1471, J1478, J1479	\$277,185	\$4,000,000
Macomb NE - Macomb W 138 kV Terminal Equipment Upgrade	25.00	0.2050	J1471, J1478, J1479	\$14,275	\$250,000

10.4.34 J1479

J1479 Deliverable (NRIS) Amount in 2024 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	82.93	0.1283	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$74,545	\$3,000,000
J1338 POI - J1363 POI 345 kV Rebuild	73.28	0.1283	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$397,574	\$16,000,000
Denmark - Burlington 161 kV Line Rebuild	80.97	0.5206	J1471, J1478, J1479	\$4,882,222	\$13,850,000
Hamilton 138/69 kV Transformer Upgrade	88.79	0.1357	J1471, J1478, J1479	\$1,182,829	\$4,000,000
Macomb NE - Macomb W 138 kV Terminal Equipment Upgrade	100.00	0.2565	J1471, J1478, J1479	\$71,444	\$250,000

10.4.35 J1487

J1487 Deliverable (NRIS) Amount in 2023 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
J1363 POI - J1352 POI 345 kV Rebuild	0.00	0.0507	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$35,349	\$3,000,000
Raun-Remsen 345 kV Structure Replacements	24.27	0.1066	J1221, J1270, J1298, J1314, J1315, J1321, J1325, J1349, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1474, J1487	\$56,627	\$1,500,000
BVR CRK - Grimes 345 Line Rebuild	57.69	0.0970	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$879,735	\$26,000,000
J1181 POI - Hazelton 345 Structure Remediation	57.69	0.1629	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$652,511	\$15,500,000
Raun-S3451 345 kV Rebuild	60.20	0.0874	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417 , J1426, J1445, J1446, J1456, J1474, J1487	\$151,479	\$5,000,000
J1413 POI - J1181 POI 345 kV Structure Replacements	70.43	0.1629	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1445, J1446, J1456, J1474, J1487	\$252,585	\$6,000,000
Hazelton - Arnold 345 kV Structure Replacements	91.51	0.1566	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1445, J1446, J1456, J1474, J1487	\$33,801	\$800,000

J1487 Deliverable (NRIS) Amount in 2023 case: (Conditional on ERIS and IC upgrades and case assumptions)	0 MW (0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	Upgrade Costs Allocated to Project	Total Cost of Upgrade
Morgan Valley-Tiffin 345 Structure Replacements	94.42	0.1384	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1359, J1365, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$27,434	\$900,000
Arnold - J1284 POI 345 kV Structure Replacements	99.52	0.1408	J1221, J1270, J1298, J1312, J1314, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1413, J1416, J1417, J1426, J1438, J1444, J1445, J1446, J1456, J1474, J1487	\$33,452	\$1,000,000
J1338 POI - J1363 POI 345 kV Rebuild	113.73	0.0507	J1218, J1229, J1298, J1359, J1365, J1413, J1416, J1432, J1438, J1444, J1471, J1477, J1478, J1479, J1487	\$188,530	\$16,000,000
Austin - Adams 161 kV Rebuild	116.42	0.0841	J1298, J1487	\$3,820,247	\$15,750,000
Bondurant - Montezuma 345 kV Structure Replacements	120.00	0.0621	J1221, J1270, J1312, J1313, J1315, J1321, J1325, J1328, J1349, J1371, J1395, J1416, J1417, J1426, J1445, J1446, J1456, J1487	\$29,853	\$1,000,000

Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection project was performed for this System Impact Study.

No SNUs were identified in this study.

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Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

12.1 Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

12.2 ERIS Network Upgrades Proposed for DPP West Area Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the MISO ERIS analysis, LPC analyses, and the affected system analysis. The ERIS network upgrades include voltage network upgrades and thermal network upgrades identified in the MISO steady-state analysis, network upgrades identified in the Local Planning Criteria analysis and affected system analysis, voltage network upgrades identified in the MWEX voltage stability analysis, stability network upgrades identified in the MISO transient stability analysis, and short circuit network upgrades identified in the MISO short circuit analysis. The total costs of ERIS network upgrades for the 2024 scenario are summarized in Table 12-1.

Table 12-1: Summary of ERIS Network Upgrades

Category of Network Upgrades	Cost (\$)
Base Case Network Upgrades	\$210,829,263
Network Upgrades Identified in MWEX Voltage Stability analysis	\$0
Additional Thermal Network Upgrades Identified in MISO Steady-State Analysis	\$185,843,690
Additional Reactive Power Network Upgrades for Voltage Constraints	\$4,000,000
Network Upgrades Identified in Stability Analysis	\$0
Network Upgrades Identified in Short Circuit Analysis	\$0
Network Upgrades Identified in GRE LPC Analysis	\$3,000,000
Network Upgrades Identified in OTP LPC Analysis	\$35,000,000
Network Upgrades Identified in DPC LPC Analysis	\$0
Network Upgrades Identified in CIPCO affected system	\$0

Category of Network Upgrades	Cost (\$)
Network Upgrades Identified in MPC affected system	\$60,000,000
Network Upgrades Identified in PJM affected system	\$56,780,000
Network Upgrades Identified in AECI affected system	\$1,911,723
Network Upgrades Identified in SPP affected system	\$398,305,056
Shared Network Upgrades	\$0
Total	\$955,669,732

ERIS network upgrades are listed below.

Table 12-2: Network Upgrades Required for Mitigating Voltage Collapse and Severe Thermal Overloads

NUs	Needs	Miles	Cost (\$)
Hazel Creek-Scott County 345 kV	Base Case NU	115	\$210,829,263
2×75 MVar switched capacitor at Cedar Mountain 345 kV (615647)	DPP 2018 Apr. Ph3 NU		\$3,000,000 ¹

Note 1: The cost is currently assigned to projects in DPP 2018 Apr. Phase 3 cycle. The NU cost may be assigned to DPP 2019 projects if projects in DPP 2018 Apr. are withdrawn.

Table 12-3: Network Upgrades Required for MWEX Voltage Stability

NUs	Miles	Cost (\$)
No MWEX NUs		\$0

Table 12-4: Additional Thermal Network Upgrades in MISO Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
Montgomery-J1145 POI 345 kV	Ameren	Rebuild 28 mi of 345 kV line for 3000A SE. Exclusively tied to H104/H105 through the TCA study process. \$42M	\$0
Amoco-Niota 69 kV	Ameren	Replace (1) 72 MVA switch with 1200 A switch.	\$55,000
Niota 161-69 kV xfmr	Ameren	Replace with new 112 MVA 161/69 kV Transformer.	\$4,000,000
Wilmarth-Sheas Lake 345 kV	XEL	Rebuild 24.7 mi of 345 kV to bundled 954 ACSS	\$61,750,000
Blue Lake-Scott Co 345 kV	XEL	Rebuild 8.1 mi of 345 kV to bundled 954 ACSS	\$20,250,000
Blue Lake-Scott Co 345 kV (P2-3)	XEL	Add an extra breaker at Chub Lake 345 kV to eliminate the P2-3 contingency	\$820,000
Lyon County-Hazel Creek 345 kV	XEL	Upgrade Substation Equipment	\$5,000,000

Constraint	Owner	Mitigation	Cost (\$)
Helena-Sheas Lake 345 kV	XEL	Rebuild 7.5 mi of 345 kV to bundled 954 ACSS	\$18,750,000
Pulaski-Dickinson 115 kV	XEL GRE	XEL: Upgrade Substation Equipment. \$1M GRE: GRE owns a switch at Dickinson rated to 239 MVA Summer/Winter. \$0	\$1,000,000
Scott County 345-115-34.5 kV xfmr #9	XEL	Add 3rd Scott County 345/115 672 MVA transformer	\$15,000,000
Crystal Foods Tap-Arlington 69 kV	XEL	Rebuild 6.9 miles to 477 ACSR	\$4,830,000
Gibbon-Cairo 69 kV	XEL	Rebuild 8.1 miles to 477 ACSR	\$7,000,000
Murphy Crk-J1298 POI	SMMPA	Rebuild 16 miles of Murphy Creek-Hayward 161 kV line. New rating 406 MVA	\$12,000,000
Dickinson-Dickinson S 115 kV	GRE	Re-conductor to 795 DRAKE ACSS and replace wave trap	\$152,690
Canby-Granites Falls 115 kV	OTP WAPA	OTP: Structure replacements. \$2M WAPA: NU is not required unless it is identified as constraint in affected system study.	\$2,000,000
Wahpeton-Fergus Falls 230 kV	OTP MRES	OTP: Structure replacement. \$750,000. 2017 Aug. Ph3 OTP LPC upgrade MRES: Terminal rating at Fergus Falls is sufficient. \$0	\$0 ¹
Walters-Alden Junction 69 kV	ITCM	Rebuild 6 miles of 69 kV. New rating of 157 MVA	\$6,000,000
Alden Junction-Albert Lea Westside 69 kV	ITCM	Rebuild 7 miles of 69 kV. New rating of 157 MVA	\$7,000,000
PCI J2-Oakridge 69 kV	ITCM CIPCO	CIPCO: Not a constraint. \$0 ITCM: Relay thermal, replace sub taps (.08 miles), new rating 143 MVA. \$450K	\$450,000
Hayward-J1298 161 kV	ITCM SMMPA	ITCM: Not a constraint. ITCM Facilities at 308 MVA. \$0 SMMPA: Rebuild 16 miles of Murphy Creek-Hayward 161 kV line. New rating 406 MVA. \$12M	Incl.
Grand Mound-Maquoketa 161 kV	CIPCO ITCM	CIPCO: Not a constraint, rating can be increased to 237 MVA at no cost. ITCM: Not an ITCM constraint. ITCM facilities rated at 366 MVA. \$0	\$0
Denmark-Burlington 161 kV #1	ITCM	Rebuild 13 miles of 161 kV and bus conductor at Denmark and Burlington. New line rating would be 371 MVA	\$16,500,000
Denmark-Burlington 161 kV #2	ITCM	Line is sag limited. Replace 2 spans to increase rating to 257 MVA	\$650,000
Apple River 161-69 kV xfmr #2	DPC	Upgrade Apple River Transformer #2	\$2,636,000
Wabaco-Rochester 161 kV	DPC	A third party will fund the Rochester-Wabaco 161kV rebuild. \$11M. MTEP Appendix A project	\$0

Note 1: Costs of Network Upgrades required in DPP 2017 Aug West Ph3 OTP LPC are currently assigned to projects in DPP 2017 Aug. Phase 3 cycle. The NUs cost may be assigned to DPP 2019 projects if projects in DPP 2017 Aug. are withdrawn.

Table 12-5: Additional Reactive Power NUs Required for Voltage Constraints

Network Upgrades	Owner	Cost (\$)
Add 2×40 Mvar switched capacitor at Audubon 230 kV (620336)	OTP	\$4,000,000

Table 12-6: Network Upgrades Required for Transient Stability

Mitigation	Owner	Cost (\$)
J1432 is required to meet HVRT requirement	J1432	TBD
J1446 is required to meet LVRT requirement	J1446	TBD
J1456 is required to resolve transient high voltage violation at Harvey 230 kV	J1456	TBD

Table 12-7: Network Upgrades in Short Circuit Analysis

Constraint	Owner	Mitigation	Cost (\$)
No additional NUs			\$0

Table 12-8: J1221 GRE Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Wing River 230-115-13.8 kV xfmr	GRE	Replacement of the Wing River xfmr	\$3,000,000

Table 12-9: J1426 GRE Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No J1426 GRE LPC NUs			\$0

Table 12-10: OTP Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Voltage collapse, thermal and voltage constraints in OTP	OTP	Build a 2nd Astoria-Brookings County 345 kV line	\$35,000,000
High voltage at J1456 POI 230 kV	OTP	1×15 Mvar switched reactor at J1456 POI 230 kV	\$2,500,000 ¹

Note 1: The cost of \$2.5 M of the reactor is accounted for in the interconnection substation

Table 12-11: DPC Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No DPC LPC NUs			\$0

Table 12-12: CIPCO Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
No CIPCO AFS constraints			\$0

Table 12-13: MPC Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Center-Roughrider 230	MPC	Resag to 100°	\$500,000
Drayton 230-115 kV xfmr	MPC	Replace transformer.	\$2,000,000
Prairie-Lake Ardoch 230	OTP MPC	MPC: Resag to 100° OTP: Mitigation not required for ASA	\$500,000
Transient Instability	MPC	+50/-0 MVar SVC at Winger 230	\$15,000,000
Transient Instability	MPC	2x30 MVar MSC at Winger 230	\$2,000,000
Transient Instability	MPC	+50/-0 MVar SVC at Prairie 230	\$15,000,000
Transient Instability	MPC	5x30 MVar MSC at Prairie 230	\$5,000,000
Transient Instability	MPC	+100/-0 MVar SVC at Frontier 230	\$20,000,000

Table 12-14: PJM Affected System Network Upgrades

Constraint	Owner	Mitigation	Total Cost (\$)	J1432 Cost (\$)
GARDEN PR; R - SILVER LK; R 345 kV	PJM	re-conductor the line, mitigate sag limits, upgrade relay package, replace station conductor at both line terminals, replace 2-345 kV breakers and CT's at TSS 138 Silver Lake	\$32,400,000	\$1,580,000
CHERRY VA; B - GARDEN PR; R 345 kV	PJM	install a new 345kV line from the AB1-089 interconnection substation to TSS 144 Wayne station	\$165,000,000	\$41,540,000
NELSON ; B - ELECT JCT; B 345 kV	PJM	perform sag mitigation on a portion of the line section along with re-conductoring on a different section, upgrade station conductor at both line terminals, replace both line motor operated disconnect switches and line	\$10,500,000	\$1,740,000

Constraint	Owner	Mitigation	Total Cost (\$)	J1432 Cost (\$)
		current transformers, upgrade line relay schemes at both terminals		
BYRON ; B - CHERRY VA; B 345 kV	PJM	mitigate sag on the line and replace a 345kV circuit breaker at Cherry Valley	\$12,200,000	\$970,000
BYRON ; B - CHERRY VA; B 345 kV	PJM	reconductor the line, station conductor, relay scheme upgrade and replace 3-345kV circuit breakers at Cherry Valley	\$23,400,000	\$10,950,000

Table 12-15: AECI Affected System Network Upgrades

Network Upgrade	Project	Total Cost
AECI AFS NU	J1218	\$195,760
AECI AFS NU	J1229	\$48,940
AECI AFS NU	J1315	\$543,195
AECI AFS NU	J1456	\$271,588
AECI AFS NU	J1471	\$532,261
AECI AFS NU	J1478	\$61,450
AECI AFS NU	J1479	\$258,529

Table 12-16: SPP Constraints Network Upgrades

Interconnection Request	ERIS MW	NRIS MW	ERIS	NRIS	Total	ERIS Total	NRIS Total	Total
J1190	200	200	\$0	\$0	\$0	\$275,600,356	\$122,704,700	\$398,305,056
J1218	200	200	\$0	\$0	\$0			
J1221	150	150	\$0	\$1,084,499	\$1,084,499			
J1229	50	50	\$0	\$0	\$0			
J1270	200	200	\$2,205,552	\$3,571,286	\$5,776,838			
J1298	150	150	\$7,876,897	\$12,313,729	\$20,190,626			
J1312	50	50	\$81,710,893	\$27,660,352	\$109,371,245			
J1313	100	100	\$0	\$922,138	\$922,138			
J1314	100	100	\$3,193,893	\$581,007	\$3,774,900			
J1315	600	600	\$46,029,408	\$10,825,040	\$56,854,448			
J1321	65	65	\$702,286	\$1,152,995	\$1,855,281			
J1325	80	80	\$8,284,359	\$1,952,466	\$10,236,825			

Interconnection Request	ERIS MW	NRIS MW	ERIS	NRIS	Total	ERIS Total	NRIS Total	Total
J1328	100	100	\$9,446,750	\$1,873,915	\$11,320,665			
J1349	335	335	\$13,472,065	\$6,304,841	\$19,776,906			
J1359	95	95	\$0	\$380,782	\$380,782			
J1365	169.4	169.4	\$420,151	\$1,009,394	\$1,429,545			
J1371	150	150	\$15,505,450	\$3,032,778	\$18,538,228			
J1395	200	200	\$17,757,399	\$3,816,570	\$21,573,969			
J1413	235	215	\$1,028,155	\$1,026,944	\$2,055,099			
J1416	149	149	\$8,306,024	\$18,883,512	\$27,189,536			
J1417	75	75	\$3,002,249	\$7,345,648	\$10,347,897			
J1426	100	100	\$4,202,720	\$684,361	\$4,887,081			
J1432	250	250	\$0	\$0	\$0			
J1438	100	100	\$441,237	\$478,230	\$919,467			
J1444	200	200	\$0	\$826,139	\$826,139			
J1445	100	100	\$4,239,986	\$684,738	\$4,924,724			
J1446	150	150	\$6,092,136	\$1,025,837	\$7,117,973			
J1456	300	300	\$0	\$2,476,406	\$2,476,406			
J1468	100	100	\$3,767,009	\$659,226	\$4,426,235			
J1471	200	200	\$0	\$0	\$0			
J1474	200	200	\$5,722,187	\$1,083,107	\$6,805,294			
J1477	50	50	\$0	\$0	\$0			
J1478	25	25	\$0	\$0	\$0			
J1479	100	100	\$0	\$0	\$0			
J1485	250	0	\$25,842,417	\$0	\$25,842,417			
J1487	120	120	\$6,351,133	\$11,048,760	\$17,399,893			

Table 12-17: Shared Network Upgrades

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total Network Upgrade Cost (\$)	Cost Responsibility
No SNUs					\$0

12.3 Cost Allocation Methodology

12.3.1 Thermal Network Upgrade Cost Allocation

The costs of thermal Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case. For thermal constraints identified in the shoulder peak scenario, the MW impact is calculated using the shoulder peak post-DPP case. The MW impact on constraints identified in the summer peak scenario is calculated using the summer peak post-DPP case. With all Group Study generation projects dispatched in the Study Case, all thermal constraints will be identified and a distribution factor from each project on each constraint will be obtained.

Thermal NU cost will be allocated based on the pro rata share of the MW impact on all constraints from each project, where $MW\ impact = DF * Gen\ Output$ of the project in the model where the constraint occurs. If the Network Upgrade alleviates multiple constrained facilities the cost is allocated based on the sum of the highest MW contribution on all of the constrained elements for the DPP project under contingency. The methodology to determine the cost allocation of thermal NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max}(\text{Proj.A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj.i MW contribution on constraint})} \right)$$

12.3.2 Voltage Network Upgrade Cost Allocation

Voltage NU cost allocation will be determined by the pro rata share of the voltage impact each project has on the most constrained bus under the most constraining contingency. The voltage impact of each project will be calculated by locking all voltage regulating equipment in the model and then backing out each project one at a time to identify each project's impact to the constraint. In severe instances of voltage collapse where projects cannot be backed out one at a time, they will be added one at a time to determine their impact to the constraint.

12.3.3 Transient Stability Network Upgrade Cost Allocation

Transient stability driven Network Upgrades will be cost allocated based on the pro rata share of the total MW request of all the projects causing instability. The project(s) causing instability will be determined by backing out each project one at a time to identify each project's impact to the constraint.

12.4 Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

For each thermal constraint, the maximum MW contribution (increasing flow) from each DPP project is calculated. MW contribution from one DPP project is set as zero if the project doesn't violate DPP reliability criteria for a constrained element.

For voltage network upgrades, generators which degrade voltage at the most constrained bus under the most constraining contingency will be responsible for mitigating these constraints.

Transient stability Network Upgrades are allocated based on projects causing instability. If multiple projects are causing instability, cost allocation will be based on pro rata share of total MW of all projects causing instability.

Additional NRIS Network Upgrades are allocated to the impacting NRIS projects. ERIS Network Upgrades will be allocated to the impacting projects only based on the ERIS results.

The calculated DF results, voltage impact, and MW contribution on each constraint are in Appendix K.1 for the 2024 scenario.

Finally, the cost allocation for each NU is calculated based on the contribution of each generating facility, as detailed in Appendix K.2 for the 2024 scenario.

Assuming all generating facilities in the DPP 2019 West Area group advance, a summary of the costs for total NUs (NUs for ERIS, NRIS, and Interconnection Facilities) allocated to each generating facility is listed in Table 12-18.

Table 12-18: Summary of Total NU Costs Allocated to Each Generation Project

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J1190	200	\$144,940	\$725	0.01%
J1218	200	\$17,408,170	\$87,041	1.53%
J1221	150	\$17,740,206	\$118,268	1.56%
J1229	50	\$227,043	\$4,541	0.02%
J1270	200	\$31,171,056	\$155,855	2.74%
J1298	150	\$39,659,435	\$264,396	3.49%
J1312	50	\$111,168,572	\$2,223,371	9.77%
J1313	100	\$1,527,003	\$15,270	0.13%
J1314	100	\$13,045,256	\$130,453	1.15%
J1315	600	\$142,447,821	\$237,413	12.53%
J1321	65	\$4,870,597	\$74,932	0.43%
J1325	80	\$26,488,486	\$331,106	2.33%
J1328	100	\$18,636,027	\$186,360	1.64%
J1349	335	\$103,442,731	\$308,784	9.10%
J1359	95	\$1,873,623	\$19,722	0.16%
J1365	169.4	\$28,509,049	\$168,294	2.51%
J1371	150	\$37,881,177	\$252,541	3.33%
J1395	200	\$43,315,403	\$216,577	3.81%
J1413	235	\$24,103,721	\$102,569	2.12%
J1416	149	\$37,307,218	\$250,384	3.28%

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J1417	75	\$12,530,791	\$167,077	1.10%
J1426	100	\$11,447,263	\$114,473	1.01%
J1432	250	\$71,002,795	\$284,011	6.24%
J1438	100	\$8,410,264	\$84,103	0.74%
J1444	200	\$9,793,912	\$48,970	0.86%
J1445	100	\$12,520,915	\$125,209	1.10%
J1446	150	\$31,078,587	\$207,191	2.73%
J1456	300	\$90,383,299	\$301,278	7.95%
J1468	100	\$28,120,134	\$281,201	2.47%
J1471	200	\$27,215,254	\$136,076	2.39%
J1474	200	\$23,890,408	\$119,452	2.10%
J1477	50	\$160,255	\$3,205	0.01%
J1478	25	\$2,095,764	\$83,831	0.18%
J1479	100	\$12,972,756	\$129,728	1.14%
J1485	250	\$45,566,606	\$182,266	4.01%
J1487	120	\$49,140,840	\$409,507	4.32%
Total/Average	5698.4	\$1,137,297,377	\$217,394	100.00%

Appendix A

Model Development

A.1 DPP 2019 Generation Projects

Table A-1: DPP 2019 West Area Projects

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1190	IA	Clinton	ITCM	Calamus East 161 kV	200	200	Solar	NRIS
J1218	IA	Muscatine	MEC	Sub 92 345 kV	200	200	Solar	NRIS
J1221	MN	Wadena	GRE	Wing River-Riverton 230 kV	150	150	Solar	NRIS
J1229	IA	Muscatine	MEC	Sub 92 345 kV	50	50	Battery	NRIS
J1270	IA	Palo Alto	MEC	Kossuth - O'Brien 345 kV	200	200	Wind	NRIS
J1298	MN	Freeborn	SMMPA	Hayward - Murphy Creek 161 kV	150	150	Solar	NRIS
J1312	SD	McCook	XCEL	Grant 115 kV	50	50	Solar	NRIS
J1313	IA	Greene	ITCM	Karma 161 kV	100	100	Solar	NRIS
J1314	WI	Polk	DPC	Apple River 161 kV	100	100	Solar	NRIS
J1315	MN	Cottonwood, Murray, Redwood	XCEL	Lyon County - Cedar Mountain 345 kV	600	600	Wind	NRIS
J1321	IA	Palo Alto	MEC	O'Brien - Kossuth 345 kV	65	65	Battery	NRIS
J1325	MN	Rock	ITCM	Magnolia 161 kV	80	80	Solar	NRIS
J1328	SD	Grant	OTP	Big Stone South 345 kV	100	100	Solar	NRIS
J1349	MN	Cottonwood	XCEL	Crandall 345 kV	335	335	Hybrid	NRIS
J1359	IA	Linn	ITCM	Oakridge 69 kV	95	95	Gas	NRIS
J1365	IA	Tama	ITCM	Traer 161 kV	169.4	169.4	Wind	NRIS
J1371	MN	Yellow Medicine	OTP	Astoria 345 kV	150	150	Wind	NRIS
J1395	MN	Lyon	XCEL	Nest Lake - Lyon County 345 kV	200	200	Solar	NRIS
J1413	IA	Howard, Mitchell	ITCM	Hazleton - Mitchell County 345 kV	235	215	Solar	NRIS

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1416	IA	Worth	ITCM	Worth County - Glenworth 161 kV	149	149	Solar	NRIS
J1417	IA	Worth	ITCM	Ledyard - Colby 345 kV	75	75	Battery	NRIS
J1426	MN	Benton	GRE	Benton County 115 kV	100	100	Solar	NRIS
J1432	IA	Clinton	ITCM	Salem - Rock Creek 345 kV	250	250	Wind	NRIS
J1438	IA	Fayette	ITCM	Windsor - Hazelton 161 kV	100	100	Solar	NRIS
J1444	IA	Linn	ITCM	Duane Arnold 345 kV	200	200	Solar	NRIS
J1445	MN	Benton	XCEL	Mayhew Lake 115 kV	100	100	Solar	NRIS
J1446	MN	Wright	XCEL	Lake Pulaski 115 kV	150	150	Battery	NRIS
J1456	ND	Sheridan	OTP	Harvey - Underwood 230 kV	300	300	Wind	NRIS
J1468	MN	Dakota	XCEL	West Hastings 115 kV	100	100	Battery	NRIS
J1471	IA	Lee	ITCM	Wever - Burlington Generator 161 kV	200	200	Solar	NRIS
J1474	WI	Chippewa	XCEL	Pine Lake - Eagle Point 115 kV	200	200	Solar	NRIS
J1477	IA	Scott	MEC	Sub 56 161 kV (J1131)	50	50	Battery	NRIS
J1478	IA	Des Moines	ITCM	Huntwoods 69 kV (J1135)	25	25	Battery	NRIS
J1479	IA	Des Moines	ITCM	Burlington 161 kV	100	100	Battery	NRIS
J1485	SD	Deuel	OTP	Astoria 345 kV (J493/J510)	250	0	Wind	ERIS
J1487	MN	Freeborn	ITCM	Freeborn 161 kV	120	120	Wind	NRIS

Table A-2: Generator, Collector System and Substation Transformer Modeling for DPP West Area Projects

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1190	80 SMA 2.5 MW Inverter MVA = 2.75 MVA each	PF: 0.909 Qmin = - 91.7 MVAR Qmax = +91.7 MVAR 220 MVA	R = 0.0022, X = 0.0024, B = NA	0.55 - 34.5 kV YG - D MVA = 2.75 MVA Z% = 5.75, X/R = 8 #Units = 80 DETC 2.5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 132 MVA Z% = 8.5, X/R = 32 DETC 2.5% w 5 taps	R = 0.0001, X = 0.0004, B = NA
J1218	54 TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	2× PF: 0.90 Qmin = - 49.42 MVAR Qmax = +49.42 MVAR 113.4 MVA	Collector 1: R = 0.00049, X = 0.00057, B = 0.00281 1×3 MVAR cap bank Collector 2: R = 0.00049, X = 0.00057, B = 0.00281 1×3 MVAR cap bank	2× 0.63 - 34.5 kV YG - D MVA = 113.4 MVA Z% = 5.75, X/R = 10.0 #Units = 54 DETC 2.5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	NA
J1221	40TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	PF: 0.90 Qmin = - 73.22 MVAR Qmax = +73.22 MVAR 168.0 MVA	R = 0.00065, X = 0.00080, B = 0.00416 1×4 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10.0 #Units = 40 DETC 2.5% w 5 taps	34.5 - 230 - 13.8 kV YG - YG - D MVA = 102 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	R = 0.000112, X = 0.000468, B = 0.000409
J1229	14 TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	PF: 0.90 Qmin = - 25.63 MVAR Qmax = +25.63 MVAR 58.8 MVA	R = 0.00186, X = 0.00218, B = 0.00150 1×2 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10.0	34.5 - 345 - 13.8 kV YG - YG - D MVA = 36 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	NA

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
				#Units = 14 DETC 2.5% w 5 taps		
J1270	49 Vestas V136 MVA = 5.1 MVA each	PF: 0.95 Qmin = - 65.73 MVAR Qmax = +65.73 MVAR 211 MVA	R = 0.00239, X = 0.00382, B = 0.04383 2×13 MVAR cap bank	0.72 - 34.5 kV YG - D MVA = 5.10 MVA Z% = 9.90, X/R = 12.375 #Units = 49 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 144 MVA Z% = 9.5, X/R = 52.80 DETC 10% w 33 taps	R = 0.00020, X = 0.001208, B = 0.010995
J1298	51 PE HEM 3 MW Inverter MVA = 3.3 MVA each	PF: 0.909 Qmin = - 70.15 MVAR Qmax = +70.15 MVAR 168.3 MVA	R = 0.003403, X = 0.003126, B = 0.005843	0.60 - 34.5 kV YG - D MVA = 3.3 MVA Z% = 8, X/R = 10.5 #Units = 51 DETC 2.5% w 5 taps	34.5 - 161 YG - YG - D MVA = 100 MVA Z% = 9.0, X/R = 50 DETC 2.5% w 5 taps	R = 0.00010, X = 0.00020, B = 0.00010
J1312	16 PE HEMK 3.432 MW Inverter MVA = 3.432 MVA each	PF: 0.95 Qmin = - 18.048 MVAR Qmax = +18.048 MVAR 54.91 MVA	R = 0.006143, X = 0.008598, B = 0.009158 2×7 MVAR cap bank	0.66 - 34.5 kV YG - D MVA = 4.0 MVA Z% = 9, X/R = 25 #Units = 16 DETC 2.5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 47 MVA Z% = 7.5, X/R = 25 DETC 10% w 33 taps	R = 0.000210, X = 0.000603, B = 0.00029
J1313	30 PE HEMK 3.59 MW Inverter MVA = 3.59 MVA each	PF: 0.95 Qmin = - 33.629 MVAR Qmax = +33.629 MVAR 107.7 MVA	R = 0.003435, X = 0.004876, B = 0.018044 2×14 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 4.0 MVA Z% = 5.75, X/R = 8 #Units = 30 DETC 2.5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 100 MVA Z% = 7.0, X/R = 25 DETC 10% w 33 taps	R = 0.0001070, X = 0.0003080, B = 0.0000700
J1314	30 PE HEMK 3.59 MW Inverter MVA = 3.59 MVA each	PF: 0.95 Qmin = - 33.629 MVAR Qmax = +33.629 MVAR 107.7 MVA	R = 0.003435, X = 0.004876, B = 0.018044 1×28 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 4.0 MVA Z% = 9, X/R = 25	34.5 - 161 - 13.8 kV YG - YG - D MVA = 100 MVA	R = 0.0001070, X = 0.0003080, B = 0.0000700

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
				#Units = 30 DETC 2.5% w 5 taps	Z% = 7.0, X/R = 25 DETC 10% w 33 taps	
J1315	174 Vesta V126 3.45 MW MVA = 3.45 MVA each	6× PF: +0.90/ -0.92 Qmin = - 42.599 MVAR Qmax = +48.432 MVAR 111.1 MVA	6× R = 0.0000613, X = 0.000883, B = 0.11121 1×25 MVAR cap bank	6× 0.69 - 34.5 kV YG - D MVA = 116 MVA Z% = 5.7, X/R = 8 #Units = 174 DETC 2.5% w 5 taps	6× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 75 MVA Z% = 9.0, X/R = 60 DETC 10% w 33 taps	R = 0.001461, X = 0.00419, B = 0.163137
J1321	18 TMEIC 3.78 MW Inverter MVA = 4.2 MVA each	PF: 0.90 Qmin = - 32.95 MVAR Qmax = +32.95 MVAR 75.6 MVA	R = 0.00152, X = 0.00170, B = 0.00193 1×2 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.2 MVA Z% = 5.75, X/R = 10 #Units = 18 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 45 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	NA
J1325	24 PE HEMK 3.59 MW Inverter MVA = 3.59 MVA each	2× PF: 0.95 Qmin = - 13.452 MVAR Qmax = +13.452 MVAR 43.08 MVA	2× R = 0.003435, X = 0.004876, B = 0.022654 1×26 MVAR cap bank	2× 0.69 - 34.5 kV YG - D MVA = 120 MVA Z% = 5.75, X/R = 8.0 #Units = 60 DETC 2.5% w 5 taps	2× 34.5 - 161 - 13.8 kV YG - YG - D MVA = 100 MVA Z% = 9.0, X/R = 25 DETC 10% w 33 taps	R = 0.000107, X = 0.000308, B = 0.00012
J1328	28 SMA 3.52 MW Inverter MVA = 4.4 MVA each	PF: 0.8 Qmin = - 73.92 MVAR Qmax = +73.92 MVAR 123.2 MVA	R = 0.0035, X = 0.0031, B = 0.0102	0.66 - 34.5 kV YG - D MVA = 4.4 MVA Z% = 6.5, X/R = 8.58 #Units = 28 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 100 MVA Z% = 9.34, X/R = 13 DETC 2.5% w 5 taps	R = 0.0002, X = 0.0008, B = 0.0333 2.54 mi

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1349	17 SMA 3.52 MW Inverter MVA = 4.4 MVA each 98 2.82 MW GE WTG MVA = 3.13 MVA each	SMA: 1× PF: 0.8 Qmin = - 44.88 MVAR Qmax = +44.88 MVAR 74.8 MVA GE WTG #1 PF: 0.90 Qmin = - 65.56 MVAR Qmax = +65.56 MVAR 150.4 MVA GE WTG #2 PF: 0.90 Qmin = - 68.291 MVAR Qmax = +68.291 MVAR 156.66 MVA	Solar: R = 0.005, X = 0.0037, B = 0.0055 Wind: Col1: R = 0.0120, X = 0.0105, B = 0.0409 1x15 MVAR Cap Col2: R = 0.0141, X = 0.0109, B = 0.0443 1x15 MVAR Cap	SMA: 1× 0.66 - 34.5 YG - D MVA = 4.4 MVA Z% = 6.5, X/R = 8.58 #Units = 17 DETC 2.5% w 5 taps Wind #1 0.69 - 34.5 YG - D MVA = 156 MVA Z% = 6, X/R = 7.50 Wind #2 0.69 - 34.5 YG - D MVA = 162.5 MVA Z% = 6, X/R = 7.50 #Units = 98 DETC 2.5% w 5 taps	SMA: 1× 34.5 - 345-13.8 kV YG - YG-D MVA = 48 MVA Z% = 7.50, X/R = 13 DETC 2.5% w 5 taps Wind: 2× 34.5 - 345-13.8 kV YG - YG-D MVA = 100 MVA Z% = 8.85, X/R = 39.3 DETC 2.5% w 5 taps	R = 0.001053, X = 0.009609, B = 0.162 19 mi
J1359	Siemens ABB 2 60 MW Gas Units MVA = 70.59 MVA each	2× PF: +0.85/ -0.90 Qmin = - 29.06 MVAR Qmax = +37.18 MVAR 70.59 MVA	NA	NA	2× 12.5 - 69 kV YG - D MVA = 50 MVA Z% = 12.0, X/R = 23.70 DETC 2.5% w 5 taps	NA

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1365	77 Vestas 2.2 MW V120 WTG MVA = 2.326 MVA each	PF: 0.95 Qmin = - 55.679 MVAR Qmax = +55.679 MVAR 179.1 MVA	R=0.00595, X=0.01015, B=0.03735	0.69 - 34.5 kV YG - D MVA = 2.326 MVA Z% = 9.62, X/R = 10.9 #Units = 77 DETC 2.5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 107.50 MVA Z% = 9.0, X/R = 45 DETC 2.5% w 5 taps	R=0.000033, X=0.000138, B=0.00005
J1371	37 Vestas V150 4.2 MW MVA = 5.1 MVA each	Qmin = - 61.4 MVAR Qmax = +77.9 MVAR 188.7 MVA	R=0.00355, X=0.00501, B=0.01390 2×15 MVAR Cap bank	0.72 - 34.5 kV YG - D MVA = 4.70 MVA Z% = 9.0, X/R = 11.25 #Units = 37 DETC 2.5% w 5 taps	34.5 - 345 - 13.2 kV YG - YG - D MVA = 101 MVA Z% = 10.0, X/R = 43 LTC 10% w 33 taps	R=0.00088, X=0.00694, B=0.0685
J1395	70 HEMK PE 3.3 MW Inverter MVA = 3.3 MVA each	2× Qmin = - 55.7759 MVAR Qmax = +55.7759 MVAR 115.5 MVA	Collector 1: R=0.00303, X=0.00239, B=0.01192 Collector 2: R=0.00463, X=0.00366, B=0.01496	2× 0.66 - 34.5 kV YG - D MVA = 126 MVA Z% = 5.75, X/R = 9.70 #Units = 70 DETC 2.5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 75 MVA Z% = 12.0, X/R = 45 DETC 10% w 33 taps	R=0.00001, X=0.00006, B=0.00067
J1413	63 TMEIC Ninja 3.78 MW Inverter MVA = 4.2 MVA each	2× PF: 0.90 Qmin = - 57.36 MVAR Qmax = +57.36 MVAR 131.59 MVA	2× R=0.00509, X=0.00607, B=0.15801 1×4 MVAR cap bank	2× 0.63 - 34.5 kV YG - D MVA = 168 MVA Z% = 5.75, X/R = 10 #Units = 80 DETC 2.5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 105 MVA Z% = 9, X/R = 45 DETC 2.5% w 5 taps	NA

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1416	40 TMEIC Ninja 3.78 MW Inverter MVA = 4.2 MVA each	PF: 0.90 Qmin = - 73.23 MVAR Qmax = +73.23 MVAR 168 MVA	R=0.00701, X=0.00810, B=0.08127 1×4 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10 #Units = 40 DETC 2.5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 105 MVA Z% = 9, X/R = 45 DETC 2.5% w 5 taps	NA
J1417	20 TMEIC Ninja 3.78 MW Inverter MVA = 4.2 MVA each	PF: 0.90 Qmin = - 36.61 MVAR Qmax = +36.61 MVAR 84 MVA	R=0.00454, X=0.00548, B=0.20033	0.63 - 34.5 kV YG - D MVA = 4.20 MVA Z% = 5.75, X/R = 10 #Units = 20 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 54 MVA Z% = 9, X/R = 35 DETC 2.5% w 5 taps	R=0.00005, X=0.0007, B=0.00624
J1426	27 TMEIC 3.71 MW Inverter MVA = 4.05 MVA each	PF: 0.916 Qmin = - 43.796 MVAR Qmax = +43.796 MVAR 109.35 MVA	R=0.00074, X=0.00136, B=0.003941 2×7 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 4.05 MVA Z% = 5.75, X/R = 7 #Units = 27 DETC 2.5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 35 DETC 2.5% w 5 taps	NA
J1432	83 GE 3.03 MW WTG MVA = 3.36 MVA each	PF: 0.90 Qmin = - 121.8 MVAR Qmax = +121.8 MVAR 278.88 MVA	R=0.00268, X=0.00395, B=0.0466 2×15 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 3.30 MVA Z% = 5.75, X/R = 7.5 #Units = 83 DETC 2.5% w 5 taps	34.5 - 345 - 13.8 kV YG - YG - D MVA = 165 MVA Z% = 8.5, X/R = 45 DETC 2.5% w 5 taps	R=0.000002, X=0.00002, B=0.00017
J1438	27 TMEIC 4.05 MVA inverters Total MVA = 108 MVA	PF: 0.926 Qmin = - 40.792 MVAR Qmax = +40.792 MVAR 108 MVA	R = 0.0013, X = 0.0012, B = 0.0025 2×7 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 109.35 (4.05*27) Z% = 5.75, X/R = 7 #Units =27 DETC 5% w 5 taps	34.5 - 161 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 40 DETC 5% w 5 taps	R = 0.000025, X = 0.000146, B = 0.000036

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1444	54 TMEIC 4.05 MVA inverters Total MVA = 216 MVA	PF: 0.926 Qmin = - 81.584 MVAR Qmax = +81.584 MVAR 216 MVA	R = 0.00037, X = 0.00068, B = 0.00394 2×14 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 218.7 (4.05*54) Z% = 5.75, X/R = 7 #Units =54 DETC 5% w 5 taps	34.5 -345 - 13.8 kV YG - YG - D MVA = 135 MVA Z% = 8.5, X/R = 45 DETC 5% w 5 taps	R = 0.00003, X = 0.00023, B = 0.00232
J1445	27 TMEIC 4.05 MVA inverters Total MVA = 109 MVA	PF: 0.917 Qmin = - 43.37 MVAR Qmax = +43.37 MVAR 109 MVA	R = 0.00074, X = 0.00136, B = 0.00197 2×7 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 109.35 (4.05*27) Z% = 5.75, X/R = 7 #Units =27 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 35 DETC 5% w 5 taps	R = 0.000048, X = 0.000286, B = 0.000019
J1446	126 PE 1.29 MVA inverters Total MVA = 162 MVA	PF: 0.926 Qmin = - 61.188 MVAR Qmax = +61.188 MVAR 162 MVA	R = 0.00138, X = 0.00253, B = 0.001 2×11 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 157.5 (2.5*63) Z% = 6.84, X/R = 7.5 #Units =126 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 105 MVA Z% = 8, X/R = 35 DETC 5% w 5 taps	NA

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1456	150 Vestas 2.0 MVA WTG Total MVA = 105.26*3 MVA	3× PF: 0.95 Qmin = - 32.8684 MVAR Qmax = +32.8684 MVAR 105.26 MVA	Collector 1: R = 0.0052, X = 0.00601, B = 0.16775 1×10 MVAR cap bank 1×(-5) MVAR reactor bank Collector 2: R = 0.00395, X = 0.00454, B = 0.13017 1×10 MVAR cap bank 1×(-5) MVAR reactor bank Collector 3: R = 0.0023, X = 0.00261, B = 0.07866 1×10 MVAR cap bank 1×(-5) MVAR reactor bank	3× 0.69 - 34.5 kV YG - D MVA = 103 (2.06*50) Z% = 9.8, X/R = 10.5 #Units =150 DETC 5% w 5 taps	3× 34.5 - 230 - 13.2 kV YG - YG - D MVA = 75 MVA Z% = 9.0, X/R = 35 DETC 10% w 33 taps	R = 0.00001, X = 0.00008, B = 0.00018
J1468	84 PE 1.29 MVA inverters Total MVA = 108.4 MVA	PF: 0.9 Qmin = - 48.432 MVAR Qmax = +48.432 MVAR 108.4 MVA	R = 0.00138, X = 0.00253, B = 0.001 1×14 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 105 (2.5*42) Z% = 6.84, X/R = 7.5 #Units =84 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 69 MVA Z% = 8, X/R = 35 DETC 5% w 5 taps	NA

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1471	54 TMEIC 4.05 MVA inverters Total MVA = 216 MVA	PF: 0.926 Qmin = - 81.584 MVAR Qmax = +81.584 MVAR 216 MVA	R = 0.00037, X = 0.00068, B = 0.00394 2×14 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 218.7 (4.05*54) Z% = 5.75, X/R = 7 #Units =54 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 138 MVA Z% = 8.5, X/R = 40 DETC 5% w 5 taps	R = 0.00004, X = 0.00021, B = 0.0001
J1474	54 TMEIC 4.05 MVA inverters Total MVA = 216 MVA	PF: 0.926 Qmin = - 81.584 MVAR Qmax = +81.584 MVAR 216 MVA	R = 0.00138, X = 0.00253, B = 0.001 2×14 MVAR cap bank	0.63 - 34.5 kV YG - D MVA = 218.7 (4.05*54) Z% = 5.75, X/R = 7 #Units =54 DETC 5% w 5 taps	34.5 - 115 - 13.8 kV YG - YG - D MVA = 138 MVA Z% = 8.5, X/R = 40 DETC 5% w 5 taps	NA
J1477	17 PE FS3000CU15 Inverter Total MVA = 58.33 MVA	PF: 0.86 Qmin = - 30.06 MVAR Qmax = +30.06 MVAR 58.33 MVA	R = 0.00776, X = 0.00951, B = 0.02263	0.69 - 34.5 kV YG - D MVA = 59.5 (3.5*17) Z% = 5.75, X/R = 11 #Units = 17 DETC 5% w 5 taps	34.5 - 161 kV YG - D MVA = 99 MVA Z% = 9.5, X/R = 32.5 DETC 5% w 5 taps	NA
J1478	9 PE FS3000CU15 Inverter Total MVA = 29.17 MVA	PF: 0.86 Qmin = - 15.03 MVAR Qmax = +15.03 MVAR 29.17 MVA	R = 0.00776, X = 0.00951, B = 0.02263	0.69 - 34.5 kV YG - D MVA = 31.5 (3.5*9) Z% = 5.75, X/R = 11 #Units = 9 DETC 5% w 5 taps	34.5 - 69 kV YG - D MVA = 51 MVA Z% = 8.5, X/R = 31 DETC 5% w 5 taps	NA
J1479	34 PE FS3000CU15 Inverter Total MVA = 116.67 MVA	PF: 0.86 Qmin = - 60.099 MVAR Qmax = +60.099 MVAR 116.67 MVA	R = 0.00776, X = 0.00951, B = 0.02263	0.69 - 34.5 kV YG - D MVA = 119 (3.5*34) Z% = 5.75, X/R = 11 #Units = 34 DETC 5% w 5 taps	34.5 - 161 kV YG - D MVA = 66 MVA Z% = 9.5, X/R = 32.5 DETC 5% w 5 taps	R = 0.00217, X = 0.0091, B = 0.011315

MISO Project #	Turbine / Inverter	Reactive Power Capability (Mvar)	Collector System	Pad Mount xfmr	Substation xfmr	Gen Tie
J1485	100 2.52 MW GE turbines Total MVA = 138.89*2 MVA (2.78 MVA each)	2× PF: 0.90 Qmin = - 60.5403 MVAR Qmax = +60.5403 MVAR 138.89 MVA	2× R = 0.000008, X = 0.000094, B = 0.000907 1×3 MVAR cap bank	2× 0.69 - 34.5 kV YG - D MVA = 140 (2.8*50) Z% = 5.75, X/R = 10 #Units = 50*2 DETC 5% w 5 taps	2× 34.5 - 345 - 13.8 kV YG - YG - D MVA = 84 MVA Z% = 9.0, X/R = 45 DETC 5% w 5 taps	R = 0.00508, X = 0.00607, B = 0.000907
J1487	49 2.52 MW GE turbines Total MVA = 137.2 MVA (2.8 MVA each)	PF: 0.90 Qmin = - 59.804 MVAR Qmax = +59.804 MVAR 137.2 MVA	R = 0.006707, X = 0.006266, B = 0.061328 1×11 MVAR cap bank	0.69 - 34.5 kV YG - D MVA = 147 (3*49) Z% = 5.75, X/R = 7 #Units = 49 DETC 5% w 5 taps	34.5 - 161 kV YG - YG MVA = 80 MVA Z% = 10.0, X/R = 36 DETC 5% w 5 taps	R = 0.002238, X = 0.019604, B = 0.020399

Table A-3: DPP 2019 Central Area Projects

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1191	MO	Boone	CWLD	Bolstad 69 kV	64	64	Solar	NRIS
J1197	IL	LaSalle	AMIL	Oglesby 138 kV	75	75	Battery	NRIS
J1198	IL	Jasper	AMIL	Newton 345 kV	51.8	51.8	Hybrid	NRIS
J1199	IL	Fulton	AMIL	Duck Creek 345 kV	20	20	Hybrid	NRIS
J1200	IL	Putnam	AMIL	Hennepin 138 kV	24.2	24.2	Hybrid	NRIS
J1201	IL	Montgomery	AMIL	Coffeen 345 kV	44.2	44.2	Hybrid	NRIS
J1202	IL	Randolph	AMIL	Baldwin Plant 345 kV Switchyard - Baldwin Unit 2	68.4	68.4	Hybrid	NRIS
J1204	IL	Cumberland	AMIL	Casey West 345 kV	199.3	199.3	Solar	NRIS
J1208	IL	Saline	AMIL	Muddy - Crab Orchard 138 kV	80	80	Solar	NRIS
J1209	IL	Saline	AMIL	Muddy - Crab Orchard 138 kV	80	80	Solar	NRIS
J1213	MO	Iron	AMMO	Taum Sauk 138kV Switchyard, Bus 1 & 2	60	60	Hydro	NRIS
J1216	IL	Washington	AMIL	Ashley 138 kV	250	250	Solar	NRIS
J1222	MO	Ralls	AMMO	Spencer Creek 345 kV	200	200	Solar	NRIS
J1225	IL	Washington	AMIL	Ashley-Turkey Hill Line 1482, between the Okawville and North Nashville substations	160	160	Solar	NRIS
J1227	IN	Spencer	SIGE	Dubois - FB Culley 138 kV	125	125	Solar	NRIS
J1231	KY	Lyon	BREC	Barkley - Caldwell 161 kV	125	125	Solar	NRIS
J1232	IL	Coles	AMIL	Chesterville - Mattoon West 138 kV	99	99	Solar	NRIS
J1233	IN	Pike	SIGE	Cato 138 kV	32.5	32.5	Battery	NRIS
J1234	IN	Knox	DEI	Edwardsport - Washington Mun, Linton 138 kV	100	100	Solar	NRIS
J1235	IN	Knox	DEI	Edwardsport - Washington Mun, Linton 138 kV	50	50	Solar	NRIS
J1241	IL	Jefferson	AMIL	West Mt Vernon - Xenia 345 kV	200	200	Solar	NRIS
J1242	IL	Hancock	AMIL	Hamilton Line 796 Tap to Audrian 69 kV Line	25	25	Solar	External NRIS
J1243	IL	McDonough	AMIL	Macomb West #2 - Blandinsville Tap 69 kV	25	25	Solar	External NRIS
J1254	IN	Knox	DEI	Merom - Gibson - Wheatland 345 kV	150	150	Solar	NRIS

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1263	IL	Clark	AMIL	Casey - Kansas West 345 kV	408.6	408.6	Hybrid	NRIS
J1265	IN	LaGrange	NIPSCO	Wolcottville - South Milford 69 kV	43	43	Solar	NRIS
J1266	IL	Marion	AMIL	Kinmundy - Salem 138 kV	99	99	Solar	NRIS
J1268	MO	Lincoln	AMMO	Pike - Troy 161 kV	150	150	Solar	NRIS
J1269	IN	Warrick	SIGE	Elliott 69kV	100	100	Hybrid	NRIS
J1271	IN	Vanderburgh	SIGE	Oak Grove 138 kV	100	100	Battery	NRIS
J1272	IN	Vanderburgh	SIGE	Angel Mound 138 kV	50	50	Battery	NRIS
J1274	IN	Posey	SIGE	Point 69 kV	100	100	Hybrid	NRIS
J1275	KY	Henderson	BREC	Henderson 161 kV	100	100	Solar	NRIS
J1276	IN	Posey	SIGE	Point 69 kV	100	100	Hybrid	NRIS
J1289	IL	Morgan	AMIL	Turner 345 kV	200	200	Wind	NRIS
J1295	IN	Gibson	DEI	Gibson - Francisco 345 kV	280	280	Solar	NRIS
J1296	IN	Gibson	SIGE	Francisco 138 kV	200	200	Solar	NRIS
J1299	MO	Scott	AMMO	Miner - Kelso 161 kV	200	200	Solar	NRIS
J1301	IL	Washington	AMIL	Prest 138 kV	150	150	Solar	NRIS
J1302	IL	Saline	AMIL	Muddy - Norris City 138 kV	150	150	Solar	NRIS
J1303	IL	Jackson	SIPC	Campbell Hill - Jackson 161 kV	150	150	Solar	NRIS
J1306	IL	Perry	AMIL	Pickneyville - St John 230 kV	200	200	Solar	NRIS
J1308	IN	Posey	SIGE	AB Brown - Gibson 345 kV	300	300	Solar	NRIS
J1311	IL	St. Clair	AMIL	Fayetteville Bee Hollow Road 138 kV	150	150	Solar	NRIS
J1318	IN	Rush	DEI	Rushville 69 kV	25	25	Solar	NRIS
J1323	IN	LaGrange	NIPSCO	Lagrange 138 kV	25	25	Battery	NRIS
J1330	IL	Randolph	AMIL	Baldwin 345 kV	200	200	Solar	NRIS
J1332	IN	Jasper	NIPSCO	Schahfer 345 kV	200	200	Solar	NRIS
J1333	IN	Jasper, Starke	NIPSCO	Schahfer - Burr Oak 345 kV	199.5	199.5	Hybrid	NRIS
J1334	IN	Jasper, Starke	NIPSCO	Schahfer - Burr Oak 345 kV	199.5	199.5	Hybrid	NRIS

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1335	IN	Jasper, Starke	NIPSCO	Schahfer - Burr Oak 345 kV	199.5	199.5	Hybrid	NRIS
J1336	IN	Jasper, Starke	NIPSCO	Schahfer 345 kV	199.5	199.5	Hybrid	NRIS
J1338	MO	Audrain	AMMO	Spencer Creek - Montgomery 345 kV	150	150	Wind	NRIS
J1339	IN	Jasper, Starke	NIPSCO	Schahfer 345 kV	199.5	199.5	Hybrid	NRIS
J1340	IN	Jasper, Starke	NIPSCO	Schahfer 345 kV	199.5	199.5	Hybrid	NRIS
J1342	IN	Starke County	NIPSCO	Starke 138 kV	99	99	Solar	NRIS
J1348	IN	Vermillion	DEI	Cayuga - Sugar Creek 345 kV	200	200	Solar	NRIS
J1352	MO	Montgomery	AMMO	Montgomery 345 kV	100	100	Solar	NRIS
J1353	IN	Lake	NIPSCO	Green Acres 138 kV	150	150	Battery	NRIS
J1354	IN	Decatur	HE	Decatur County 138 kV	146	146	Battery	NRIS
J1355	IN	LaPorte	NIPSCO	Stillwell 138 kV	160	160	Battery	NRIS
J1356	IN	Martin	DEI	Whitfield 138 kV	160	160	Solar	NRIS
J1357	IN	Madison	DEI	Madison 138 kV	150	150	Battery	NRIS
J1358	IN	LaPorte	NIPSCO	Luchtman 138 kV	125	125	Battery	NRIS
J1360	IL	Piatt	AMIL	Goose Creek 345 kV	300	300	Wind	NRIS
J1363	MO	Audrain	AMMO	Spencer Creek - Montgomery 345 kV	150	150	Solar	NRIS
J1369	IN	Dubois	SIGE	Dubois 138 kV	100	100	Solar	NRIS
J1376	MO	Cooper	AMMO	Barnett - Georgetown Overton 161 kV	100	100	Wind	NRIS
J1378	IN	Fountain	DEI	Veedersburg West 230 kV	200	200	Solar	NRIS
J1381	IN	Howard	DEI	Greentown 230 kV	200	200	Solar	NRIS
J1382	IL	Clark	AMIL	Hutsonville - Neoga 138 kV	150	150	Solar	NRIS
J1383	IL	Fulton	AMIL	Ipava 138 kV	150	150	Solar	NRIS
J1385	IN	Hendricks	DEI	Stilesville 138 kV	100	100	Solar	NRIS
J1386	IN	Lake	NIPSCO	Green Acres 138 kV	100	100	Solar	NRIS
J1387	IN	Jasper	NIPSCO	Circuit 6958 69 kV Line	60	60	Solar	ERIS
J1388	IN	Gibson	DEI	Oakland City - Old Ben Coal 138 kV	150	150	Solar	NRIS
J1390	IN	Bartholomew	DEI	Gallagher Station - Columbus 230 kV	150	150	Solar	NRIS

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1391	IN	Dubois	SIGE	Duff 138 kV	100	100	Solar	NRIS
J1392	IN	LaGrange	NIPSCO	Lagrange 138 kV	100	100	Solar	NRIS
J1393	IN	Pike	SIGE	Cato 138 kV	130	130	Solar	NRIS
J1396	IN	Pike	HE	Ratts-Victory 161 kV	150	150	Solar	NRIS
J1398	IN	Pike	HE	Ratts 161 kV	150	150	Solar	NRIS
J1405	IN	Greene	DEI	Sanborn - Linton 138 kV	140	140	Solar	NRIS
J1407	IN	Porter	NIPSCO	South Valparaiso 138 kV	100	100	Solar	NRIS
J1408	IL	Macoupin	AMIL	Virden South 138 kV	100	100	Hybrid	NRIS
J1409	KY	Henderson	BREC	Henderson 161 kV	25	25	Battery	NRIS
J1419	IN	Greene	DEI	Edwardsport - Amo 345 kV	250	250	Solar	NRIS
J1422	IL	Edwards	AMIL	Albion - Olney 138 kV	150	150	Hybrid	NRIS
J1447	IN	Porter	NIPSCO	Schahfer - Green Acres 345 kV	250	250	Solar	NRIS
J1448	KY	Caldwell	BREC	Hopkins - Caldwell County - Barkely 161 kV	150	150	Solar	NRIS
J1450	KY	Webster	BREC	Hopkins - Reid 161 kV	150	150	Solar	NRIS
J1453	IL	Mason	AMIL	Havana - Shockey 138 kV	188.3	188.3	Wind	NRIS
J1454	IL	Mason	AMIL	Mason 138 kV	58.5	58.5	Wind	NRIS
J1457	IN	Vermillion	DEI	Sugar Creek - Cayuga 345 kV	200	200	Solar	NRIS
J1464	IL	Macoupin, Morgan, Sangamon	AMIL	Meredosia - Austin 345 kV	592.8	592.8	Solar	NRIS
J1466	KY	Henderson	BREC	Reid 161 kV	250	250	Solar	NRIS
J1470	IN	Benton, Warren	DEI	Cayuga CT 345 kV	30	30	Wind	NRIS Only
J1475	IL	Clark	AMIL	Casey West - Sullivan 345 kV	75	75	Solar	NRIS
J1481	IN	Howard	DEI	Greentown 230 kV	200	200	Solar	NRIS
J1482	IN	White	NIPSCO	Monticello - Springboro 138 kV	150	150	Solar	NRIS
J1488	MO	Randolph	AMMO	McCredie - Montgomery 345 kV	500	500	HVDC	NRIS
J1490	MO	Randolph	AMMO	McCredie - Montgomery 345 kV	1000	1000	HVDC	NRIS

Table A-4: DPP 2019 Michigan Area Projects

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1192	MI	Eaton	METC	Oneida 138 kV	35	35	Solar	NRIS
J1194	MI	Midland, Saginaw	METC	Murphy 138 kV	225.2	225.2	Wind	NRIS
J1196	MI	Sanilac	ITCT	Greenwood - Rapson (Banner) 345 kV	200	200	Solar	NRIS
J1203	MI	Genesee	METC	Thetford 345 kV	125	125	Solar	NRIS
J1207	MI	Branch	METC	Newton 138 kV	50	49.5	Gas	NRIS
J1210	MI	Iosco	METC	Whittemore - Twining 138 kV	50	50	Solar	NRIS
J1224	MI	Washtenaw	ITCT	Majestic - Milan 345 kV	170	170	Solar	NRIS
J1226	MI	Lenawee	METC	Morocco 345 kV	200	200	Solar	NRIS
J1248	MI	Calhoun	METC	Marshall - Black Stone 138 kV	100	100	Solar	NRIS
J1255	MI	Ionia	METC	Marquette - Vergennes 138 kV	100	100	Solar	NRIS
J1262	MI	Wayne	ITCT	Waterman 120 kV	25	0	Gas	ERIS
J1292	MI	Ionia	METC	Thompson Road 138 kV	198	198	Wind	NRIS
J1297	MI	Shiawassee	METC	Cornell - Bell Road 138 kV	170	170	Solar	NRIS
J1307	MI	Eaton	METC	Oneida - Battle Creek 345 kV	150	150	Solar	NRIS
J1310	MI	Jackson	METC	Blackstone - Marshall 138 kV	125	125	Solar	NRIS
J1319	MI	Huron	ITCT	Rapson 120 kV	117.5	117.5	Wind	NRIS
J1320	MI	Branch	METC	Coldwater - Batavia 138 kV	150	150	Solar	NRIS
J1329	MI	Washtenaw	ITCT	Majestic - Milan 345 kV	20	20	Battery Storage	NRIS
J1331	MI	St. Clair	ITCT	Greenwood 120 kV	100	100	Solar	NRIS
J1350	MI	Monroe	METC	Milan - Monroe - Morocco 345 kV Point	150	150	Solar	NRIS
J1375	MI	Muskegon	METC	Du Pont - Cobb -White Lake 138 kV	100	100	Solar	NRIS
J1379	MI	Montcalm	METC	Eureka - Vestaburg 138 kV	200	200	Solar	NRIS
J1389	MI	Saginaw	METC	Murphy Station - Nelson Road 345 kV	120	120	Solar	NRIS
J1399	MI	Ingham	METC	Delhi - Churchill - Tompkins 138 kV	90	90	Solar	NRIS
J1401	MI	Branch	METC	Batavia - Wagner 138 kV	150	150	Solar	NRIS

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1403	MI	Ionia	METC	Marquette 138 kV	200	200	Solar	NRIS
J1430	MI	Calhoun	METC	Verona - Foundary 138 kV	100	100	Solar	NRIS
J1472	MI	Jackson	METC	Leoni - Parr Raod 138 kV	100	100	Solar	NRIS

Table A-5: DPP 2019 ATC Area Projects

MISO Project #	State	County	Trans. Owner	Point of Interconnection	ERIS Output	NRIS Output	Fuel Type	Service Type
J1214	WI	Dane	ATC	Rockdale 345 kV	300	300	Solar	NRIS
J1244	MI	Houghton	ATC	Winona - Toivola - Portage Tap - Atlantic Mine 69 kV	38.4	38.4	Wind	NRIS
J1251	MI	Marquette	ATC	Arnold - Perkins 138 kV	100	100	Solar	NRIS
J1253	WI	Fond du Lac	ATC	South Fond Du Lac to Fitzgerald 345 kV	100	100	Solar	NRIS
J1304	WI	Rock	ATC	Paddock - Townline Road 138 kV	65	65	Solar	NRIS
J1305	WI	Green	ATC	Albany - Bass Creek 138 kV	49.9	49.9	Solar	NRIS
J1316	WI	Kenosha	ATC	Paris 138 kV	50	50	Battery	NRIS
J1326	WI	Dane	ATC	Rockdale 345 kV	75	75	Battery	NRIS
J1345	WI	Dodge	ATC	North Beaver Dam 69 kV	25	25	Battery	NRIS
J1370	MI	Chippewa	ATC	9 Mile SW STA - Dafter 69 kV	50	50	Gas	NRIS
J1374	WI	Grant	ATC	ATC Ebenezer Sub on Lancaster-Eden 138 kV	67.2	67.2	Wind	NRIS
J1377	WI	Rock	ATC	Wempletown - Rockdale 345 kV	98.56	98.56	Solar	NRIS
J1410	WI	Dane	ATC	N. Madison 345 kV	300	300	Solar	NRIS
J1411	WI	Dane	ATC	N. Madison 345 kV	75	75	Battery	NRIS
J1460	WI	Rock	ATC	Sunrise 138 kV	200	200	Solar	NRIS
J1483	WI	Iowa	ATC	Hill Valley 345 kV	99	0	Wind	NRIS

A.2 DPP 2017 August West Area Phase 3 Network Upgrades

Table A-6: DPP 2017 August West Phase 3 NUs

Network Upgrade	Owner
Sheyenne-Mapleton 115 kV Rebuild	XEL, OTP
Crookston-Falconer 115 kV Jumper Replacement	OTP
New J628 POI– Prairie 230 kV 2nd Circuit	GRE
J628 POI-Prairie 230 kV Rebuild	GRE
Helena-Chub Lake 2nd Circuit	GRE/CAPX
Big Stone-Big Stone South 230 kV Circuit 1 Rebuild	OTP
Big Stone-Big Stone South 230 kV Circuit 2 Rebuild	OTP
Big Stone-Blair 230 kV Rebuild	OTP/NWE
Adams 345/161 kV Transformer Upgrade	XEL
161 kV Reconfigure into J898 POI Substation	DPC

A.3 Model Review Comments

Table A-7: Model Review Comments

Company	Python/ Idev File Name	2024 SH Benchmark	2024 SH Study Discharge	2024 SH Study Charge	2024 SPK Benchmark	2024 SPK Study Discharge
PJM	Master_Remove PJM withdrawn prjs.py	X	X	X	X	X
SPP	Master_Remove SPP withdrawn prjs.py	X	X	X	X	X
SPP	DISIS-2017-001_West_Master.py	X	X	X	X	X
SPP	SH-Dispatch DISIS-2017-001.py	X	X	X		
SPP	PK-Dispatch DISIS-2017-001.py				X	X
ICs	J1190_upd.py	X	X	X	X	X
ICs	J1221_POI_WinRiver_to_Riverton.py	X	X	X	X	X
ICs	J1298_POI_change.py	X	X	X	X	X
ICs	J1312_upd.py	X	X	X	X	X
ICs	J1366_Out_of_Service.py	X	X	X	X	X
ICs	J1343_Out_of_Service.py	X	X	X	X	X
ICs	J1344_Out_of_Service.py	X	X	X	X	X
ICs	Bus_Code.py	X	X	X	X	X
ICs	J1313_upd.py	X	X	X	X	X
ICs	J1314_upd.py	X	X	X	X	X
ICs	J1315_upd.py	X	X	X	X	X
ICs	J1325_upd.py	X	X	X	X	X
ICs	J1337_upd.py	X	X	X	X	X
ICs	J1193_Out_of_Service.py	X	X	X	X	X
ICs	J1480_Out_of_Service.py	X	X	X	X	X
ICs	Update Vref_347832.py	X	X	X	X	X
ICs	correct J1262.py	X	X	X	X	X
ICs	J1371 correction.py	X	X	X	X	X
ICs	J1371 Dispatch_SH.py	X	X	X		
ICs	DPP-2019-C1_West_J1395_2020-10-22.idv	X	X	X	X	X
ICs	PK-J1395 Dispatch.py				X	X
ICs	DPP-2019-C1_West_J1456_2020-10-22.idv	X	X	X	X	X
ICs	J1467_PF_upd.py	X	X	X	X	X
ICs	J1468_PF_upd.py	X	X	X	X	X
ICs	SUM_PGEN_PMIN_Fix.py	X	X	X	X	X

Company	Python/ Idev File Name	2024 SH Benchmark	2024 SH Study Discharge	2024 SH Study Charge	2024 SPK Benchmark	2024 SPK Study Discharge
ICs	MBASE_fix.py	X	X	X	X	X
ICs	Rating_Fix.py	X	X	X	X	X
ICs	LowVoltage_fix.py	X	X	X	X	X
MEC	MEC_DPP_2019_West_Ph1_SH_Updates.py	X	X	X		
MEC	MEC_DPP_2019_West_Ph1_SUM_Updates.py				X	X
MEC	Add J449 Offline.idv	X	X	X	X	X
MEC	RMV Duplicated J614.py	X	X	X	X	X
MEC	ADD J530 Offline.idv	X	X	X	X	X
MEC	Correct J534 area.py	X	X	X	X	X
MEC	Correct J1341 gentie.py	X	X	X	X	X
MEC	Correct 631144-41814-44137-631139.py	X	X	X	X	X
MEC	RMV prior DPP NUs.py	X	X	X	X	X
MEC	dispatch Gen-SH Dis Study.py		X			
MEC	dispatch Gen-SH Ch Study.py			X		
MEC	dispatch Gen-SH Bench.py	X				
MEC	dispatch Gen-PK Study.py					X
MEC	dispatch Gen-PK Bench.py				X	
MDU	DPP19-West-Ph1-2024SH_MDU-Updates.idv	X	X	X		
MDU	DPP19-West-Ph1-2024SUM_MDU-Updates.idv				X	X
MDU	DPP-2019-West-Ph1_MDU-Updates.con	X	X	X	X	X
ATC	Remove_SFL_FIZ.idv	X	X	X	X	X
GRE	Correct J897 POI.py	X	X	X	X	X
SPTI	J1359_Control_Mode_Change.py	X	X	X	X	X
SPTI	Correct J1122 branchs.py	X	X	X	X	X
SPTI	Correct J1122 modeling.py	X	X	X	X	X
SPTI	Correct J1468 Modeling.py	X	X	X	X	X
SPTI	Dispatch J1468_SH-Discharge.py		X			
SPTI	Dispatch J1468_SH-Charge.py			X		
MPC	MPC-DPP-2019-Cycle-Phase 1 Updates.idv	X	X	X	X	X
MRES	2019 West Hutch Voltages.idv	X	X	X	X	X
MRES	MRES OJ Line Rating Change.idv	X	X	X	X	X
OTP	OTP_2019_West_Ph1_corrections.idv	X	X	X	X	X

Company	Python/ Idev File Name	2024 SH Benchmark	2024 SH Study Discharge	2024 SH Study Charge	2024 SPK Benchmark	2024 SPK Study Discharge
AMRN	Ameren Correction.py	X	X	X	X	X
ATC	20231231_CHC_River_Crossing.idv	X	X	X	X	X
ITCM	ITCM Corrections.idv	X	X	X	X	X
AMRN	Ameren Ratings.idv	X	X	X	X	X
CIPCO	CIPCO Corrections.py	X	X	X	X	X
SPTI	Correct MBase.py	X	X	X	X	X
SPTI	RMV PJM Dup-WD.py	X	X	X	X	X
SPTI	Dispatch Correction_SH.py	X	X	X		
SPTI	Dispatch Correction_PK.py				X	X
Changes applied to Phase 2 study						
ICs	Move J1221 POI.py	X	X	X	X	X
ICs	Move J1365 POI.py	X	X	X	X	X
ICs	Add Cap J1371.py	X	X	X	X	X
ICs	DPP-2019-C1_West_J1395_2021-01-20.idv	X	X	X	X	X
ICs	DPP-2019-C1_West_J1456_2021-03-10.idv	X	X	X	X	X
ICs	Update J1328 Model.py	X	X	X	X	X
ICs	Update J1349 Model.py	X	X	X	X	X
OTP	OTP model updates_v33.idv	X	X	X	X	X
OTP	Audubon-Erie ratings v33.idv	X	X	X	X	X
MISO	Correct Split Rock-White ratings.py	X	X	X	X	X
ICs	J926 Solar_PK.py				X	X
ICs	J926 Solar_SH.py	X	X	X		
CIPCO	Liberty - Dundee Rating Correction.idv	X	X	X	X	X
MPC	MPC-fix-praicans.idv	X	X	X	X	X
ICs	Update J1135 POI.py	X	X	X	X	X
MISO	RMV Withdrawn_DPP2017.py	X	X	X	X	X
MISO	RMV Withdrawn_DPP18-West.py	X	X	X	X	X
MISO	RMV Withdrawn_DPP18-Central-MI.py	X	X	X	X	X
MISO	Update MW Reduction_PK.py				X	X
MISO	Update MW Reduction_SH.py	X	X	X		
MISO	RMV Withdrawn_DPP19.py	X	X	X	X	X
MISO	Remove DPP 2017Aug Ph2 NUs.py	X	X	X	X	X

Company	Python/ Idev File Name	2024 SH Benchmark	2024 SH Study Discharge	2024 SH Study Charge	2024 SPK Benchmark	2024 SPK Study Discharge
MISO	Remove DPP 2018Apr Ph2 NUs.py	X	X	X	X	X
MISO	Add DPP2017Aug Ph3 NUs.py	X	X	X	X	X
MISO	RMV Withdrawn_PJM.py	X	X	X	X	X
MISO	RMV Withdrawn_SPP.py	X	X	X	X	X
MISO	RMV SPP NUs.py	X	X	X	X	X
OTP	MR xfmr1 replacement impedance_V33.py	X	X	X	X	X
MPC	MPC-fixrtngs-DPP19_West-Ph2_V33.idv	X	X	X	X	X
MEC	MEC_DPP_2019_West_Ph2_Updates.py	X	X	X	X	X
MISO	J1446_SH Charge_Dispatch.py			X		
MISO	J1446_SH Discharge_Dispatch.py		X			

A.4 MISO North as the Study Sink

Table A-8: MISO North as the Study Sink

Area #	Area Name	Area #	Area Name
207	HE	608	MP
208	DEI	613	SMMPA
210	SIGE	615	GRE
216	IPL	620	OTP
217	NIPS	627	ALTW
218	METC	633	MPW
219	ITC	635	MEC
295	WEC	661	MDU
296	MIUP	663	BEPC-MISO
314	BREC	680	DPC
333	CWLD	694	ALTE
356	AMMO	696	WPS
357	AMIL	697	MGE
360	CWLP	698	UPPC
361	SIPC	701	MISO Prior
600	Xcel		

A.5 PJM Market as PJM Projects Sink

Table A-9: PJM Market as PJM Projects Sink

Area #	Area Name	Area #	Area Name
201	AP	229	PPL
202	ATSI	230	PECO
205	AEP	231	PSE&G
206	OVEC	232	BGE
209	DAY	233	PEPCO
212	DEO&K	234	AE
215	DLCO	235	DP&L
222	CE	236	UGI
225	PJM	237	RECO
226	PENELEC	320	EKPC
227	METED	345	DVP
228	JCP&L		

A.6 SPP Market as SPP Projects Sink

Table A-10: SPP Market as SPP Projects Sink

Area #	Area Name	Area #	Area Name
515	SWPA	541	KCPL
520	AEPW	542	KACY
523	GRDA	544	EMDE
524	OKGE	545	INDN
525	WFEC	546	SPRM
526	SPS	640	NPPD
527	OMPA	645	OPPD
531	MIDW	650	LES
534	SUNC	652	WAPA
536	WERE	659	BEPC-SPP

A.7 Contingency Files used in Steady-State Analysis

Table A-11: List of Contingencies used in Steady-State Analysis

Contingency File Name	Description	Shoulder	Peak
Automatic single element contingencies	Single element outages at buses 60 kV and above in the study region	x	x
CC Bipole Events.con	Specified category P1, P7 contingencies in GRE Coal Creek	x	x
HVDC_Red_2024SH.con	Contingencies with HVDC reduction	x	
HVDC_Red_2024PK.con	Contingencies with HVDC reduction		x
MEC-DPP2019WESTPH1-Cat P1-2020.11.06.con	Specified category P1 contingencies in MEC	x	x
MEC-DPP2019WESTPH1-Cat P2-2020.11.06.con	Specified category P2 contingencies in MEC	x	x
MEC-DPP2019WESTPH1-Cat P5-2020.11.06.con	Specified category P5 contingencies in MEC	x	x
MEC-DPP2019WESTPH1-Cat P7-2020.11.06.con	Specified category P7 contingencies in MEC	x	x
MISO19_2024_SUM_TA_P1_ATC.con	Specified category P1 contingencies in ATC	x	x
MISO19_2024_SUM_TA_P1_IOWA.con	Specified category P1 contingencies in Iowa	x	x
MISO19_2024_SUM_TA_P1_MINN-DAKS.con	Specified category P1 contingencies in Minnesota, Dakotas	x	x
MISO19_2024_SUM_TA_P1_P2_P4_P5_NoLoadLoss.con	Specified category P1, P2, P4, P5 contingencies in MISO CLASSIC	x	x
MISO19_2024_SUM_TA_P2_P4_P5_P7_LoadLoss.con	Specified category P2, P4, P5, P7 contingencies in MISO Classic	x	x
MISO19_2024_SUM_TA_P1_AMEREN.con	Specified category P1 contingencies in Ameren	x	x
2020_RTEP_Single_2017Aug-Updated-3-4-19.con	Specified single contingencies in PJM	x	x
2020_RTEP_Bus_AC2-updated-3-4-19.con	Specified bus contingencies in PJM	x	x
2020_RTEP_Line_FB_2017Aug-updated-3-4-19.con	Specified breaker failure contingencies in PJM	x	x
2020_RTEP_Tower_AC2-updated-3-4-19.con	Specified common structure contingencies in PJM	x	x
160303-KACY_P1.con	Specified category P1 contingencies in KACY	x	x
160303-KACY_P2.con	Specified category P2 contingencies in KACY	x	x
AECI-AMMO.CON	Specified contingencies between AECI and AMMO	x	x
KCPL_P1.con	Specified category P1 contingencies in KCPL	x	x
KCPL_P2.con	Specified category P2 contingencies in KCPL	x	x
KCPL_P4.con	Specified category P4 contingencies in KCPL	x	x

Contingency File Name	Description	Shoulder	Peak
KCPL_P5.con	Specified category P5 contingencies in KCPL	x	x
KCPL_P7.con	Specified category P7 contingencies in KCPL	x	x

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Model Data

B.1 Power Flow Model Data

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B.2 Dynamic Model Data

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B.3 2024 Slider Diagrams

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Appendix

C

Reactive Power Requirement Analysis Results (FERC Order 827)

Table C-1: Reactive Power Requirement Analysis Results

Project #	TYPE	HV Side Bus #	Lagging Power Factor Results				Leading Power Factor Results				Inverter Inherent Power Factor	Shunt Compensation
			MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?		
J1190	Solar	41903	196.70	50.92	0.9681	No	196.28	-138.72	-0.8166	Yes	SMA 2.5 MW Inverter PF: +/- 0.909	None
J1218	Solar	42299	198.06	64.25	0.9512	No	197.82	-146.60	-0.8034	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	2×3 MVAR
J1221	Solar	42306	148.45	46.40	0.9545	No	148.26	-109.57	-0.8042	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×4 MVAR
J1229	Battery	42299	49.50	17.74	0.9414	Yes	49.44	-37.24	-0.7988	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×2 MVAR
J1270	Wind	42703	197.18	50.73	0.9685	No	196.98	-110.11	-0.8729	Yes	Vestas 5.1 MVA WTG PF: +/- 0.95	2×13 MVAR
J1298	Solar	41283	147.59	36.15	0.9713	No	147.14	-111.39	-0.7973	Yes	PE HEM 3.0 MW Inverter PF: +/- 0.909	None
J1312	Solar	43123	49.49	24.53	0.8960	Yes	49.49	-25.83	-0.8865	Yes	PE HEMK 3.432 MW Inverter PF: +/- 0.95	2×7 MVAR
J1313	Solar	43133	98.65	49.84	0.8926	Yes	98.70	-44.76	-0.9107	Yes	PE HEMK 3.59 MW Inverter PF: +/- 0.95	2×14 MVAR
J1314	Solar	43143	98.97	46.97	0.9034	Yes	98.98	-48.23	-0.8990	Yes	PE HEMK 3.59 MW Inverter PF: +/- 0.95	1×28 MVAR
J1315	Wind	43213	593.94	382.02	0.8410	Yes	594.12	-305.46	-0.8893	Yes	Vestas 3.45 MW WTG PF: + 0.90/ - 0.92	6×25 MVAR
J1321	Battery	43213	64.34	21.77	0.9472	Yes	64.25	-48.59	-0.7976	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×2 MVAR
J1325	Solar	43256	79.44	77.28	0.7168	Yes	79.58	-26.56	-0.9486	Yes	PE HEMK 3.59 MW Inverter PF: +/- 0.95	2×26 MVAR
J1328	Solar	43283	96.26	54.62	0.8697	Yes	95.83	-97.14	-0.7023	Yes	SMA 3.52 MW Inverter PF: +/- 0.8	None
J1349	Hybrid	43499	326.20	145.05	0.9137	Yes	324.82	-249.61	-0.7929	Yes	SMA 3.52 MW Inverter PF: +/- 0.8 GE 2.82 MW WTG PF: +/- 0.90	2×15 MVAR
J1365	Wind	43653	165.65	16.01	0.9954	No	165.24	-100.24	-0.8550	Yes	Vestas 2.2 MW WTG PF: +/- 0.95	None
J1371	Wind	43713	152.52	65.83	0.9181	Yes	152.27	-107.24	-0.8176	Yes	Vestas 4.2 MW WTG PF: +0.893 /-0.93	2×15 MVAR
J1395	Solar	43956	199.41	65.33	0.9503	No	199.13	-163.90	-0.7721	Yes	PE HEMK 3.3 MW Inverter PF: +/- 0.90	None
J1413	Solar	44136	231.54	111.48	0.9010	Yes	231.58	-122.74	-0.8836	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	2×4 MVAR
J1416	Solar	44163	145.93	53.68	0.9385	Yes	145.40	-102.39	-0.8176	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	1×4 MVAR
J1417	Battery	44173	73.95	40.35	0.8778	Yes	74.07	-30.25	-0.9258	Yes	TMEIC 3.78 MW Inverter PF: +/- 0.90	None
J1426	Solar	44263	98.69	-38.25	0.9324	Yes	98.56	-64.66	-0.8361	Yes	TMEIC 3.71 MW Inverter PF: +/- 0.916	2×7 MVAR
J1432	Wind	44323	245.25	100.12	0.9258	Yes	244.70	-179.27	-0.8067	Yes	GE 3.03 MW PF: +/- 0.90	2×15 MVAR
J1438	Solar	44383	98.70	35.34	0.9415	Yes	98.59	-61.09	-0.8500	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.926	2×7 MVAR

Table C-1: Reactive Power Requirement Analysis Results

			Lagging Power Factor Results				Leading Power Factor Results					
Project #	TYPE	HV Side Bus #	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?	Inverter Inherent Power Factor	Shunt Compensation
J1444	Solar	44443	197.55	68.55	0.9447	Yes	197.29	-125.45	-0.8439	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.926	2×14 MVAR
J1445	Solar	44453	98.69	37.66	0.9343	Yes	98.56	-64.36	-0.8373	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.917	2×7 MVAR
J1446	Battery	44463	147.68	51.82	0.9436	Yes	147.42	-94.86	-0.8409	Yes	PE 1.29 MVA Inverter PF: +/- 0.926	2×11 MVAR
J1456	Wind	44673	294.92	99.21	0.9478	Yes	294.44	-133.06	-0.9113	Yes	Vestas 2.0 MW WTG PF: +/- 0.95	3×10/-5 MVAR
J1468	Battery	44683	98.42	40.55	0.9246	Yes	98.33	-72.04	-0.8067	Yes	PE 1.29 MVA Inverter PF: +/- 0.90	1×14 MVAR
J1471	Solar	44713	197.51	71.59	0.9401	Yes	197.23	-124.58	-0.8455	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.926	2×14 MVAR
J1474	Solar	44743	197.11	70.72	0.9413	Yes	196.93	-124.38	-0.8455	Yes	TMEIC 4.05 MVA Inverter PF: +/- 0.926	2×14 MVAR
J1477	Battery	41313	49.37	25.76	0.8866	Yes	49.35	-34.42	-0.8202	Yes	PE 3.43 MVA Inverter PF: +/- 0.86	None
J1478	Battery	41353	24.75	14.29	0.8660	Yes	24.76	-15.51	-0.8475	Yes	PE 3.24 MVA Inverter PF: +/- 0.86	None
J1479	Battery	44793	97.98	37.84	0.9328	Yes	97.77	-74.98	-0.7935	Yes	PE 3.43 MVA Inverter PF: +/- 0.86	None
J1485	Wind	44856	247.68	75.08	0.9570	No	247.40	-182.34	-0.8050	Yes	GE 2.52 MW WTG PF: +/- 0.90	2×3 MVAR
J1487	Wind	44873	120.76	47.19	0.9314	Yes	120.38	-87.63	-0.8085	Yes	GE 2.52 MW WTG PF: +/- 0.90	1×11 MVAR

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2024 Summer Shoulder Contingency Analysis Results

D.1 Stage-1 Constraints in 2024 Summer Shoulder (SH) Charging Condition

Table D-1: Stage-1 2024 SH Charging System Intact Thermal Constraints

Table D-2: Stage-1 2024 SH Charging System Intact Voltage Constraints

Table D-3: Stage-1 2024 SH Charging Category P1 Thermal Constraints

Table D-4: Stage-1 2024 SH Charging Category P1 Voltage Constraints

Table D-5: Stage-1 2024 SH Charging Category P2-P7 Thermal Constraints

Table D-6: Stage-1 2024 SH Charging Category P2-P7 Voltage Constraints

Table D-7: Stage-1 2024 SH Charging Non-Converged Contingencies

Table D-8: Stage-1 2024 SH Charging Non-Converged Contingencies DCCC Results

Table D-9: Stage-1 2024 SH Charging Thermal Constraints with Maximum Screened Loading

Table D-10: Stage-1 2024 SH Charging Voltage Constraints with Worst Screened Voltage

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D.2 Stage-2 (with Base Case NUs) Constraints in 2024 Summer Shoulder (SH) Discharging Condition

Table D-11: Stage-2 2024 SH Discharging System Intact Thermal Constraints

Table D-12: Stage-2 2024 SH Discharging System Intact Voltage Constraints

Table D-13: Stage-2 2024 SH Discharging Category P1 Thermal Constraints

Table D-14: Stage-2 2024 SH Discharging Category P1 Voltage Constraints

Table D-15: Stage-2 2024 SH Discharging Category P2-P7 Thermal Constraints

Table D-16: Stage-2 2024 SH Discharging Category P2-P7 Voltage Constraints

Table D-17: Stage-2 2024 SH Discharging Non-Converged Contingencies

Table D-18: Stage-2 2024 SH Discharging Non-Converged Contingencies DCCC Results

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D.3 Stage-2 (with Base Case NUs) Constraints in 2024 Summer Shoulder (SH) Charging Condition

Table D-19: Stage-2 2024 SH Charging System Intact Thermal Constraints

Table D-20: Stage-2 2024 SH Charging System Intact Voltage Constraints

Table D-21: Stage-2 2024 SH Charging Category P1 Thermal Constraints

Table D-22: Stage-2 2024 SH Charging Category P1 Voltage Constraints

Table D-23: Stage-2 2024 SH Charging Category P2-P7 Thermal Constraints

Table D-24: Stage-2 2024 SH Charging Category P2-P7 Voltage Constraints

Table D-25: Stage-2 2024 SH Charging Non-Converged Contingencies

Table D-26: Stage-2 2024 SH Charging Non-Converged Contingencies DCCC Results

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2024 Summer Peak Contingency Analysis Results

E.1 Constraints in 2024 Summer Peak (SPK) Discharging Condition

Table E-1: 2024 SPK Discharging System Intact Thermal Constraints

Table E-2: 2024 SPK Discharging System Intact Voltage Constraints

Table E-3: 2024 SPK Discharging Category P1 Thermal Constraints

Table E-4: 2024 SPK Discharging Category P1 Voltage Constraints

Table E-5: 2024 SPK Discharging Category P2-P7 Thermal Constraints

Table E-6: 2024 SPK Discharging Category P2-P7 Voltage Constraints

Table E-7: 2024 SPK Discharging Non-Converged Contingencies

Table E-8: 2024 SPK Discharging Non-Converged Contingencies DCCC Results

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Local Planning Criteria Analysis Results

F.1 GRE Local Planning Criteria Analysis Results

F.1.1 J1221 GRE Local Planning Criteria Analysis

Below is the J1221 GRE local planning criteria analysis report.

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F.1.2 J1426 GRE Local Planning Criteria Analysis

Below is the J1426 GRE local planning criteria analysis report.

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F.2 OTP LPC Analysis

Below is the OTP local planning criteria analysis report.

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F.3 DPC LPC Analysis

Below is the DPC local planning criteria analysis report.

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Affected System Contingency Analysis Results

G.1 CIPCO Affected System Analysis Results

Table G-1: 2024 SPK Discharging CIPCO Affected System Analysis Results

Table G-2: 2024 SH Charging CIPCO Affected System Analysis Results

Table G-3: 2024 SH Discharging CIPCO Affected System Analysis Results

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G.2 MPC Affected System Analysis Results

Below is the Affected System Analysis report provided by MPC.

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G.3 PJM Affected System Study Results

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G.4 AECI Affected System Study Results

Below is the AECI affected system study report provided by AECI.

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G.5 SPP Affected System Study Results

Below is the SPP affected system study report provided by SPP.

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Transient Stability Results

H.1 2024 Summer Peak Stability Results

Stability simulation was performed in the DPP 2019 Phase 2 2024 summer peak (SPK) stability model. The stability model includes the Hazel Creek-Scott County 345 kV Base Case NU and 2×75 MVar capacitor bank at Cedar Mountain 345 kV (DPP 2018 April voltage NU).

H.1.1 2024 SPK Stability Summary

Summer peak stability study results are summarized in **Table H-1**.

Table H-1: 2024 Summer Peak Phase 2 Stability Analysis Results Summary

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H.1.2 2024 SPK Stability Plots

Plots of stability simulations for 2024 summer peak Phase 2 study case are in separate files which are listed below:

AppendixH1_2024SPK_DPP 2019-West_Ph2_Study_Plots.zip

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H.2 2024 Summer Shoulder Stability Results

Stability simulation was performed in the DPP 2019 Phase 2 2024 summer shoulder (SH) stability model. The stability model includes the Hazel Creek-Scott County 345 kV Base Case NU and 2×75 MVar capacitor bank at Cedar Mountain 345 kV (DPP 2018 April voltage NU).

H.2.1 2024 SH Stability Summary

Summer shoulder stability study results are summarized in Table H-2.

Table H-2: 2024 Summer Shoulder Phase 2 Stability Analysis Results Summary

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H.2.2 2024 SH Stability Plots

Plots of stability simulations for 2024 summer shoulder Phase 2 study case are in separate files which are listed below:

AppendixH2_2024SH_DPP 2019-West_Ph2_Study_Plots.zip

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MWEX Voltage Study Details

Below is the MWEX voltage stability study report provided by ATC.

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Short Circuit Analysis

- J.1 J1190 Short Circuit Study**
- J.2 J1218 Short Circuit Study**
- J.3 J1221 Short Circuit Study**
- J.4 J1229 Short Circuit Study**
- J.5 J1270 Short Circuit Study**
- J.6 J1298 Short Circuit Study**
- J.7 J1312 Short Circuit Study**
- J.8 J1313 Short Circuit Study**
- J.9 J1314 Short Circuit Study**
- J.10 J1315 Short Circuit Study**
- J.11 J1321 Short Circuit Study**
- J.12 J1325 Short Circuit Study**
- J.13 OTP Short Circuit Study (J1328, J1371, J1456, J1485)**
- J.14 J1349 Short Circuit Study**
- J.15 J1359 Short Circuit Study**
- J.16 J1365 Short Circuit Study**
- J.17 J1395 Short Circuit Study**

- J.18 J1413 Short Circuit Study**
- J.19 J1416 Short Circuit Study**
- J.20 J1417 Short Circuit Study**
- J.21 J1426 Short Circuit Study**
- J.22 J1432 Short Circuit Study**
- J.23 J1438 Short Circuit Study**
- J.24 J1444 Short Circuit Study**
- J.25 J1445 Short Circuit Study**
- J.26 J1446 Short Circuit Study**
- J.27 J1468 Short Circuit Study**
- J.28 J1471 Short Circuit Study**
- J.29 J1474 Short Circuit Study**
- J.30 J1477 Short Circuit Study**
- J.31 J1478 Short Circuit Study**
- J.32 J1479 Short Circuit Study**
- J.33 J1487 Short Circuit Study**

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2024 Cost Allocation Results

K.1 Distribution Factor (DF), Voltage Impact, and MW Contribution Results for Cost Allocation in 2024

Table K-1: Voltage Impact on Hazel Creek-Scott County 345 kV Base Case NU Cost Allocation

Table K-2: Distribution Factor and MW Contribution on Constraints for Other Thermal NU Cost Allocation

Table K-3: Voltage Impact on MISO Voltage NUs Cost Allocation

Table K-4: Voltage Impact on OTP LPC Voltage NUs Cost Allocation

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K.2 Cost Allocation Details

Table K-5: Network Upgrades Cost Allocation in 2024

Table K-5: Network Upgrades Cost Allocation in 2024

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